KINETIC ACCELERATION OF NON-EQUILIBRIUM STEADY STATES

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Abstract. We study convergence to non-equilibrium steady states (NESS) for dissipative evolutions on a Hilbert space generated by $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\rm skew} + \eta\mathcal{V}_{\rm pert}$, where $\gamma > 0$ is a friction parameter, $\mathcal{R} \le 0$ is a self-adjoint dissipator, $\mathcal{V}_{\rm skew}$ is skew-adjoint, and $\eta \ge 0$ measures the departure from detailed balance. Our main question is: when does kinetic acceleration (lifting) survive, and how does the non-equilibrium strength η reduce the rate?

Within the Dolbeault–Mouhot–Schmeiser (DMS) hypocoercivity framework [14, 13], we build a modified Lyapunov functional and obtain an explicit η-dependent rate together with a computable certified threshold $\eta^*(\gamma)$. At the optimal friction $\gamma_{\rm opt} \asymp \sqrt{\lambda(Q_O)}$ (where $\lambda(Q_O)$ is the spectral gap of the reversible backbone Q_O in the effective generator $\mathcal{L}_O$), the threshold scales as $\eta^* \asymp \sqrt{\lambda(Q_O)}$ and the rate improves from the overdamped diffusive scale $\mathcal{O}(\lambda(Q_O)/\gamma)$ to the kinetic scale $\mathcal{O}(\sqrt{\lambda(Q_O)})$. An independent overdamped ceiling, valid under the additional assumption that the semigroup is a contraction in the chosen Hilbert norm (automatic at equilibrium, model-dependent for $\eta \ne 0$), shows that this acceleration is genuinely a finite-friction effect and that the $\sqrt{\lambda(Q_O)}$ scaling is exponent-optimal whenever that ceiling applies. Therefore, systems with larger reversible gap can tolerate stronger non-equilibrium driving while keeping accelerated convergence. We also propose a simple algorithmic lift from a given $\mathcal{L}_O$ to a full generator $\mathcal{L}$. Applications are given to forced underdamped Langevin dynamics. A boundary-driven quantum toy model with a two-channel coupling is included as a finite-dimensional illustration that satisfies all structural assumptions, including kernel compatibility, and validates the full $\sqrt{\lambda(Q_O)}$ acceleration.

1. Introduction

1.1. **Motivation and main question.** Many systems of interest—forced diffusions [22, 32], boundary-driven quantum chains [27], and actively controlled open quantum systems [23]—operate away from detailed balance. Their generators contain a non-reversible component of strength η and converge to a non-equilibrium steady state (NESS). For reversible systems ($\eta = 0$), lifting to a larger phase space can accelerate mixing [11, 16, 6, 30]: at optimal intermediate friction, the rate improves from the diffusive scale to its square root (quadratic speedup). More recently, non-reversible lifting mechanisms beyond the reversible-target setting have also been studied from a different viewpoint in Markov-process theory [15]. In this paper we ask:

> For a system that is η away from equilibrium, under what conditions does the kinetic acceleration (lifting) persist, and how does the convergence rate degrade as a function of η?

Concretely, we consider generators on a Hilbert space $\mathcal{H}$ of the form

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\rm skew} + \eta\mathcal{V}_{\rm pert}, \tag{1}$$

where $\gamma > 0$ is a friction (scale-separation) parameter, $\mathcal{R} \le 0$ is self-adjoint with nontrivial kernel $\mathcal{H}_S := \ker(\mathcal{R})$, $\mathcal{V}_{\rm skew}$ is skew-adjoint, and $\eta\mathcal{V}_{\rm pert}$ is a non-reversible perturbation whose strength $\eta \ge 0$ is *given by the physics*. In the singular limit $\gamma \to \infty$, adiabatic elimination [25, 38] reduces the dynamics to an effective slow generator

$$\mathcal{L}_O = -P_S\mathcal{V}\mathcal{R}^+\mathcal{V}P_S$$

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acting on $\mathcal{H}_S$, where P_S is the orthogonal projection onto $\mathcal{H}_S$ and $\mathcal{R}^+$ is the Moore–Penrose pseudoinverse. This overdamped limit gives a decay rate of order $\lambda(\mathcal{L}_O)/\gamma$, i.e., a diffusive bottleneck. We then ask: (i) whether the quadratic acceleration at finite friction survives the non-equilibrium perturbation, and (ii) how large η can be before the accelerated regime breaks down.

1.2. **Main results.** The decomposition (1) isolates a self-adjoint *reversible backbone*

$$Q_O := -P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} P_S, \tag{2}$$

which captures the reversible skeleton of the slow dynamics. Let $\lambda(Q_O)$ denote its spectral gap on the mean-zero slow subspace.

Within the DMS hypocoercivity framework [14, 13], we construct a modified Lyapunov functional $\mathcal{E}_{\alpha,\varpi}(f) = \frac{1}{2}\|f\|^2 + \alpha\,\Re\langle \mathcal{A}_\varpi f, f\rangle$ with an explicit corrector operator $\mathcal{A}_\varpi$ adapted from [22, 7], and establish the following.

Theorem A (Theorem 5.1). *Under Assumptions 4.2–4.4 (and Assumption 4.5 in the unbounded setting), for every $\gamma > 0$ and $\eta \le \eta^*(\gamma)$,*

$$\|e^{t\mathcal{L}} f\| \le C_*(\gamma)\, e^{-\lambda_\eta t}\|f\|, \qquad \forall\, f \in \mathcal{H}_0 \text{ (mean-zero subspace)},\ t \ge 0,$$

with an explicit rate $\lambda_\eta \ge (\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma))/(1+\delta_*(\gamma))$. Here $\Lambda_{\mathrm{hypo}}(\gamma)$ is the unperturbed hypocoercive rate, determined by the spectral gap $\lambda(Q_O)$, the fast dissipation gap λ_R, and the friction γ (see (47) for the explicit formula), while $B_{\mathrm{pert}}(\gamma)$ quantifies the destabilising effect of the non-equilibrium driving (see (48)). The distortion parameter $\delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|$ is explicit and satisfies $\delta_*(\gamma) \le 2^{-1/2}$, so in particular $C_*(\gamma) \le 1+\sqrt{2}$ and

$$\lambda_\eta \ge (2-\sqrt{2})(\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma)).$$

The choice $\varpi_* := \lambda(Q_O)$ is the gap-adapted optimizer of the DMS corrector parameter. The certified threshold $\eta^*(\gamma) := \Lambda_{\mathrm{hypo}}(\gamma)/(2B_{\mathrm{pert}}(\gamma))$ is fully explicit and computable.

Theorem B (Theorem 5.6). *Under Assumptions 4.2–4.4 (and Assumption 4.5 in the unbounded setting), together with the contraction hypothesis and the existence of a slow eigenfunction realizing $\lambda(Q_O)$ stated in Theorem 5.6, any admissible L^2 exponential rate ν with prefactor C satisfies the overdamped ceiling*

$$\nu \le (1+\log C)\left[\frac{(1+\kappa_{\mathrm{ell}})\lambda(Q_O)}{\gamma} + \eta(b_1{+}b_2) + \eta(a_1{+}a_2)\frac{\sqrt{\lambda(Q_O)}}{\gamma}\right].$$

In particular, for $\gamma \to \infty$ and fixed η, the rate is at most $\mathcal{O}(\gamma^{-1}) + \mathcal{O}(\eta)$: the diffusive bottleneck persists, confirming that acceleration is inherently a finite-friction phenomenon.

Taken together, Theorems A and B give a clear picture whenever the additional assumptions of Theorem B are available. The lower bound from Theorem A is fully constructive under our structural hypotheses, whereas the overdamped ceiling from Theorem B should be read as a conditional complement whose applicability depends on the ambient Hilbert norm. At intermediate friction $\gamma \asymp \sqrt{\lambda(Q_O)}$, the rate scales as $\lambda_\eta \asymp \sqrt{\lambda(Q_O)}$, which is a quadratic speedup over the diffusive scale $\lambda(Q_O)$. A concrete illustration of this mechanism—and of the gap-adapted shift that makes it work within the DMS framework—is given for the underdamped Langevin equation in Section 2, before the abstract development. This accelerated regime is guaranteed whenever $\eta \le \eta^*(\gamma)$; inside that certified regime the rate deteriorates linearly in η. Outside that regime, the lower bound from Theorem A no longer certifies acceleration, while Theorem B shows that simply moving to the overdamped regime cannot recover kinetic scaling.

Remark 1.1 (Near-optimality of the accelerated rate). Within the scope of Theorem B, the $\sqrt{\lambda(Q_O)}$ scaling achieved by Theorem A at optimal friction is *exponent-optimal*: Theorem B shows that for every $\gamma > 0$, any admissible L^2 rate satisfies $\nu = \mathcal{O}(\lambda(Q_O)/\gamma) + \mathcal{O}(\eta)$; evaluating at intermediate friction $\gamma \asymp \sqrt{\lambda(Q_O)}$ shows that no rate faster than $\mathcal{O}(\sqrt{\lambda(Q_O)})$ is achievable in this regime. Since

Theorem A also achieves $\Theta(\sqrt{\lambda(Q_O)})$ in this regime, the lower and upper bounds match in the exponent of the spectral gap. The same square-root ceiling has been established independently by Eberle–Lörler [16, 17] and by Brigati–Lörler–Wang [6], confirming that the ceiling is intrinsic to kinetic lifting. We emphasize that optimality is in the *exponent* $\lambda(Q_O)^{1/2}$; the multiplicative constants in Theorems 5.1 and 5.6 are not claimed to be sharp.

1.3. **Kinetic acceleration and non-equilibrium stability.** Our contribution has two parts. First, lifting gives a *quadratic speedup*: at optimal finite friction $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$, the rate improves from $\lambda(Q_O)$ to $\sqrt{\lambda(Q_O)}$. In such perturbative regimes, Remark 4.7 gives $\lambda(\mathcal{L}_O) \approx \lambda(Q_O)$, so this is substantial when the slow gap is small. Second, we give an explicit quantitative description of how non-equilibrium driving changes this acceleration within a DMS-type slow–fast framework built around the self-adjoint backbone Q_O; earlier analyses closest in spirit to ours [11, 16, 6, 30] focus on reversible targets.

The rate degrades *linearly* in η, with slope $B_{\mathrm{pert}}(\gamma)$. At optimal friction, Corollary 5.2 shows a key cancellation: the apparent $\lambda(Q_O)^{-1/2}$ singularity in B_{pert} is offset by the Lyapunov weight. As a result, $B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) = \mathcal{O}(1)$ and $\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}$. Interpretation: faster reversible mixing allows stronger non-equilibrium forcing while keeping kinetic acceleration.

An important subtlety is that the lifted NESS h_η and the overdamped NESS g_η may differ at finite friction: $\|P_S h_\eta - g_\eta\| = \mathcal{O}(\eta/\gamma)$ in general. However, for lifts constructed via the reverse adiabatic elimination (Section 6.1), one can enforce $P_F \mathcal{V} P_F = 0$ as a design choice, eliminating this bias entirely (Proposition 6.1). This is achievable in finite dimensions and, more generally, whenever the consistency relation can be solved within the chosen operator class. In this case $\varepsilon_* = 0$ (see Algorithm 1) and the full quadratic speedup applies for all target accuracies; a two-phase extension remains available for natural physical models where $P_F \mathcal{V} P_F \neq 0$ (Remark 6.3).

1.4. **Related work.** The hypocoercive method was pioneered by Villani [39] via modified H^1-type functionals. The L^2 approach based on modified Lyapunov functionals was developed by Dolbeault–Mouhot–Schmeiser [14, 13], who introduced the auxiliary corrector operator that restores coercivity on the degenerate subspace; this DMS framework is the methodological foundation of the present work. The adaptation to Langevin dynamics in weighted Sobolev spaces, using resolvent-type correctors, was carried out by Iacobucci–Olla–Stoltz [22], Roussel–Stoltz [36], and Cao–Lu–Wang [7]. For open quantum systems, Fang–Lu–Tong recently derived hypocoercive mixing-time bounds that separate Hamiltonian and dissipative effects [18].

Our contribution is closer in spirit to lifting and adiabatic elimination: we use an explicit slow–fast decomposition to quantify finite-friction acceleration and its stability under non-equilibrium perturbations.

The lifting perspective—embedding slow dynamics into an enlarged state space with faster mixing—originates in the Markov chain literature with Chen–Lovász–Pak [11]; fundamental limits on lifting acceleration were identified by Apers–Ticozzi–Sarlette [3]. The programme has been extended to continuous time by Eberle–Lörler [16] and Brigati–Lörler–Wang [6]; see also Eberle–Lörler [17] for an extension to non-reversible diffusions on manifolds with boundary. The quantum analogue was recently developed by Li–Lu [30]. A complementary non-reversible theory based on flow-Poincaré inequalities was developed by Eberle–Guillin–Hahn–Lörler–Michel [15]. Our framework is different in both scope and technique: it is an operator-theoretic slow–fast perturbation theory centered on the self-adjoint reversible backbone Q_O, tailored to non-equilibrium steady states generated by $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$.

In the non-equilibrium setting specifically, Dietert [12] develops L^2-hypocoercivity for kinetic equations with non-equilibrium forcing via a modified DMS approach, but focuses on specific kinetic models rather than the abstract lifting framework pursued here. Monmarché [33] establishes L^2 geometric ergodicity for the kinetic Langevin process with NESS using a Lyapunov-based technique different from the corrector approach. The present paper differs from both in providing an

abstract operator-theoretic framework applicable to arbitrary generators satisfying the structural assumptions, with an explicit dependence on the non-equilibrium strength η and a certified stability threshold.

Quantitative Zeno-limit analysis in the singular regime $\gamma \to \infty$ has recently been developed for boundary-driven quantum systems by Carlen–Huse–Lebowitz [8, 9].

1.5. **Outline.** Section 2 presents a self-contained motivating example: for the underdamped Langevin equation with convex potential, a single modification to the classical DMS corrector—replacing the fixed resolvent shift by the Poincaré gap—upgrades the convergence rate from $O(m^{5/2})$ to $O(\sqrt{m})$, recovering the kinetic square-root scaling. This concrete computation previews the mechanism that drives the abstract results. Section 3 sets up the abstract framework and adiabatic elimination. Section 4 formulates the structural assumptions and introduces the DMS corrector. Section 5 proves the main decay-rate bounds, including the stability threshold and singularity cancellation at optimal friction. Section 6 verifies the full framework for nonequilibrium Langevin dynamics, develops the reverse adiabatic elimination recipe, and studies a boundary-driven quantum toy model as a finite-dimensional illustration satisfying all structural assumptions. Section 7 discusses extensions and open problems.

2. Motivating example: the gap-adapted DMS corrector

The classical DMS hypocoercivity method [14, 13] applied to underdamped Langevin with a fixed resolvent shift $\varpi = 1$ yields a convergence rate of $O(m^{5/2})$ [7, Appendix B], far slower than the expected kinetic scaling $O(\sqrt{m})$. We show that replacing $\varpi = 1$ by the gap-adapted shift $\varpi = m$ upgrades the rate to $O(\sqrt{m})$, following the same notation as [7, Appendix B] for direct comparison.

This computation previews the abstract framework of Section 4: each corrector bound below has a counterpart in Lemma 4.11, and the gap-adapted shift motivates the general choice $\varpi_* = \lambda(Q_O)$ that drives the main results.

2.1. **Setup.** Consider the underdamped Langevin backward Kolmogorov equation on $\mathbb{R}^d_x \times \mathbb{R}^d_v$:

$$\partial_t f = \mathcal{L} f, \qquad \mathcal{L} = \mathcal{L}_{\mathrm{ham}} + \gamma \mathcal{L}_{\mathrm{FD}}, \tag{3}$$

where $\gamma > 0$, $\mathcal{L}_{\mathrm{ham}} = v \cdot \nabla_x - \nabla_x U \cdot \nabla_v$ is the skew-adjoint Hamiltonian transport and $\mathcal{L}_{\mathrm{FD}} = -\nabla_v^* \nabla_v$ is the non-positive fluctuation-dissipation operator. The equilibrium measure is $d\rho_\infty = d\mu(x)\, d\kappa(v)$ with $d\mu \propto e^{-U(x)}\, dx$ and $d\kappa = (2\pi)^{-d/2} e^{-|v|^2/2}\, dv$; all norms and inner products are in $L^2(\rho_\infty)$. Write $f_S := \Pi_v f$ and $f_F := (1 - \Pi_v) f$ for the slow (velocity-averaged) and fast components, so that $\|f\|^2 = \|f_S\|^2 + \|f_F\|^2$. The Gaussian Poincaré inequality gives the fast spectral gap $\|f_F\|^2 \le \|\nabla_v f\|^2$.

The effective slow operator on $L^2(\mu)$ is $Q := -\nabla_x^* \nabla_x$, where $\nabla_x^* = -\nabla_x \cdot + \nabla_x U \cdot$ is the $L^2(\mu)$-adjoint of ∇_x. A direct computation using $\int v_i v_j \, d\kappa = \delta_{ij}$ gives the identity

$$(\mathcal{L}_{\mathrm{ham}} \Pi_v)^* (\mathcal{L}_{\mathrm{ham}} \Pi_v) = \nabla_x^* \nabla_x = -Q. \tag{4}$$

We assume throughout that U is convex and that μ satisfies a Poincaré inequality with constant $m > 0$, so that $-Q \ge m$ on mean-zero functions.

Remark 2.1 (Q as a special case of Q_O). In the abstract framework of Section 4, the reversible backbone is $Q_O = -P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} P_S$, where $\mathcal{R}^+$ is the pseudoinverse of the fast dissipator. For underdamped Langevin, the flux $g_\psi = P_F \mathcal{V}_{\mathrm{skew}} \psi = v \cdot \nabla_x \psi$ lies entirely in the first eigenspace of the Ornstein–Uhlenbeck operator ($\mathcal{R}$ acts as $-\mathrm{id}$ there), so $-\mathcal{R}^+ g_\psi = g_\psi$ and the adiabatic elimination formula collapses to $Q_O = (\mathcal{L}_{\mathrm{ham}} \Pi_v)^* (\mathcal{L}_{\mathrm{ham}} \Pi_v) = -Q$. The operator Q in this section is therefore *identical* to Q_O; we use the simpler notation here for readability.

2.2. **The classical corrector and its bottleneck.** The DMS corrector is defined by a resolvent applied to the flux operator $(\mathcal{L}_{\mathrm{ham}}\Pi_v)^*$. In the classical formulation [13, 7], the resolvent shift is fixed at $\varpi = 1$:

$$\mathcal{A}^{\mathrm{cl}} := (1-Q)^{-1}(\mathcal{L}_{\mathrm{ham}}\Pi_v)^*. \tag{5}$$

For the corresponding Lyapunov functional $\mathsf{E}^{\mathrm{cl}}(\phi) = \frac{1}{2}\|\phi\|^2 - \varepsilon\langle \mathcal{A}^{\mathrm{cl}}\phi, \phi\rangle$, the slow coercivity—the term that transfers dissipation from the fast to the slow sector—takes the form

$$\langle \mathcal{A}^{\mathrm{cl}}\mathcal{L}_{\mathrm{ham}} f_S, f_S\rangle = \Big\langle \frac{-Q}{1+(-Q)} f_S, f_S \Big\rangle \geq \frac{m}{1+m}\|f_S\|^2. \tag{6}$$

For small m (the physically relevant regime of slow effective mixing), the right-hand side is approximately $m\|f_S\|^2$: the slow coercivity vanishes with m, which produces the entry $M_{22}^{\mathrm{cl}} \approx \varepsilon m$ in the dissipation matrix. This leads, after optimisation over the weight ε and friction $\gamma \asymp \sqrt{m}$, to a rate of order $m^{5/2}$ [7, Proposition B.3].

The origin of the problem is clear from (6): the spectral multiplier $\lambda/(1+\lambda)$ is evaluated at $\lambda = m \ll 1$, where it gives approximately m rather than a quantity of order 1.

2.3. **The gap-adapted corrector.** The fix is to choose the resolvent shift equal to the spectral gap m, so that the spectral multiplier is evaluated at its midpoint. We define the modified corrector

$$\mathcal{A}_m := (m-Q)^{-1}(\mathcal{L}_{\mathrm{ham}}\Pi_v)^* = (m + \nabla_x^*\nabla_x)^{-1}(\mathcal{L}_{\mathrm{ham}}\Pi_v)^* \tag{7}$$

and the associated Lyapunov functional $\mathsf{E}_m(\phi) := \frac{1}{2}\|\phi\|^2 - \varepsilon\,\Re\langle \mathcal{A}_m\phi, \phi\rangle$, $\varepsilon > 0$. The slow coercivity now becomes

$$\langle \mathcal{A}_m\mathcal{L}_{\mathrm{ham}} f_S, f_S\rangle = \Big\langle \frac{-Q}{m+(-Q)} f_S, f_S \Big\rangle \geq \frac{m}{m+m}\|f_S\|^2 = \frac{1}{2}\|f_S\|^2, \tag{8}$$

since $-Q \geq m$ on mean-zero functions. The resulting entry $M_{22} = \varepsilon/2$ is now independent of m—this is the key gain.

The structural properties of $\mathcal{A}_m$ needed for the full Lyapunov argument are summarised in the following lemma.

Lemma 2.2 (Corrector estimates)**.** *Let U be convex with Poincaré constant $m > 0$. Then $\mathcal{A}_m$ satisfies:*

(i) $\|\mathcal{A}_m\phi\| \leq \frac{1}{2\sqrt{m}}\|\phi\|$ *;*
(ii) $\|\mathcal{L}_{\mathrm{ham}}\mathcal{A}_m\phi\| \leq \|\phi\|$ *;*
(iii) $\|\mathcal{A}_m\mathcal{L}_{\mathrm{ham}}(1-\Pi_v)\phi\| \leq \sqrt{2}\,\|\phi\|$ (proof in Appendix F) *;*
(iv) if $\varepsilon \leq \sqrt{m}/2$ *then* $\frac{1}{4}\|\phi\|^2 \leq \mathsf{E}_m(\phi) \leq \frac{3}{4}\|\phi\|^2$.

Proof. For (i) and (ii), write $u = \mathcal{A}_m\phi$ and $w = (\mathcal{L}_{\mathrm{ham}}\Pi_v)^*\phi$, so that $(m-Q)u = w$. Taking the inner product with u: $m\|u\|^2 + \langle u, (-Q)u\rangle = \langle w, u\rangle$. By the identity (4), $\|\mathcal{L}_{\mathrm{ham}}u\|^2 = \|(-Q)^{1/2}u\|^2$ for $u \in \mathcal{H}_S$, so $\langle w, u\rangle = \langle \phi, \mathcal{L}_{\mathrm{ham}}u\rangle \leq \|\phi\|\,\|(-Q)^{1/2}u\|$ by Cauchy–Schwarz. For (i), the AM–GM inequality $m\|u\|^2 + \|(-Q)^{1/2}u\|^2 \geq 2\sqrt{m}\|u\|\|(-Q)^{1/2}u\|$ gives the bound. For (ii), the chain $\|(-Q)^{1/2}u\|^2 \leq \langle w, u\rangle \leq \|\phi\|\|(-Q)^{1/2}u\|$ yields $\|\mathcal{L}_{\mathrm{ham}}u\| \leq \|\phi\|$.

Part (iii) requires the Bochner–Weitzenböck inequality on $L^2(\mu)$ and Isserlis' moment formula; the proof is given in Appendix F. Part (iv) follows from (i) and Cauchy–Schwarz: $\varepsilon|\Re\langle \mathcal{A}_m\phi, \phi\rangle| \leq \varepsilon/(2\sqrt{m}) \cdot \|\phi\|^2 \leq \frac{1}{4}\|\phi\|^2$. □

2.4. **Convergence rate.** With these estimates, the Lyapunov argument proceeds along the same lines as [7, Appendix B]. Differentiating $\mathsf{E}_m(f)$ along the flow of (3) and collecting the interaction terms (the corrector cross-terms with $\mathcal{L}_{\mathrm{FD}}$ and $\mathcal{L}_{\mathrm{ham}}$), one arrives at

$$\frac{d}{dt}\mathsf{E}_m(f) \leq -\mathbf{w}^T\mathbf{M}\,\mathbf{w}, \qquad \mathbf{w} = \begin{pmatrix} \|f_F\| \\ \|f_S\| \end{pmatrix}, \tag{9}$$

where the symmetric 2×2 dissipation matrix is

$$\mathbf{M} = \begin{pmatrix} \gamma - \varepsilon & -\frac{\varepsilon}{2} Z(\gamma, m) \\ -\frac{\varepsilon}{2} Z(\gamma, m) & \frac{\varepsilon}{2} \end{pmatrix}, \qquad Z(\gamma, m) := \frac{\gamma}{2\sqrt{m}} + \sqrt{2}. \tag{10}$$

The entries compare with the classical DMS matrix from [7, Appendix B] as follows:

	Classical ($\varpi = 1$)	Modified ($\varpi = m$)	Effect
M_{11}	$\gamma - \varepsilon$	$\gamma - \varepsilon$	identical
M_{12}	$-(\varepsilon/2)(\gamma/2 + \sqrt{2})$	$-(\varepsilon/2)Z(\gamma, m)$	similar
M_{22}	$\varepsilon m/(1+m) \approx \varepsilon m$	$\varepsilon/2$	key gain
Rate	$O(m^{5/2})$	$O(\sqrt{m})$	

The shift ϖ enters through $\gamma\sqrt{M_R}/(2\sqrt{\varpi})$: with $M_R = 1$, this gives $\gamma/2$ (classical) versus $\gamma/(2\sqrt{m})$ (modified), producing the $Z(\gamma, m) = \gamma/(2\sqrt{m}) + \sqrt{2}$ cross-term that decouples the matrix as $m \to 0$.

Optimising ε to maximise the minimum eigenvalue of $\mathbf{M}$ gives:

Proposition 2.3. *Under the assumptions above, for any $\gamma > 0$: $\Lambda_{\mathrm{DMS},m}(\gamma) \geq \gamma/(4(Z(\gamma, m)^2 + 2))$. The optimum over γ is attained at $\gamma_{\mathrm{opt}} = 4\sqrt{m}$, yielding*

$$\Lambda_{\mathrm{DMS},m}(\gamma_{\mathrm{opt}}) \geq C_* \sqrt{m}, \qquad C_* = \frac{2 - \sqrt{2}}{8} \approx 0.073. \tag{11}$$

The full decay estimate is $\|f(t)\| \leq \sqrt{3}\, \exp\!\big(-\frac{2}{3} C_* \sqrt{m}\, t\big)\, \|f_0\|$.

Proof. Maximising $\det \mathbf{M}$ over ε gives $\varepsilon_* = \gamma/(Z^2 + 2)$ and $\det \mathbf{M}|_{\varepsilon_*} = \gamma^2/(4(Z^2 + 2))$. Since $\mathrm{Tr}(\mathbf{M})|_{\varepsilon_*} = \gamma - \varepsilon_*/2 \leq \gamma$, the eigenvalue bound $\lambda_{\min} \geq \det/\mathrm{Tr}$ yields the claimed lower bound. Moreover,

$$\frac{\varepsilon_*}{\sqrt{m}} = \frac{\gamma/\sqrt{m}}{Z(\gamma, m)^2 + 2} = \frac{2x}{(x + \sqrt{2})^2 + 2}, \qquad x := \frac{\gamma}{2\sqrt{m}},$$

and the elementary optimization

$$\sup_{x>0} \frac{2x}{(x + \sqrt{2})^2 + 2} = 1 - \frac{1}{\sqrt{2}} < \frac{1}{2}$$

shows that $\varepsilon_* \leq (1 - \frac{1}{\sqrt{2}})\sqrt{m} < \sqrt{m}/2$ for every $\gamma > 0$. Hence Lemma 2.2(iv) applies at the optimizing weight. Setting $x := \gamma/(2\sqrt{m})$ reduces the problem to maximising $h(x) = x/(2((x + \sqrt{2})^2 + 2))$; the stationarity condition gives $x^2 = 4$, hence $x_* = 2$ and $\gamma_{\mathrm{opt}} = 4\sqrt{m}$. The decay estimate follows by Grönwall's inequality using the norm equivalence from Lemma 2.2(iv). □

2.5. **From the concrete to the abstract.** The four structural properties of the corrector (Lemma 2.2(i)–(iv)) encode the slow–fast interaction without reference to the specific form of the Langevin equation. In the abstract framework (Sections 3–4), they become the standing assumptions on a general decomposition $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$: the corrector bound becomes Lemma 4.11(1) with ϖ replacing m; the regularity bounds become k_{reg} and K_{geom} (Lemma 4.11(2)–(3)); and the 2×2 dissipation matrix (10) becomes the unperturbed matrix $\mathbf{M}_0$ in Theorem 5.1. The key new ingredient is the non-equilibrium perturbation $\eta\mathcal{V}_{\mathrm{pert}}$, which modifies the dissipation matrix by a controlled correction of order η; the certified threshold $\eta^*(\gamma)$ quantifies the forcing range within which the accelerated rate is guaranteed.

3. Preliminaries

We set up the abstract Hilbert-space framework and the adiabatic-elimination reduction used throughout.

3.1. **Hilbert-space setup.** Let $\mathcal{H}$ be a complex Hilbert space with inner product $\langle\cdot,\cdot\rangle$ and norm $\|\cdot\|$. The generator $\mathcal{L}$ (possibly unbounded) generates a strongly continuous semigroup $(\mathcal{P}_t)_{t\geq 0}$ on $\mathcal{H}$, with adjoint $\mathcal{L}^\dagger$.

Concrete realizations. In the noncommutative (quantum) setting, $\mathcal{H} = L^2(\sigma)$ for a faithful state σ on a finite-dimensional space $\mathfrak{h}$, equipped with the KMS inner product $\langle X, Y\rangle_\sigma := \mathrm{tr}(\sigma^{1/2}X^\dagger\sigma^{1/2}Y)$. The generator is then a GKSL Lindbladian $\mathcal{L}(X) = i[H, X] + \sum_\alpha \mathcal{D}^\dagger[L_\alpha]X$. In the classical (PDE) setting, $\mathcal{H} = L^2(\mu)$ for a reference measure μ.

3.2. **Spectral gaps.** Let Π denote the projection onto the stationary direction and $\mathcal{H}_0 := \ker(\Pi)$ the mean-zero subspace. Decompose the generator as $\mathcal{L} = \mathcal{S} + \mathcal{A}$ with $\mathcal{S} := \frac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger) \leq 0$ (symmetric part) and $\mathcal{A} := \frac{1}{2}(\mathcal{L} - \mathcal{L}^\dagger)$ (skew part).

Definition 3.1 (Gap parameters)**.** On $\mathcal{H}_0$ we define the *coercive gap* $\lambda_S := \inf_{f\in\mathcal{H}_0\setminus\{0\}}(-\langle f, \mathcal{S}f\rangle/\|f\|^2)$ and the *spectral gap* $\lambda(\mathcal{L}) := \inf\{\Re\mu : \mu \in \mathrm{Spec}(-\mathcal{L}|_{\mathcal{H}_0})\}$.

When $\mathcal{L}$ is self-adjoint (detailed balance), $\lambda(\mathcal{L}) = \lambda_S$. In the non-reversible regime, it may happen that $\lambda_S = 0$ while $\lambda(\mathcal{L}) > 0$: the skew part $\mathcal{A}$ couples degenerate directions to dissipative ones, producing exponential relaxation by hypocoercivity [39, 14]. The standard hypocoercive estimate asserts $\|\mathcal{P}_t f\| \leq Ce^{-\lambda t}\|f\|$ for $f \in \mathcal{H}_0$; when $C > 1$ is necessary, the relaxation is *strictly hypocoercive*.

Lemma 3.2 (Coercivity characterization)**.** *Assume* $\ker(\mathcal{L}) = \mathbb{C}\mathbf{1}$*. Then:*

(i) If $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$ *and* $-\mathcal{S}|_{\mathcal{H}_0}$ *has a positive spectral gap (automatic in finite dimensions), then the semigroup is coercive (*$C = 1$*). Conversely, coercivity implies* $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$*.*

(ii) If $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$*, any exponential decay estimate requires* $C > 1$*.*

3.3. **Slow–fast decomposition and adiabatic elimination.** Consider the generator $\mathcal{L}_\gamma = \gamma\mathcal{R} + \mathcal{V}$ with self-adjoint dissipator $\mathcal{R} \leq 0$ and interaction $\mathcal{V}$. Define the *slow* and *fast* subspaces

$$\mathcal{H}_S := \ker(\mathcal{R}), \qquad \mathcal{H}_F := \mathcal{H}_S^\perp,$$

with orthogonal projections P_S, $P_F := \mathrm{id} - P_S$. We assume $\mathcal{R}$ has spectral gap $\lambda_R > 0$ on $\mathcal{H}_F$. Any $f \in \mathcal{H}$ decomposes as $f = u + v$ with $u = P_S f$ (macroscopic) and $v = P_F f$ (microscopic). In the quantitative theory below, the stationary direction is generated by a vector $\mathbf{1} \in \mathcal{H}_S$, so Π is the orthogonal projection onto $\mathbb{C}\mathbf{1}$. Because $\mathcal{H}_F \perp \mathcal{H}_S$, this implies

$$\mathcal{H}_F \subset \mathcal{H}_0, \qquad \mathcal{H}_0 = (\mathcal{H}_S \cap \mathcal{H}_0) \oplus \mathcal{H}_F.$$

In particular, if $f \in \mathcal{H}_0$, then both $f_S := P_S f$ and $f_F := P_F f$ remain in $\mathcal{H}_0$.

Definition 3.3 (Centered coupling)**.** The interaction satisfies $P_S\mathcal{V}P_S = 0$.

This centering condition ensures that the effective slow dynamics is second-order in γ^{-1}—generated by slow–fast–slow excursions—rather than driven by an $\mathcal{O}(1)$ tangential drift [32, 9].

Definition 3.4 (Effective generator)**.** Let $\mathcal{R}^+$ denote the Moore–Penrose pseudoinverse of $\mathcal{R}$, bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \leq \lambda_R^{-1}$. Under the centered coupling condition, we define

$$\mathcal{L}_O := -P_S\mathcal{V}\mathcal{R}^+\mathcal{V}P_S. \tag{12}$$

In the context of lifting, $\mathcal{L}_O$ is the effective "target" dynamics one seeks to accelerate. On the macroscopic timescale $\tau = t/\gamma$, the full evolution converges to $e^{\tau\mathcal{L}_O}$ as $\gamma \to \infty$ (quantitative Zeno-limit error bounds are given in [9]). From the rate perspective, Theorem 5.6 shows that the decay is bounded by a diffusive γ^{-1} ceiling in this overdamped regime, motivating the finite-friction analysis of Section 5.

When the interaction is decomposed as $\mathcal{V} = \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ (skew-adjoint part plus perturbation), the centering condition is enforced componentwise: $P_S\mathcal{V}_{\mathrm{skew}}P_S = 0$ and $P_S\mathcal{V}_{\mathrm{pert}}P_S = 0$.

4. The non-equilibrium lifting framework

We now state the structural assumptions used for quantitative hypocoercivity at finite friction. The goal is to isolate the ingredients that transfer dissipation from fast modes to slow modes through the DMS corrector [14, 13, 22]: fast spectral gap, centering, flux regularity, and perturbation bounds.

Remark 4.1 (Methodological context). Several approaches to hypocoercivity are available; each has its natural domain of applicability. Villani's H^1 method [39] constructs Lyapunov functionals in higher Sobolev norms and applies to kinetic Fokker–Planck equations with detailed balance; verifying the required coercivity for lifted NESS generators appears difficult. The variational approach of Albritton–Armstrong–Mourrat–Novack [2] yields sharp regularity estimates for kinetic Fokker–Planck equations but is specific to that PDE structure. For linear/Gaussian models, the matrix-Lyapunov method of Achleitner–Arnold–Carlen [1] gives optimal rates; the 2×2 matrix analysis in our warm-up (Section 2) is a special case of this technique. We adopt the DMS L^2 corrector framework because it operates directly at the level of spectral gaps and extends naturally to non-self-adjoint (NESS) settings; it has JFA precedent in the operator-theoretic analysis of Grothaus–Stilgenbauer [20], and has been applied to quantum kinetic equations (the Wigner–Fokker–Planck equation) by Arnold et al. [4]. The key novelty here is the gap-adapted resolvent shift $\varpi_* = \lambda(Q_O)$ in the corrector, which upgrades the classical DMS rate from $\mathcal{O}(\lambda^{5/2})$ to $\mathcal{O}(\sqrt{\lambda})$.

4.1. **Structural assumptions.** Let $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$ be an orthogonal slow–fast decomposition. We study generators depending on friction $\gamma > 0$ and perturbation strength $\eta \geq 0$:

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}. \tag{13}$$

The transport term is decomposed as $\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$, and P_S, P_F denote the orthogonal projections onto $\mathcal{H}_S, \mathcal{H}_F$.

The next assumption packages exactly the hypotheses needed in Section 5: fast coercivity, centered coupling, control of slow-induced fluxes, and energy-norm bounds for the perturbation.

Assumption 4.2 (Structural Conditions on the Non-Equilibrium Generator)**.** The system is defined on $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$. The generator is $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$. We assume the following structural conditions:

(1) **Microscopic Spectral Gap.** The fast dissipator $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$, with kernel $\ker(\mathcal{R}) = \mathcal{H}_S$. Moreover, it is strictly dissipative on $\mathcal{H}_F$: there exists $\lambda_R > 0$ such that

$$-\langle \mathcal{R}f, f\rangle \geq \lambda_R \|f\|^2, \qquad \forall f \in \mathcal{H}_F. \tag{14}$$

In particular, $(-\mathcal{R})^{1/2}$ is well-defined by functional calculus. This allows defining the Moore–Penrose pseudoinverse $\mathcal{R}^+$, which is a bounded self-adjoint operator on $\mathcal{H}_F$ with $-\mathcal{R}^+ \geq 0$ and $\|\mathcal{R}^+\| \leq \lambda_R^{-1}$ (the bound follows from the spectral theorem: $\mathcal{R}|_{\mathcal{H}_F} \leq -\lambda_R$ implies $\|(\mathcal{R}|_{\mathcal{H}_F})^{-1}\| \leq \lambda_R^{-1}$, valid for both bounded and unbounded self-adjoint operators without requiring compact resolvent).

(2) **Centered Coupling.** We assume that the interaction does not generate an $\mathcal{O}(1)$ drift on the slow manifold in the macroscopic time scaling $\tau = t/\gamma$. Equivalently, the full interaction $\mathcal{V} := \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ satisfies

$$P_S \mathcal{V} P_S = 0. \tag{15}$$

Since η is a free parameter in our framework, we enforce this componentwise:

$$P_S \mathcal{V}_{\text{skew}} P_S = 0, \qquad P_S \mathcal{V}_{\text{pert}} P_S = 0. \tag{16}$$

This matches the centered coupling condition (Definition 3.3) used to define the effective generator $\mathcal{L}_O$.

(3) **Fast-to-slow Regularity (Flux conditioning and flux regularity).** For $\psi \in \mathrm{dom}(Q_O) \subset \mathcal{H}_S$ (where Q_O is the reversible backbone defined in (25) below), define its *microscopic flux* induced by transport as

$$g_\psi := P_F \mathcal{V}_{\mathrm{skew}} P_S \psi \in \mathcal{H}_F. \tag{17}$$

In the bounded-operator (finite-dimensional) setting, Q_O is a bounded self-adjoint operator on $\mathcal{H}_S$ and thus $\mathrm{dom}(Q_O) = \mathcal{H}_S$. In unbounded settings, the identities below are understood on the common analytic core from Assumption 4.5 and then extended by density to the relevant energy domains. We assume a **flux conditioning bound**: there exists a constant $M_R \in (0, \infty)$ such that

$$\|g_\psi\|^2 \le M_R \langle g_\psi, -\mathcal{R}^+ g_\psi \rangle, \qquad \forall \psi \in \mathrm{dom}(Q_O) \subset \mathcal{H}_S. \tag{18}$$

Since (18) is monotone in M_R, if it holds for some constant then it also holds for any larger one; throughout, M_R may be taken to be any convenient upper bound (at the cost of more conservative constants in subsequent estimates). In the bounded-operator (finite-dimensional) setting, since $-\mathcal{R} \ge 0$ is bounded on $\mathcal{H}_F$, one may take $M_R := \|\mathcal{R}\|$ (equivalently $\|-\mathcal{R}\|$ on $\mathcal{H}_F$). In particular, for all $h \in \mathcal{H}_F$,

$$\langle (-\mathcal{R})h, h \rangle \le M_R \|h\|^2. \tag{19}$$

We also assume a **flux regularity bound**: applying the fast–fast component of the transport to the corrected flux is controlled by the induced slow operator. Concretely, there exists $\kappa_{\mathrm{ell}} > 0$ such that for all $\psi \in \mathrm{dom}(Q_O)$,

$$\|P_F \mathcal{V}_{\mathrm{skew}} P_F \left(-\mathcal{R}^+\right) g_\psi\| \le \ \kappa_{\mathrm{ell}} \|P_S \mathcal{V}_{\mathrm{skew}} P_F (-\mathcal{R}^+) g_\psi\| \ = \ \kappa_{\mathrm{ell}} \|(-Q_O)\psi\|. \tag{20}$$

(4) **Structural Decomposition of the Perturbation.** The perturbation operator $\mathcal{V}_{\mathrm{pert}}$ admits a decomposition into a skew-symmetric part and a symmetric part, each controlled in the $(-\mathcal{R})^{1/2}$-energy norm $\|\cdot\|_{(1)}$:

$$\mathcal{V}_{\mathrm{pert}} = \mathcal{K}_{\mathrm{pert}} + \mathcal{S}_{\mathrm{pert}}, \tag{21}$$

satisfying the following conditions:

(a) **Skew Perturbation Control:** The operator $\mathcal{K}_{\mathrm{pert}}$ is skew-symmetric, i.e., for all $f \in \mathrm{dom}(\mathcal{V}_{\mathrm{pert}})$:

$$\Re \langle \mathcal{K}_{\mathrm{pert}} f, f \rangle = 0. \tag{22}$$

Moreover, $\mathcal{K}_{\mathrm{pert}}$ is relatively bounded with respect to $(-\mathcal{R})^{1/2}$: there exist constants $a_1, b_1 \ge 0$ such that for all $f \in \mathrm{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{K}_{\mathrm{pert}} f\| \le a_1 \|(-\mathcal{R})^{1/2} f\| + b_1 \|f\|. \tag{23}$$

(b) **Symmetric Perturbation Control:** The operator $\mathcal{S}_{\mathrm{pert}}$ is symmetric and defined at least on the energy domain $\mathrm{dom}((-\mathcal{R})^{1/2})$. There exist constants $a_2, b_2 \ge 0$ such that for all $f \in \mathrm{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{S}_{\mathrm{pert}} f\| \le a_2 \|(-\mathcal{R})^{1/2} f\| + b_2 \|f\|. \tag{24}$$

Equivalently, $\mathcal{S}_{\mathrm{pert}}$ is bounded as a map from $(\mathrm{dom}((-\mathcal{R})^{1/2}), \|\cdot\|_{(1)})$ to $\mathcal{H}$. (In PDE models, $\mathcal{S}_{\mathrm{pert}}$ is often an unbounded multiplier on plain L^2; (24) is the standard replacement allowing it to be controlled by the fast dissipation.)

Under Assumption 4.2, we recall the effective generator from Definition 3.4, $\mathcal{L}_O = -P_S \mathcal{V} \mathcal{R}^+ \mathcal{V} P_S$. To construct the Lyapunov functional, we isolate the reversible contribution to the effective diffusion via the **reversible backbone** Q_O:

$$Q_O := -P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} P_S. \tag{25}$$

From the skew-adjointness of $\mathcal{V}_{\text{skew}}$ and the negativity of $\mathcal{R}^+$, it follows that for any $f \in \mathcal{H}_S$,

$$\langle f, Q_O f\rangle = -\langle f, P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S f\rangle = -\langle \mathcal{V}_{\text{skew}}^\dagger f, \mathcal{R}^+\mathcal{V}_{\text{skew}} f\rangle = \langle \mathcal{V}_{\text{skew}} f, \mathcal{R}^+\mathcal{V}_{\text{skew}} f\rangle \le 0. \tag{26}$$

Thus, Q_O is a self-adjoint, dissipative operator on $\mathcal{H}_S$.

Remark 4.3 (Well-definedness of Q_O in unbounded settings). In finite dimensions the composition $Q_O = -P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S$ is automatic, but in infinite-dimensional kinetic models $\mathcal{V}_{\text{skew}}$ is an unbounded differential operator and one must check that $\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S\psi \in \text{dom}(\mathcal{V}_{\text{skew}})$. Assumption 4.5 isolates this bookkeeping: one works first on a common core $\mathcal{C}$ (e.g. Schwartz functions in the Langevin example), where the compositions are well defined, and then extends the resulting identities to the required energy spaces by density.

Assumption 4.4 (Kernel-Compatible Spectral Gap of Effective Dynamics)**.** We assume that the stationary direction is the whole kernel of the reversible backbone on the slow space:

$$\ker(Q_O) = \mathbb{C}\mathbf{1} \subset \mathcal{H}_S.$$

Equivalently, on the mean-zero slow subspace $\mathcal{H}_{S,0} := \mathcal{H}_S \cap \mathcal{H}_0$, the operator Q_O admits a **spectral gap** $\lambda(Q_O) > 0$, i.e.

$$-\langle f, Q_O f\rangle \ge \lambda(Q_O)\|f\|^2, \qquad \forall f \in \mathcal{H}_{S,0}. \tag{27}$$

Consequently, for any shift parameter $\varpi > 0$, the operator $(\varpi - Q_O)$ is strictly positive definite on $\mathcal{H}_S$ and invertible, with operator norm $\|(\varpi - Q_O)^{-1}\| \le \varpi^{-1}$.

In unbounded settings, we additionally assume (as part of the model-dependent analytic hypotheses) that $(\varpi - Q_O)^{-1}$ maps $\mathcal{H}_S$ into $\text{dom}(Q_O)$.

Assumption 4.5 (Analytic Core in Unbounded Settings)**.** This assumption is vacuous in finite dimensions. In unbounded settings, we fix a dense subspace

$$\mathcal{C} \subset \text{dom}(\mathcal{L}) \cap \text{dom}((-\mathcal{R})^{1/2})$$

such that:

(1) $\mathcal{C}$ is invariant under P_S, P_F, $\mathcal{R}^+$, $\mathcal{V}_{\text{skew}}$, and $\mathcal{V}_{\text{pert}}$, and $\mathcal{C} \cap \mathcal{H}_S$ is a core for Q_O;
(2) for every $\varpi > 0$, the resolvent $(\varpi - Q_O)^{-1}$ maps $\mathcal{H}_S$ into $\text{dom}(Q_O)$ and preserves $\mathcal{C} \cap \mathcal{H}_S$;
(3) all operator compositions appearing in the definitions of Q_O, $\mathcal{A}_\varpi$, and in the Lyapunov and singular-value computations of Appendices B–D are well defined on $\mathcal{C}$, and the corresponding bilinear forms extend continuously from $\mathcal{C}$ to the energy spaces on which the theorems are stated.

Remark 4.6 (Why kernel compatibility is needed). The quantitative estimate in Theorem 5.1 is an L^2 decay statement on the whole mean-zero space $\mathcal{H}_0$. For that reason, the reversible backbone must have no neutral directions there: if $\ker(Q_O) \cap \mathcal{H}_0 \neq \{0\}$, then the DMS coercivity term controls only the quotient by those extra zero modes, and one cannot conclude exponential decay on all of $\mathcal{H}_0$ without adding further structure. Section 6.3 provides a finite-dimensional illustration of this obstruction.

Remark 4.7 (Perturbative comparison: $\mathcal{L}_O$ vs. Q_O). By the expansion (74),

$$\mathcal{L}_O = Q_O + \eta\,\mathcal{B}_1 + \eta^2\,\mathcal{B}_2, \tag{28}$$

with model-dependent bounded operators $\mathcal{B}_1, \mathcal{B}_2$ on $\mathcal{H}_S$ in the finite-dimensional setting (and relatively bounded perturbations on the common core in the unbounded examples). Hence

$$\delta_\eta := \|\mathcal{L}_O - Q_O\| \le c_1\eta + c_2\eta^2 \tag{29}$$

for suitable constants $c_1, c_2 > 0$. Assume that the stationary eigenvalue 0 of $-Q_O$ is isolated. Assume also that the low-lying spectrum of $-Q_O$ near $\lambda(Q_O)$ is discrete, as is automatic in finite dimensions and in many compact-resolvent settings. If

$$\delta_\eta \le \theta\,\lambda(Q_O), \qquad 0 < \theta < 1, \tag{30}$$

then bounded-perturbation theory for the self-adjoint operator Q_O implies that the spectrum of $-\mathcal{L}_O$ stays within the δ_η-neighborhood of the spectrum of $-Q_O$; see [24, Chapter V] and, in finite dimensions, [5, III.6.3]. In particular, every nonzero spectral point $z \in \sigma(-\mathcal{L}_O|_{\mathcal{H}_{S,0}})$ satisfies

$$\Re z \ge \lambda(Q_O) - \delta_\eta \ge (1-\theta)\lambda(Q_O). \tag{31}$$

If moreover $\lambda(Q_O)$ is attained by an eigenvalue of $-Q_O$, then there exists a spectral point $z_\eta \in \sigma(-\mathcal{L}_O|_{\mathcal{H}_{S,0}})$ with $|z_\eta - \lambda(Q_O)| \le \delta_\eta$, giving $\lambda(\mathcal{L}_O) \le \Re z_\eta \le (1+\theta)\lambda(Q_O)$, where $\lambda(\mathcal{L}_O) := \inf\{\Re z : z \in \sigma(-\mathcal{L}_O|_{\mathcal{H}_{S,0}})\}$. Therefore, in this perturbative isolated-spectrum regime,

$$(1-\theta)\lambda(Q_O) \le \lambda(\mathcal{L}_O) \le (1+\theta)\lambda(Q_O),$$

and the scaling statement $\lambda_\eta \asymp \sqrt{\lambda(Q_O)}$ can equally be read as $\lambda_\eta \asymp \sqrt{\lambda(\mathcal{L}_O)}$.

Remark 4.8 (Scope: self-adjointness excludes Liouvillian Exceptional Points). Because Q_O is self-adjoint, it is normal and cannot exhibit defective eigenvalues or exceptional points. Assumption 4.4 therefore asks only for a positive spectral gap above the stationary mode; no separate diagonalizability or completeness hypothesis is needed. Possible Liouvillian exceptional points belong to the non-self-adjoint effective generator $\mathcal{L}_O$ (or to the full lifted dynamics), not to Q_O itself [35].

4.2. **The DMS corrector.** To close the hypocoercive estimate, we introduce a bounded corrector that solves a shifted slow-space Poisson problem. Following the Dolbeault–Mouhot–Schmeiser (DMS) approach [14, 13], extended to L^2 settings by Iacobucci–Olla–Stoltz [22] and Cao–Lu–Wang [7], this operator captures the response of the slow component to fast transport excursions and provides the cross term that restores coercivity at the Lyapunov level. This is the abstract counterpart of the concrete corrector $\mathcal{A}_m$ from Section 2, with the resolvent shift ϖ replacing the Poincaré gap m.

Definition 4.9 (DMS Corrector)**.** Let $\varpi > 0$ be a shift parameter, to be optimized in Section 5; it is distinct from the reference measures μ_0, μ_η appearing in the examples. We define the **corrector** $\mathcal{A}_\varpi : \mathcal{H} \to \mathcal{H}_S$ by

$$\mathcal{A}_\varpi := (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+) P_F. \tag{32}$$

The key obstruction to a direct energy estimate is that dissipation is degenerate on $\mathcal{H}_S$: the plain energy $\frac{1}{2}\|f\|^2$ cannot produce a spectral gap when $\ker(\mathcal{R})$ is non-trivial.

Proposition 4.10 (Why the plain energy method fails)**.** *Assume $\eta = 0$ and $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}}$ with $\mathcal{V}_{\text{skew}}$ skew-adjoint and $\ker(\mathcal{R}) = \mathcal{H}_S \neq \{0\}$. Then there is no constant $\lambda > 0$ such that*

$$\frac{d}{dt}\frac{1}{2}\|f_t\|^2 \le -\lambda\,\|f_t\|^2 \qquad \textit{for all solutions } \partial_t f_t = \mathcal{L} f_t. \tag{33}$$

Proof. For any $\phi \in \mathcal{H}_S$ with $\phi \neq 0$, one has $\mathcal{R}\phi = 0$ and $\Re\langle \mathcal{V}_{\text{skew}}\phi, \phi\rangle = 0$, hence

$$\frac{d}{dt}\frac{1}{2}\|f_t\|^2\Big|_{t=0,\, f_0=\phi} = \Re\langle \mathcal{L}\phi, \phi\rangle = \gamma\langle \mathcal{R}\phi, \phi\rangle + \Re\langle \mathcal{V}_{\text{skew}}\phi, \phi\rangle = 0. \tag{34}$$

If such a $\lambda > 0$ existed, we would get $0 \le -\lambda\|\phi\|^2 < 0$, a contradiction. □

Proposition 4.10 confirms that the energy $\frac{1}{2}\|f\|^2$ alone is degenerate on $\mathcal{H}_S$. The remedy is to augment it with a cross term involving the corrector $\mathcal{A}_\varpi$; differentiating $\Re\langle \mathcal{A}_\varpi f, f\rangle$ along the flow generates a slow coercive contribution tied to $-Q_O$, together with controlled remainders from the fast and perturbative components.

The following lemma establishes the critical boundedness properties of $\mathcal{A}_\varpi$, including the control over non-equilibrium perturbations. The reader may compare with the concrete Langevin analogues in Lemma 2.2.

Lemma 4.11 (Structural Estimates for the DMS Corrector)**.** *Let $\varpi > 0$ be the resolvent shift parameter. The corrector $\mathcal{A}_\varpi := (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+) P_F$ satisfies the following structural estimates, where Q_O is the reference effective dynamics.*

Under Assumption 4.2, together with Assumption 4.5 in the unbounded setting, the operator $\mathcal{A}_\varpi$ satisfies the following structural estimates, with constants depending solely on the structural parameters $(M_R, \lambda_R, \kappa_{\text{ell}}, a_1, b_1, a_2, b_2)$ and not *on ϖ:*

(1) ***Corrector Boundedness:***

$$\|\mathcal{A}_\varpi\| \le k_{\mathcal{A}} \varpi^{-1/2}, \quad \text{with } k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}. \tag{35}$$

(2) ***Regularity Estimates (Forward and Dual):*** *The corrector acts as a bounded regularizer for the skew-symmetric transport in both directions:*

$$\|\mathcal{V}_{\text{skew}} \mathcal{A}_\varpi\| \le k_{\text{reg}}, \quad \text{with } k_{\text{reg}} := \sqrt{M_R/\lambda_R}, \tag{36a}$$

$$\|\mathcal{A}_\varpi \mathcal{V}_{\text{skew}} P_F\| \le k_{\text{dual}}, \quad \text{with } k_{\text{dual}} := \kappa_{\text{ell}}. \tag{36b}$$

The dual estimate (36b) *is a direct consequence of the Macroscopic Flux Regularity (Assumption 4.2.3).*

(3) ***Perturbation Control:*** *The interaction with the non-equilibrium perturbation is controlled in both directions:*

$$\|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\varpi\| \le k_P(\varpi), \tag{37a}$$

$$\|\mathcal{A}_\varpi \mathcal{V}_{\text{pert}}\| \le \tilde{k}_P(\varpi). \tag{37b}$$

with control constants

$$k_P(\varpi) := (b_1 + b_2) k_{\mathcal{A}}\, \varpi^{-1/2}, \tag{38}$$

$$\tilde{k}_P(\varpi) := \Big((b_1 + b_2) k_{\mathcal{A}} + \tfrac{a_1 + a_2}{2} \Big) \varpi^{-1/2}. \tag{39}$$

Moreover, the centered perturbation coupling in Assumption 4.2.2 implies

$$P_S \mathcal{V}_{\text{pert}} P_S = 0 \qquad \Longrightarrow \qquad P_S \mathcal{V}_{\text{pert}}^\dagger P_S = 0, \tag{40}$$

and therefore

$$\mathcal{V}_{\text{pert}}\, \mathcal{H}_S \subset \mathcal{H}_F, \qquad \mathcal{V}_{\text{pert}}^\dagger\, \mathcal{H}_S \subset \mathcal{H}_F. \tag{41}$$

Proof. See Appendix B. □

4.3. **The modified Lyapunov functional.** We define a modified squared norm that explicitly incorporates the correlation between slow and fast degrees of freedom mediated by the corrector $\mathcal{A}_\varpi$.

Definition 4.12 (Modified Lyapunov functional [14, 13])**.** For any $f \in \mathcal{H}$ and a parameter $\alpha \in (0, 1)$, we define the functional:

$$\mathcal{E}_{\alpha,\varpi}(f) := \frac{1}{2}\|f\|^2 + \alpha \Re\langle \mathcal{A}_\varpi f, f\rangle. \tag{42}$$

This functional defines a norm equivalent to the underlying Hilbert norm, provided the weight α is sufficiently small relative to the bounded solver $\mathcal{A}_\varpi$.

Proposition 4.13 (Norm equivalence)**.** *If the parameter α satisfies $\alpha\|\mathcal{A}_\varpi\| \le 1/2$, then for all $f \in \mathcal{H}$:*

$$\frac{1}{2}(1 - 2\alpha\|\mathcal{A}_\varpi\|)\|f\|^2 \le \mathcal{E}_{\alpha,\varpi}(f) \le \frac{1}{2}(1 + 2\alpha\|\mathcal{A}_\varpi\|)\|f\|^2. \tag{43}$$

Proof. The proof follows directly from the Cauchy–Schwarz inequality $|\Re\langle\mathcal{A}_\varpi f, f\rangle| \le \|\mathcal{A}_\varpi\|\|f\|^2$ and the triangle inequality. □

5. Convergence bounds for the lifted semigroup

We now derive quantitative decay bounds for the semigroup on the mean-zero subspace. The strategy has two parts: a constructive lower bound from a 2×2 dissipation matrix, and a complementary overdamped upper ceiling from singular values. Together they identify the finite-friction acceleration window.

5.1. Lower bound via modified hypocoercivity.

This subsection gives an explicit lower bound on the decay rate by jointly optimizing the Lyapunov parameters (α, ϖ). The optimization is fully algebraic and returns both the friction-dependent profile and the admissible perturbation range.

The complete matrix estimates and parameter optimization are deferred to Appendix C; the explicit formulas used throughout the sequel are stated below.

Notation for the rate functions. The following structural scalars, depending only on the parameters in Assumption 4.2, are fixed throughout this section:

$$\Theta := \frac{\sqrt{M_R}}{2}, \qquad K_{\text{geom}} := \kappa_{\text{ell}}, \qquad b_{2,\text{eff}} := b_2 + \frac{a_2}{2}(1 + \lambda_R), \tag{44}$$

where κ_{ell} is the flux regularity constant from (20). Interpretation: Θ measures slow–fast flux amplitude, K_{geom} collects transport geometry in the cross term (the forward regularity constant k_{reg} enters only through M_{11} and the trace, see Appendix C), and $b_{2,\text{eff}}$ records the fast-mode amplification from $\|(-\mathcal{R})^{1/2}f\|^2 \ge \lambda_R\|f_F\|^2$.

For each friction $\gamma > 0$, define the *coupling scale*

$$Z(\gamma) := \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}, \tag{45}$$

which measures effective slow–fast interaction strength as a function of γ, and set:

$$\alpha_*(\gamma) := \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}, \tag{46}$$

$$\Lambda_{\text{hypo}}(\gamma) := \frac{\frac{1}{4}\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}, \tag{47}$$

$$B_{\text{pert}}(\gamma) := b_{2,\text{eff}} + \alpha_*(\gamma)\, C_{\text{sing}}\, \lambda(Q_O)^{-1/2}, \tag{48}$$

where $C_{\text{sing}} := 2(b_1 + b_2)\, k_{\mathcal{A}} + (a_1 + a_2)$ is the perturbation interaction constant (see Appendix C for the derivation), with $k_{\mathcal{A}}$ the resolvent norm bound from Lemma 4.11. Here $\alpha_*(\gamma)$ is the optimal Lyapunov mixing weight, $\Lambda_{\text{hypo}}(\gamma)$ is the lower-rate profile, and $B_{\text{pert}}(\gamma)$ quantifies perturbation destabilization. Although B_{pert} contains an apparent $\lambda(Q_O)^{-1/2}$ factor, it is cancelled at optimal friction because $\alpha_*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}$ (Corollary 5.2).

With the gap-adapted choice $\varpi_* := \lambda(Q_O)$, define the associated Lyapunov-distortion parameter and norm-equivalence prefactor by

$$\delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|, \tag{49}$$

$$C_*(\gamma) := \sqrt{\frac{1 + \delta_*(\gamma)}{1 - \delta_*(\gamma)}}. \tag{50}$$

Main result. The structural properties established in Appendix C yield an explicit norm equivalence with distortion parameter $\delta_*(\gamma) < 1$ for all $\gamma > 0$. In fact the appendix proves the uniform bound $\delta_*(\gamma) \le 2^{-1/2}$, hence $C_*(\gamma) \le 1 + \sqrt{2}$.

Theorem 5.1 (Explicit quantitative hypocoercivity)**.** *Suppose Assumptions 4.2 and 4.4 hold, and in the unbounded setting also Assumption 4.5. Set the explicit certified threshold*

$$\eta^*(\gamma) := \frac{\Lambda_{\mathrm{hypo}}(\gamma)}{2B_{\mathrm{pert}}(\gamma)}.$$

For any $\gamma > 0$, if $\eta \le \eta^*(\gamma)$, then $e^{t\mathcal{L}}$ is exponentially stable on $\mathcal{H}_0$:

$$\|e^{t\mathcal{L}} f\| \le C_*(\gamma)\, e^{-\lambda_\eta t}\|f\| \le (1+\sqrt{2})\, e^{-\lambda_\eta t}\|f\|, \quad \forall t \ge 0,\ \forall f \in \mathcal{H}_0. \tag{51}$$

The global decay rate λ_η satisfies the strict lower bound:

$$\lambda_\eta \ge \frac{\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma)}{1+\delta_*(\gamma)} \ge (2-\sqrt{2})\left(\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma)\right). \tag{52}$$

Sketch of Proof. The full computation is in Appendix C. With the resolvent shift fixed at $\varpi_* = \lambda(Q_O)$, the Lyapunov derivative satisfies $\frac{d}{dt}\mathcal{E}_{\alpha,\varpi_*} \le -\mathbf{v}^T\mathbf{M}(\alpha)\mathbf{v}$ for the vector $\mathbf{v} = (\|P_F f\|, \|P_S f\|)^T$ and a 2×2 positive-definite dissipation matrix $\mathbf{M}(\alpha)$. Maximising over α yields the optimal weight $\alpha_*(\gamma)$ from (46) and the rate $\Lambda_{\mathrm{hypo}}(\gamma)$. The distortion parameter $\delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|$ satisfies

$$\delta_*(\gamma) \le \frac{\sqrt{\lambda_R}}{\sqrt{M_R}\big(K_{\mathrm{geom}} + \sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}}\big)} \le 2^{-1/2} \qquad \forall\, \gamma > 0, \tag{53}$$

by the solver bound $\|\mathcal{A}_{\varpi_*}\| \le (2\sqrt{\lambda_R})^{-1}\lambda(Q_O)^{-1/2}$, the formula for α_* in (46), and the elementary extremum $\sup_{x>0} x\big[(x+K_{\mathrm{geom}})^2 + 2k_{\mathrm{reg}}\big]^{-1} = 1/\big(2(K_{\mathrm{geom}} + \sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}})\big)$. This gives the explicit norm equivalence encoded by $C_*(\gamma)$, together with the uniform simplification $C_*(\gamma) \le 1+\sqrt{2}$ and the lower bound in (52). □

Regime summary. The formula (47) has two clear regimes. For $\gamma \to \infty$, the denominator behaves like $Z(\gamma)^2 \sim \gamma^2\Theta^2\lambda(Q_O)^{-1}$, so $\Lambda_{\mathrm{hypo}}(\gamma) \asymp \lambda(Q_O)\,\gamma^{-1}$ (diffusive). For small or order-one γ, the denominator is dominated by the constant part $K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}$, so $\Lambda_{\mathrm{hypo}}(\gamma) \asymp \gamma$. Hence the maximum is attained at intermediate friction and scales as $\max_\gamma \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}$.

Corollary 5.2 (Kinetic acceleration and singularity cancellation)**.** *Assume the setting of Theorem 5.1. Assume furthermore that the structural constants Θ, K_{geom}, k_{reg}, λ_R, $b_{2,\mathrm{eff}}$, and C_{sing} are independent of $\lambda(Q_O)$ (in particular, they remain $\mathcal{O}(1)$ as $\lambda(Q_O) \downarrow 0$), and that $b_{2,\mathrm{eff}} > 0$.*

Then the unperturbed rate function $\Lambda_{\mathrm{hypo}}(\gamma)$ defined in (47) attains its global maximum at

$$\gamma_{\mathrm{opt}} = \frac{\sqrt{\lambda(Q_O)}}{\Theta}\sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}} \asymp \sqrt{\lambda(Q_O)}. \tag{54}$$

At this friction scale, the maximal lower bound exhibits kinetic scaling:

$$\max_{\gamma>0} \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}. \tag{55}$$

Moreover, the mixing weight in (48) satisfies

$$\alpha_*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}, \qquad \alpha_*(\gamma_{\mathrm{opt}})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1), \tag{56}$$

so the solver singularity $\lambda(Q_O)^{-1/2}$ in $B_{\mathrm{pert}}(\gamma)$ is cancelled at the optimal friction. Consequently, $B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) = \mathcal{O}(1)$ and the certified threshold satisfies

$$\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}. \tag{57}$$

Proof. Write

$$\Lambda_{\mathrm{hypo}}(\gamma) = \frac{\lambda_R}{4}\,\frac{\gamma}{\big(\gamma\Theta\lambda(Q_O)^{-1/2} + K_{\mathrm{geom}}\big)^2 + 2k_{\mathrm{reg}}}. \tag{58}$$

Set $x := \gamma\Theta\lambda(Q_O)^{-1/2}$ and $C_0 := 2k_{\mathrm{reg}}$. Then $\Lambda_{\mathrm{hypo}}(\gamma) = \frac{\lambda_R}{4\Theta}\sqrt{\lambda(Q_O)}\,\frac{x}{(x+K_{\mathrm{geom}})^2+C_0}$. A direct derivative computation shows that the function $x \mapsto \frac{x}{(x+K_{\mathrm{geom}})^2+C_0}$ is maximized at $x = \sqrt{K_{\mathrm{geom}}^2 + C_0}$, giving the stated γ_{opt} and the scaling $\max_\gamma \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}$.

For the perturbation term, by (48), $B_{\mathrm{pert}}(\gamma) = b_{2,\mathrm{eff}} + \alpha_*(\gamma)\, C_{\mathrm{sing}}\, \lambda(Q_O)^{-1/2}$. At $\gamma = \gamma_{\mathrm{opt}}$ we have $x = \mathcal{O}(1)$ and $Z(\gamma_{\mathrm{opt}}) = x + K_{\mathrm{geom}} = \mathcal{O}(1)$, hence $\alpha_*(\gamma_{\mathrm{opt}}) = \frac{\gamma_{\mathrm{opt}}\lambda_R}{Z(\gamma_{\mathrm{opt}})^2+2k_{\mathrm{reg}}} = \mathcal{O}(\sqrt{\lambda(Q_O)})$, which implies $\alpha_*(\gamma_{\mathrm{opt}})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1)$ and therefore $B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) = \mathcal{O}(1)$. Finally, $\eta^*(\gamma) = \Lambda_{\mathrm{hypo}}(\gamma)/(2B_{\mathrm{pert}}(\gamma))$ yields $\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}$. □

The certified threshold $\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}$ admits a transparent interpretation: the dimensionless ratio $\eta/\sqrt{\lambda(Q_O)}$ compares forcing to the kinetic relaxation scale, and the condition $\eta \le \eta^*$ requires this ratio to be $\mathcal{O}(1)$. This perturbative character is shared by prior approaches: Iacobucci–Olla–Stoltz [22] and Roussel–Stoltz [36] require the forcing to be small relative to the spectral gap, and Eberle–Lörler [16, 17] impose analogous contractivity conditions. The advantage of our threshold is that it is *explicit* and *computable* from the structural constants alone. The numerical experiments in both examples (Figures 1 and 3) show that acceleration persists well beyond η^*; for the Langevin rotor with $\lambda(Q_O) \approx 10^{-4}$, the fitted exponent remains close to 1/2 at $\eta = 1$, a factor of ∼100 beyond the certified range.

Remark 5.3 (Exact recovery of the warm-up constants). For underdamped Langevin on the torus (Section 2) with *no* perturbation ($\eta = 0$), the structural constants are $\lambda_R = M_R = 1$, hence $\Theta = \frac{1}{2}$, $k_{\mathrm{reg}} = 1$, and $\kappa_{\mathrm{ell}} = \sqrt{2}$ (from the Bochner–Weitzenböck bound). With $K_{\mathrm{geom}} = \kappa_{\mathrm{ell}} = \sqrt{2}$ and $\varpi_* = \lambda(Q_O) = m$, the abstract rate function (47) specialises to

$$\Lambda_{\mathrm{hypo}}(\gamma) = \frac{\gamma/4}{(\gamma/(2\sqrt{m}) + \sqrt{2})^2 + 2} = \frac{\gamma}{4(Z(\gamma,m)^2 + 2)},$$

with $Z(\gamma, m) = \gamma/(2\sqrt{m}) + \sqrt{2}$, exactly matching the warm-up bound in Proposition 2.3. The optimal friction is $\gamma_{\mathrm{opt}} = 2\sqrt{(\sqrt{2})^2 + 2}\,\sqrt{m} = 4\sqrt{m}$ and the optimal rate is $C_*\sqrt{m}$ with $C_* = (2-\sqrt{2})/8$, again matching the explicit warm-up computation.

Remark 5.4 (Why the gap-adapted shift matters). The $\sqrt{\lambda(Q_O)}$ scaling in Corollary 5.2 relies crucially on the gap-adapted choice $\varpi_* = \lambda(Q_O)$. This can be seen directly from the general unperturbed dissipation matrix in Appendix C. For a fixed shift $\varpi > 0$, write $\lambda := \lambda(Q_O)$ and consider

$$c_\varpi(\lambda) := \frac{\lambda}{\varpi + \lambda}, \qquad Z_\varpi(\gamma) := \gamma\Theta\varpi^{-1/2} + K_{\mathrm{geom}}. \tag{59}$$

Then the unperturbed matrix entries are

$$M^0_{11,\varpi} = \gamma\lambda_R - \alpha k_{\mathrm{reg}}, \qquad M^0_{22,\varpi} = \alpha c_\varpi(\lambda), \qquad M^0_{12,\varpi} = -\tfrac{1}{2}\alpha Z_\varpi(\gamma), \tag{60}$$

so

$$D_\varpi(\alpha) := \det \mathbf{M}_{0,\varpi} = \gamma\lambda_R c_\varpi(\lambda)\,\alpha - \Big(k_{\mathrm{reg}} c_\varpi(\lambda) + \tfrac{1}{4} Z_\varpi(\gamma)^2\Big)\alpha^2. \tag{61}$$

The maximizing weight is therefore

$$\alpha_{\varpi,*}(\gamma) = \frac{\gamma\lambda_R c_\varpi(\lambda)}{2k_{\mathrm{reg}} c_\varpi(\lambda) + \frac{1}{2} Z_\varpi(\gamma)^2}. \tag{62}$$

Now assume that ϖ stays bounded above and below away from 0, independently of λ. Then $c_\varpi(\lambda) \asymp \lambda$ as $\lambda \downarrow 0$. In the kinetic regime $\gamma \asymp \sqrt{\lambda}$ one has $Z_\varpi(\gamma) = \mathcal{O}(1)$, hence

$$\alpha_{\varpi,*}(\gamma) \asymp \gamma\lambda, \qquad D_\varpi(\alpha_{\varpi,*}) \asymp \gamma^2\lambda^2, \qquad \mathrm{Tr}(\mathbf{M}_{0,\varpi}) \asymp \gamma. \tag{63}$$

Therefore the same determinant–trace estimate used in Appendix C returns only a rate of order

$$\frac{D_\varpi(\alpha_{\varpi,*})}{\mathrm{Tr}(\mathbf{M}_{0,\varpi})} \asymp \gamma\lambda^2 \asymp \lambda^{5/2} \qquad \text{when } \gamma \asymp \sqrt{\lambda}. \tag{64}$$

By contrast, with the present choice $\varpi_* = \lambda(Q_O)$ one has $c_{\varpi_*}(\lambda) = 1/2$, while the quantity $Z(\gamma) = \gamma\Theta\lambda^{-1/2} + K_{\mathrm{geom}}$ remains $\mathcal{O}(1)$ at the optimal friction. The slow block is then of order α rather than $\alpha\lambda$, which yields $\alpha_*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda}$ and hence $\Lambda_{\mathrm{hypo}}(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda}$. In particular, the difference between the classical fixed-shift DMS scaling $\lambda(Q_O)^{5/2}$ and the present $\sqrt{\lambda(Q_O)}$ law is due to the choice of corrector/shift, not to an algebraic inconsistency. The complete explicit computation in the Langevin case with U convex is carried out in Section 2.

Define the (rigorous) L^2 relaxation time as in [30] by

$$t_{\mathrm{rel}}(\mathcal{L}) := \inf\Big\{t \geq 0\,;\ \|e^{t\mathcal{L}}f\| \leq e^{-1}\,\|f\| \text{ for all } f \in \mathcal{H}_0\Big\}. \tag{65}$$

The following holds:

Corollary 5.5 (Acceleration in the L^2 relaxation time)**.** *Assume the setting of Theorem 5.1. Then for $\eta \leq \eta^*(\gamma)$, the exponential decay estimate from Theorem 5.1 implies*

$$t_{\mathrm{rel}}(\mathcal{L}) \leq \frac{1 + \log C_*(\gamma)}{\lambda_\eta} \leq \frac{1 + \log(1+\sqrt{2})}{\lambda_\eta}. \tag{66}$$

In particular, at the optimal friction $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$ (Corollary 5.2) and for perturbations within the certified threshold, one has

$$t_{\mathrm{rel}}(\mathcal{L}) = \mathcal{O}\big(\lambda(Q_O)^{-1/2}\big). \tag{67}$$

Proof. By Theorem 5.1, for all $f \in \mathcal{H}_0$ and $t \geq 0$ one has $\|e^{t\mathcal{L}}f\| \leq C_*(\gamma)\, e^{-\lambda_\eta t}\,\|f\|$. Choose $t = (1 + \log C_*(\gamma))/\lambda_\eta$ so that $C_*(\gamma)\, e^{-\lambda_\eta t} = e^{-1}$. □

5.2. **Overdamped diffusive ceiling.** To establish that the high-friction γ^{-1} behavior is intrinsic and not an artifact of the lower-bound construction, we derive a universal upper bound on any admissible L^2 exponential rate in terms of the singular-value gap.

Define the *singular-value gap*

$$s(\mathcal{L}) := \inf_{f \in \mathcal{H}_0 \cap \mathrm{dom}(\mathcal{L}) \setminus \{0\}} \frac{\|\mathcal{L}f\|}{\|f\|}, \tag{68}$$

that is, the smallest singular value of $\mathcal{L}|_{\mathcal{H}_0}$ in finite dimensions, remaining meaningful as an infimum over the operator domain in unbounded settings. Quantities of this type are continuous-time analogues of *multiplicative reversibilizations* and the *pseudo spectral gap* used in non-reversible Markov chain theory; see, e.g., [34]. In the discrete-time finite-state setting, a closely related point of view is developed by Chatterjee, who defines a nonreversible "spectral gap" via singular values of the generator $I - P$ [10].

Theorem 5.6 (Overdamped diffusive ceiling via singular value gap)**.** *Assume Assumptions 4.2.1–4 and Assumption 4.4, and in the unbounded setting also Assumption 4.5. Assume also* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$*. Assume moreover that $\lambda(Q_O)$ is attained by a normalized eigenfunction $\phi \in \mathrm{dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0$ (automatic in finite dimensions, and in many PDE settings with compact resolvent), and that the associated first-order corrector*

$$\psi := \mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi$$

belongs to $\mathrm{dom}(\mathcal{L}) \cap \mathrm{dom}((-\mathcal{R})^{1/2})$ *with* $P_F \mathcal{V}_{\mathrm{skew}} \psi$ *well defined. Assume that* $e^{t\mathcal{L}}$ *is a contraction semigroup on* $\mathcal{H}_0$*, i.e.* $\|e^{t\mathcal{L}} f\| \le \|f\|$ *for all* $t \ge 0$ *and* $f \in \mathcal{H}_0$*. Let* $C \ge 1$ *and* $\nu > 0$ *be such that*

$$\|e^{t\mathcal{L}} f\| \le C\, e^{-\nu t}\, \|f\|, \qquad \forall t \ge 0,\ \forall f \in \mathcal{H}_0. \tag{69}$$

Then for every $\gamma > 0$*,*

$$\nu \le (1 + \log C)\, s(\mathcal{L}) \le (1 + \log C) \left[\frac{(1 + \kappa_{\mathrm{ell}})\, \lambda(Q_O)}{\gamma} + \eta\, (b_1 + b_2) + \eta\, (a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma} \right], \tag{70}$$

where κ_{ell} *is the flux-regularity constant from Assumption 4.2.3 and* a_1, a_2, b_1, b_2 *are from Assumption 4.2.4. In particular, for* $\eta \le c/\gamma$ *one obtains the diffusive overdamped scaling* $\nu = \mathcal{O}(\gamma^{-1})$ *as* $\gamma \to \infty$*.*

Sketch of Proof. The proof is given in Appendix D. The key point is that one can test the singular value gap on a *first-order corrector* $f_\gamma := \phi - \gamma^{-1}\psi$ built from a slow eigenfunction $Q_O \phi = -\lambda(Q_O)\phi$ and $\psi := \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} \phi$. The additional domain hypothesis ensures that $f_\gamma \in \mathrm{dom}(\mathcal{L})$ and that all terms below are well defined. The flux regularity bound (20) controls the only nontrivial fast remainder term $P_F \mathcal{V}_{\mathrm{skew}} \psi$ without ever invoking $\|\mathcal{V}_{\mathrm{skew}}\|$. The perturbation contribution is controlled by the $(-\mathcal{R})^{1/2}$-energy bounds in Assumption 4.2.4 and the explicit identity $\|(-\mathcal{R})^{1/2}\psi\|^2 = \lambda(Q_O)$. The first inequality $\nu \le (1 + \log C)\, s(\mathcal{L})$ then follows from Lemma D.1, which uses the contraction bound $\|e^{t\mathcal{L}} f\| \le \|f\|$ on $[0, (\log C)/\nu]$ and the exponential decay on $[(\log C)/\nu, \infty)$. □

Theorem 5.6 shows that in the overdamped regime the best achievable L^2 exponential decay rate (with any finite prefactor C) is bounded above by $(1 + \log C)$ times the macroscopic diffusion scale $\lambda(Q_O)/\gamma$, up to explicit perturbative contributions. Combined with Theorem 5.1, this confirms that the accelerated $\sqrt{\lambda(Q_O)}$ regime must occur at *intermediate* friction $\gamma \asymp \sqrt{\lambda(Q_O)}$, rather than in the large-friction limit.

At equilibrium ($\eta = 0$), the contraction hypothesis holds unconditionally: $\mathrm{Re}\langle \mathcal{L} f, f\rangle = \gamma \langle \mathcal{R} f, f\rangle \le 0$ by skew-adjointness of $\mathcal{V}_{\mathrm{skew}}$, so $e^{t\mathcal{L}}$ is a contraction semigroup on $\mathcal{H}_0$. Together with Theorem 5.1, this gives a matching characterization at equilibrium: the exponent 1/2 in the $\sqrt{\lambda(Q_O)}$ scaling is optimal. For $\eta \ne 0$, the generator is generically not dissipative in $L^2(\mu_0)$; however, recent works [12, 33] establish L^2 convergence in stationary-state-adapted norms where contraction holds by construction. Reformulating the ceiling in such a norm would make the hypothesis automatic, at the cost of modifying the structural constants; this gap is one of reference-measure choice rather than a fundamental obstruction (see Open Problem 3).

Corollary 5.7 (Diffusive ceiling for the L^2 relaxation time)**.** *Assume the setting of Theorem 5.6. Define* $t_{\mathrm{rel}}(\mathcal{L})$ *as in Corollary 5.5. Then the singular-value gap satisfies the explicit ceiling*

$$s(\mathcal{L}) \le \frac{(1 + \kappa_{\mathrm{ell}})\, \lambda(Q_O)}{\gamma} + \eta\, (b_1 + b_2) + \eta\, (a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{71}$$

Moreover, the relaxation time is bounded below in terms of the singular-value gap (see Lemma D.2):

$$t_{\mathrm{rel}}(\mathcal{L}) \ge \frac{1}{2\, s(\mathcal{L})}. \tag{72}$$

Consequently, if $\eta \le c/\gamma$*, then* $s(\mathcal{L}) = \mathcal{O}(\gamma^{-1})$ *as* $\gamma \to \infty$ *and hence*

$$t_{\mathrm{rel}}(\mathcal{L}) = \Omega(\gamma) \qquad (\gamma \to \infty). \tag{73}$$

In particular, combining this ceiling with Corollary 5.5 shows that, at optimal friction, the relaxation-time upper bound is of kinetic order $t_{\mathrm{rel}}(\mathcal{L}) = \mathcal{O}(\lambda(Q_O)^{-1/2})$*.*

6. Examples

This section has two goals. First, we give a direct construction that starts from a target slow generator $\mathcal{L}_O$ and builds a lifted generator $\mathcal{L}$ whose adiabatic elimination returns $\mathcal{L}_O$. Second, we apply this construction to two examples: forced underdamped Langevin dynamics (Section 6.2), which satisfies the full kernel-compatible framework of Sections 4–5, and a finite-dimensional boundary-driven model (Section 6.3), which also satisfies all structural assumptions including kernel compatibility and provides a compact algebraic illustration of the framework.

6.1. **Reverse adiabatic elimination: from effective dynamics to lifted dynamics.** Input: a non-equilibrium effective generator $\mathcal{L}_O$ on the slow space $\mathcal{H}_S$. Output: a lifted generator $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ on $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$ such that adiabatic elimination (Definition 3.4) gives back $\mathcal{L}_O$. We call this construction *reverse adiabatic elimination.* The procedure has three steps.
Perturbation expansion of $\mathcal{L}_O$. By Definition 3.4, $\mathcal{L}_O = -P_S\mathcal{V}\mathcal{R}^+\mathcal{V}P_S$ with $\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$. Expanding in powers of η gives

(74)
$$\mathcal{L}_O = \underbrace{-P_S\mathcal{V}_{\text{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\text{skew}}\,P_S}_{Q_O}\ \underbrace{-\,\eta\,P_S\big(\mathcal{V}_{\text{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\text{pert}}\ +\ \mathcal{V}_{\text{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\text{skew}}\big)P_S}_{\text{cross terms } \mathcal{O}(\eta)}\ \underbrace{-\,\eta^2\,P_S\mathcal{V}_{\text{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\text{pert}}\,P_S}_{\text{pure perturbation } \mathcal{O}(\eta^2)}.$$

The first term is the reversible backbone Q_O (the $\eta = 0$ limit). The second and third terms are the $\mathcal{O}(\eta)$ and $\mathcal{O}(\eta^2)$ corrections. In the boundary-driven example, all these terms are written explicitly in (89).
Step 1: Identify Q_O. Set $\eta = 0$ in the slow model. The resulting operator is Q_O. When the kernel-compatible assumption $\ker(Q_O) = \mathbb{C}\mathbf{1}$ holds, its spectral gap $\lambda(Q_O)$ is the key scale in Theorem 5.1 (both for the accelerated rate and for the admissible perturbation size).
Step 2: Choose fast space and coupling to match Q_O. The adiabatic-elimination identity implies

$$Q_O = -T^*\,(\mathcal{R}_F)^{-1}\,T, \qquad T := P_F\mathcal{V}_{\text{skew}}P_S, \tag{75}$$

where $\mathcal{R}_F := \mathcal{R}|_{\mathcal{H}_F} < 0$. So, after choosing $\mathcal{H}_F$ and $\mathcal{R}_F$, we pick $T : \mathcal{H}_S \to \mathcal{H}_F$ to satisfy $T^*\mathcal{R}_F^{-1}T = -Q_O$. A universal choice is $\mathcal{H}_F = \mathcal{H}_S$, $\mathcal{R}_F = -\lambda_R\,\mathrm{id}$, $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$. Then we set $\mathcal{R} := 0 \oplus \mathcal{R}_F$ and $\mathcal{V}_{\text{skew}} := \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$.
Step 3: Determine $\mathcal{V}_{\text{pert}}$ from consistency. With $\mathcal{R}$ and $\mathcal{V}_{\text{skew}}$ fixed, we choose $\mathcal{V}_{\text{pert}}$ so that adiabatic elimination exactly reproduces $\mathcal{L}_O$ whenever the resulting consistency problem is solvable in the chosen class of operators. This is equivalent to the consistency relation

$$\mathcal{L}_O - Q_O\ =\ -\eta\,P_S\big(\mathcal{V}_{\text{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\text{skew}}\big)P_S\ -\ \eta^2\,P_S\mathcal{V}_{\text{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\text{pert}}\,P_S, \tag{76}$$

with the centering constraint $P_S\mathcal{V}_{\text{pert}}P_S = 0$. In the Langevin example, $\mathcal{V}_{\text{pert}}P_S = 0$, so the formula simplifies strongly. In the boundary-driven example, all terms are present, and we write them explicitly in (89). In finite dimensions this is an algebraic problem. In infinite-dimensional settings, solvability is model-dependent and should be verified together with the required domain properties. The resulting lifted generator $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ automatically satisfies centering. To apply Theorem 5.1, one must in addition verify the remaining structural bounds and the kernel-compatible gap assumption $\ker(Q_O) = \mathbb{C}\mathbf{1}$. We now state the key structural advantage: for a broad class of designed lifts, the projection bias can be removed exactly.

Proposition 6.1 (Unbiasedness of designed lifts)**.** *Consider a lift $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ constructed via the reverse adiabatic elimination recipe above, with $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$ (Step 2) and $\mathcal{V}_{\text{pert}}$ satisfying the centering condition $P_S\mathcal{V}_{\text{pert}}P_S = 0$ and the consistency condition* (76)*. If additionally $P_F\mathcal{V}_{\text{pert}}P_F = 0$ (the perturbation has no fast-to-fast component), then:*

(i) $P_F\mathcal{V}P_F = 0$;
(ii) the Schur complement (78) *satisfies $\gamma\,S(\gamma) = \mathcal{L}_O$ for all $\gamma > 0$;*

(iii) whenever the lifted and effective stationary states are well defined, the lifted NESS satisfies $P_S h_\eta = g_\eta$ for all $\gamma > 0$, i.e., the projection bias $\Delta_{\text{bias}}(\gamma) \equiv 0$.

The condition $P_F \mathcal{V}_{\text{pert}} P_F = 0$ is a free design choice within this parametrization: it does not appear in the consistency condition (76) (which constrains only $P_F \mathcal{V}_{\text{pert}} P_S$ and $P_S \mathcal{V}_{\text{pert}} P_F$), so once the consistency problem is solvable it can be enforced without affecting the effective dynamics.

Proof. (i) Since $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$ is block-off-diagonal, $P_F \mathcal{V}_{\text{skew}} P_F = 0$. Combined with $P_F \mathcal{V}_{\text{pert}} P_F = 0$, we get $P_F \mathcal{V} P_F = P_F(\mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}) P_F = 0$. (ii) With $P_F \mathcal{V} P_F = 0$, the Schur complement (78) reduces to $S(\gamma) = -P_S \mathcal{V}(\gamma \mathcal{R}_F)^{-1} \mathcal{V} P_S = \gamma^{-1} \mathcal{L}_O$. (iii) Since $\gamma S(\gamma) = \mathcal{L}_O$, the NESS equation $S(\gamma) u = 0$ becomes $\mathcal{L}_O u = 0$, so $u = g_\eta$. □

Proposition 6.1 gives a clean design rule: if we enforce $P_F \mathcal{V}_{\text{pert}} P_F = 0$, then the projected lifted steady state equals the overdamped steady state for all γ. This does *not* remove non-equilibrium effects, because $\mathcal{L}_O$ still depends on the off-diagonal perturbation blocks $P_S \mathcal{V}_{\text{pert}} P_F$ and $P_F \mathcal{V}_{\text{pert}} P_S$. So non-self-adjoint slow dynamics remain fully allowed.

Remark 6.2 (Kernel consistency and invariant states). We verify that the lifted and effective dynamics have the same equilibrium kernel at $\eta = 0$, and hence the same stationary state whenever that kernel is one-dimensional.

Equilibrium case ($\eta = 0$): exact kernel identity for all $\gamma > 0$. Let $f \in \ker(\mathcal{L}|_{\eta=0}) = \ker(\gamma \mathcal{R} + \mathcal{V}_{\text{skew}})$. Taking $\operatorname{Re}\langle f, \mathcal{L} f \rangle = \gamma \langle f, \mathcal{R} f \rangle = 0$ (using skew-adjointness of $\mathcal{V}_{\text{skew}}$) forces $f \in \ker(\mathcal{R}) = \mathcal{H}_S$. Then $\mathcal{V}_{\text{skew}} f = 0$, which entails $Tf = P_F \mathcal{V}_{\text{skew}} P_S f = 0$ and thus $Q_O f = -T^* \mathcal{R}_F^{-1} T f = 0$. Conversely, if $f \in \ker(Q_O) \subset \mathcal{H}_S$, then

$$0 = -\langle f, Q_O f \rangle = \langle Tf, (-\mathcal{R}_F)^{-1} T f \rangle,$$

and since $(-\mathcal{R}_F)^{-1} > 0$ on $\mathcal{H}_F$, this implies $Tf = 0$. Because $\mathcal{V}_{\text{skew}}$ is block-off-diagonal in the reverse adiabatic-elimination construction, $\mathcal{V}_{\text{skew}} f = Tf = 0$, hence $(\gamma \mathcal{R} + \mathcal{V}_{\text{skew}}) f = 0$. Therefore

$$\ker(\mathcal{L}|_{\eta=0}) = \ker(Q_O), \qquad \forall\, \gamma > 0. \tag{77}$$

Under Assumption 4.4, this common kernel is exactly $\mathbb{C}\mathbf{1}$. This is the analogue of [30, Lemma 2.10, eq. (2.17)].

Non-equilibrium case ($\eta \neq 0$). When $\eta \neq 0$ and $P_F \mathcal{V} P_F \neq 0$, the NESS $h_\eta \in \ker(\mathcal{L})$ acquires a fast component $P_F h_\eta = \mathcal{O}(\eta/\gamma)$, and the slow marginal $u = P_S h_\eta$ satisfies the Schur complement equation

$$S(\gamma)\, u = 0, \qquad S(\gamma) := -P_S \mathcal{V} \big(\gamma \mathcal{R}_F + P_F \mathcal{V} P_F\big)^{-1} \mathcal{V}\, P_S. \tag{78}$$

Expanding in a Neumann series:

$$\gamma\, S(\gamma) = \mathcal{L}_O + \gamma^{-1}\, P_S \mathcal{V}\, \mathcal{R}^+\, (P_F \mathcal{V} P_F)\, \mathcal{R}^+\, \mathcal{V}\, P_S + \mathcal{O}(\gamma^{-2}). \tag{79}$$

Thus $P_S h_\eta \to g_\eta$ as $\gamma \to \infty$ (the Zeno/adiabatic limit). The Schur correction is of order $\mathcal{O}(\gamma^{-1})$. Under the standard perturbative setting (simple isolated zero mode and smooth dependence on η), this gives $\Delta_{\text{bias}}(\gamma) := \|P_S h_\eta - g_\eta\| = \mathcal{O}(\eta/\gamma)$. However, when $P_F \mathcal{V} P_F = 0$ (as guaranteed by Proposition 6.1 for designed lifts), the expansion collapses to $\gamma\, S(\gamma) = \mathcal{L}_O$ and $\Delta_{\text{bias}}(\gamma) \equiv 0$.

Since the lifted dynamics converges to h_η, the approximation error for the overdamped NESS g_η satisfies the triangle inequality

$$\|P_S e^{t\mathcal{L}} f - g_\eta\|_{\mathcal{H}_S} \;\le\; \underbrace{C_*(\gamma)\, e^{-\lambda_\eta t}\, \|f - h_\eta\|}_{\text{convergence error}} \;+\; \underbrace{\|P_S h_\eta - g_\eta\|_{\mathcal{H}_S}}_{\text{bias } \Delta_{\text{bias}}(\gamma)}. \tag{80}$$

For lifts satisfying the conditions of Proposition 6.1, $\Delta_{\text{bias}}(\gamma) = 0$ and the accelerated mixing time is

$$T_{\text{mix}}(\varepsilon) \;=\; \mathcal{O}\Big(\frac{\log(1/\varepsilon)}{\sqrt{\lambda(Q_O)}}\Big), \qquad \forall\, \varepsilon > 0. \tag{81}$$

Algorithm 1 summarises the procedure.

Algorithm 1 Accelerated NESS Approximation via Lifting

Require: Target effective generator $\mathcal{L}_O$ on $\mathcal{H}_S$, target accuracy $\varepsilon > 0$
Ensure: Approximation of g_η (overdamped NESS) to accuracy ε

Step 1: Construct lifted dynamics (Section 6.1)
Factor $Q_O = -T^*\mathcal{R}_F^{-1}T$ and set $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$
Choose $\mathcal{V}_{\text{pert}}$ satisfying (76) with $P_F\mathcal{V}_{\text{pert}}P_F = 0$ ▷ Zero bias (Prop. 6.1)

Step 2: Compute optimal friction
$\gamma_{\text{opt}} \leftarrow \sqrt{\lambda(Q_O)} \cdot \Theta^{-1}\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}}}$
$\lambda_\eta \leftarrow \big(\Lambda_{\text{hypo}}(\gamma_{\text{opt}}) - \eta\, B_{\text{pert}}(\gamma_{\text{opt}})\big)/(1 + \delta_*(\gamma_{\text{opt}}))$

Step 3: Run lifted dynamics (rate $\sim \sqrt{\lambda(Q_O)}$)
$t_{\text{run}} \leftarrow \log\big(C_*(\gamma_{\text{opt}})\|f_0\|/\varepsilon\big)/\lambda_\eta$
return $P_S\big(e^{t_{\text{run}}\mathcal{L}}h_0\big)$ ▷ ε-approximation of g_η

Remark 6.3 (Two-phase extension for general models). For natural physical models where $P_F\mathcal{V}P_F \neq 0$ (and the conditions of Proposition 6.1 do not hold), a nonzero accuracy floor $\varepsilon_* := \Delta_{\text{bias}}(\gamma_{\text{opt}}) = \mathcal{O}(\eta/\sqrt{\lambda(Q_O)})$ may arise. Then the error bound (80) cannot go below ε_* using only lifted dynamics. A direct remedy is two-phase evolution: first run the lifted dynamics at γ_{opt} down to scale ε_*, then switch to $\mathcal{L}_O$ to reach the final target ε. The total time is at most the overdamped baseline and is strictly smaller whenever $\lambda_\eta > \lambda(\mathcal{L}_O)$. This extension is not needed for either of the examples in this paper (Remark 6.6).

Remark 6.4 (Connection to the literature and structural constants). Our construction has the same lifting spirit as previous acceleration methods in Markov-chain and MCMC settings; see [37, 28, 21, 30]. The present formulation is fully continuous-time and expressed in the operator framework of Assumption 4.2. For the universal algorithmic lift ($\mathcal{H}_F = \mathcal{H}_S$, $\mathcal{R}_F = -\lambda_R\,\text{id}$, $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$), the structural constants are fully determined by λ_R and Q_O: one has $M_R = \lambda_R$ (the conditioning number equals the spectral gap), $\kappa_{\text{ell}} = 0$ (since $P_F\mathcal{V}_{\text{skew}}P_F = 0$ by the block-off-diagonal construction), and $k_{\text{reg}} = 1$. In the Langevin lift (Section 6.2), $\lambda_R = M_R = 1$ and $\kappa_{\text{ell}} > 0$ depends on the geometry of U, so the physics-adapted lift has a richer geometry but larger coupling constants. Both lifts achieve the same $\sqrt{\lambda(Q_O)}$ scaling for the rate; the table in Section 6.2 confirms that the algorithmic constants are competitive in practice.

6.2. **Example 1: Classical Nonequilibrium Langevin Dynamics.** We now instantiate the framework on forced underdamped Langevin dynamics on $\mathbb{T}^d \times \mathbb{R}^d$.

Setup. We consider a particle with conservative potential $U \in C^\infty(\mathbb{T}^d)$ and non-conservative force $F \in C^\infty(\mathbb{T}^d; \mathbb{R}^d)$. We adopt the notation (q, p) for position and momentum coordinates; the warm-up (Section 2) used (x, v). Let μ_0 be the Gibbs measure of the unperturbed dynamics:

$$d\mu_0(q,p) = Z^{-1}e^{-\beta(\frac{1}{2}|p|^2 + U(q))}dq\,dp, \tag{82}$$

with factorization $d\mu_0 = d\nu(q)\, d\nu_p(p)$. We work in $\mathcal{H} = L^2(\mu_0)$. For $\eta \neq 0$, convergence to the NESS is measured in $L^2(\mu_0)$ toward the density h_η. All identities are first checked on the core $\mathcal{C}_{\mathrm{Lan}} := C^\infty(\mathbb{T}^d) \otimes \mathcal{S}(\mathbb{R}^d)$ and then extended by density.

Generator decomposition. With slow projection $P_S f = \Pi_p f := \int_{\mathbb{R}^d} f(\cdot, p)\, d\nu_p(p)$, the generator decomposes as $\mathcal{L} = \mathcal{V}_{\mathrm{skew}} + \gamma\mathcal{R} + \eta\mathcal{V}_{\mathrm{pert}}$:

$$\mathcal{V}_{\mathrm{skew}} := p \cdot \nabla_q - \nabla U(q) \cdot \nabla_p, \qquad \mathcal{R} := -p \cdot \nabla_p + \beta^{-1}\Delta_p, \qquad \mathcal{V}_{\mathrm{pert}} f := F(q) \cdot \nabla_p f. \tag{83}$$

Effective slow generator. Since $\mathcal{R}(p_i) = -p_i$ and $\int p_i\, d\nu_p = 0$, one computes via Definition 3.4:

$$\mathcal{L}_O \phi = \beta^{-1}\Delta_q \phi - (\nabla U - \eta F) \cdot \nabla_q \phi, \qquad \phi = \phi(q), \tag{84}$$

and the reference operator is $Q_O := \mathcal{L}_O|_{\eta=0} = \beta^{-1}\Delta_q - \nabla U \cdot \nabla_q$, self-adjoint on $L^2(\nu)$ with Dirichlet form $\langle \phi, (-Q_O)\phi\rangle_{L^2(\nu)} = \beta^{-1}\|\nabla_q \phi\|^2_{L^2(\nu)}$. Since ν satisfies a Poincaré inequality on $\mathbb{T}^d$, $\ker(Q_O) = \mathbb{C}\mathbf{1}$ and Assumption 4.4 holds on $L^2(\nu) \cap \{\int \phi\, d\nu = 0\}$.

Proposition 6.5 (Verification of Assumption 4.2)**.** *The decomposition* (83) *satisfies Assumption 4.2.1–4 in $L^2(\mu_0)$, and the core $\mathcal{C}_{\mathrm{Lan}}$ above satisfies Assumption 4.5. The corresponding structural constants are:*

(i) $\lambda_R = 1$, $M_R = 1$ *(Ornstein–Uhlenbeck spectral gap and conditioning);*
(ii) $\kappa_{\mathrm{ell}} = \sqrt{d(d+1)}\,\sqrt{\rho_{\max}/\rho_{\min}} \cdot (1 + \beta\|\nabla U\|_{L^\infty}\sqrt{C_{\mathrm{P}}(\nu)})$, *where* $\rho_{\max}/\rho_{\min} = e^{\beta(\max U - \min U)}$ *and $C_{\mathrm{P}}(\nu)$ is the Poincaré constant of ν;*
(iii) the perturbation constants (a_1, b_1, a_2, b_2) depend only on $(\beta, d, \|F\|_{L^\infty})$, independently of $\lambda(Q_O)$.

The proof (verification of skew-adjointness, centering, flux regularity via Fourier–Parseval analysis, the analytic-core property, and p-multiplier energy bounds) is given in Appendix E. Since $Q_O = \beta^{-1}\Delta_q - \nabla U \cdot \nabla_q$ is elliptic on the compact torus, its eigenfunctions are smooth; therefore the additional eigenfunction/corrector domain hypothesis in Theorem 5.6 is automatic in this example. However, we do *not* claim here that the contraction hypothesis of Theorem 5.6 holds in the fixed reference space $L^2(\mu_0)$ for all $\eta \neq 0$. For forced Langevin dynamics, the natural non-equilibrium L^2 theories are typically formulated in weighted spaces adapted to the stationary state itself (or to an equivalent stationary-state weight), rather than as a contraction statement in the equilibrium reference norm $L^2(\mu_0)$; see, for instance, Dietert [12] and Monmarché [33]. Accordingly, in this example we invoke Theorem 5.1 for the explicit lower bound and use Theorem 5.6 only as a conditional upper-ceiling statement, not as a proved property of the present $L^2(\mu_0)$ realization.

Scaling consequences. Because the structural constants do not depend on $\lambda(Q_O)$, Corollary 5.2 applies directly. Hence $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$, $\eta^*(\gamma_{\mathrm{opt}}) \asymp C(\beta, d, \|F\|_{L^\infty})\sqrt{\lambda(Q_O)}$, and $\max_\gamma \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}$. This is consistent with the classical underdamped/overdamped crossover in [22], and adds an explicit admissible range for non-equilibrium forcing.

Numerical verification. We illustrate these predictions with truncated Galerkin spectral computations in the Fourier–Hermite basis [22]. To keep orthonormality in $L^2(\mu_0)$, we use Boltzmann-weighted Fourier modes:

$$G_k(q) = \sqrt{\frac{Z_\nu}{2\pi}}\, e^{ikq + \beta U(q)/2}, \qquad Z_\nu = \int_0^{2\pi} e^{-\beta U}\, dq,$$

so that $\{\psi_{nk}(q,p) := G_k(q) H_n(p)\}$ is orthonormal in $L^2(\mu_0)$. For the rotor potential $U(q) = U_0(1 - \cos q)$, the matrix elements are (with $m = 1$):

$$\langle \psi_{n'k'}, \mathcal{L}\, \psi_{nk}\rangle = \sqrt{\beta(n+1)}\, \delta_{n',n+1}\Big[\tfrac{ik}{\beta}\delta_{kk'} + \tfrac{u_{k-k'}}{2}\Big] + \sqrt{\beta n}\, \delta_{n',n-1}\Big[\big(\tfrac{ik}{\beta} + \eta F\big)\delta_{kk'} - \tfrac{u_{k-k'}}{2}\Big] - \gamma n\, \delta_{nn'}\delta_{kk'}, \tag{85}$$

where u_j are Fourier coefficients of $U'(q)$ (for the rotor, $u_{\pm 1} = \mp i U_0/2$). The numerical setup is: $d = 1$, $\beta = 100$, $F = 0.05$, $K = 12$, $N = 24$, $U_0 \in \{0.005, 0.01, 0.02, 0.03, 0.05, 0.07, 0.08, 0.1, 0.12\}$, $\gamma \in [0.02, 4.0]$ (50 points), $\eta \in \{0, 0.1, 1\}$.

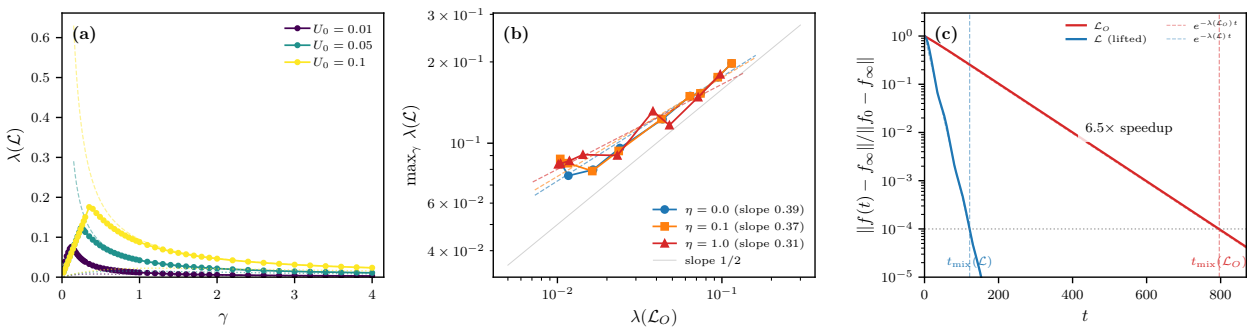

FIGURE 1. Numerical verification for Example 6.2. **(a)** Spectral gap $\lambda(\mathcal{L})$ vs. γ at $\eta = 0$ for several values of U_0; dashed curves show the overdamped ceiling $\lambda(\mathcal{L}_O)/\gamma$; dotted curves show the certified lower bound $\Lambda_{\mathrm{hypo}}(\gamma)$. **(b)** Log–log plot of $\max_\gamma \lambda(\mathcal{L})$ vs. $\lambda(\mathcal{L}_O)$ for $\eta \in \{0, 0.1, 1\}$; fitted slopes are shown in the legend. **(c)** Semigroup decay for the lifted dynamics at γ_{opt} and for $\mathcal{L}_O$ ($U_0 = 0.01$, $\eta = 0$); dashed lines show the exponential envelopes $e^{-\lambda(\mathcal{L})\,t}$ and $e^{-\lambda(\mathcal{L}_O)\,t}$. Vertical markers indicate the ε-mixing times ($\varepsilon = 10^{-4}$). Parameters: $\beta = 100$, $F = 0.05$, $K = 12$, $N = 24$ (Galerkin dimension dim $= 600$).

As a discretization check, the free case $U_0 = 0$ reproduces $\lambda_{n,k} = n\gamma + k^2/(\beta\gamma)$ [26] with relative error $\lesssim 10^{-15}$. We also compare with an unweighted Fourier–Hermite basis. Agreement is excellent for moderate βU_0, while at $\beta U_0 = 10$ the unweighted basis develops spurious unstable modes. This confirms that the weighted basis in (85) is the correct stable choice in our parameter range.

Figure 1(a) shows the expected three regimes in the truncated model: small-γ growth, large-γ diffusive ceiling, and an intermediate optimum. Figure 1(b) shows $\max_\gamma \lambda(\mathcal{L})$ versus $\lambda(\mathcal{L}_O)$ with a clearly sublinear trend. The fitted slopes are close to the predicted square-root exponent 1/2. A convergence-in-truncation study at $(K, N) \in \{(6, 12), (8, 16), (10, 20), (12, 24)\}$ confirms that spectral gaps are resolved to relative accuracy $\lesssim 10^{-4}$ at the adopted truncation. Figure 1(c) gives a representative mixing-time comparison at $\varepsilon = 10^{-4}$: for $U_0 = 0.01$, $\eta = 0$, we obtain $t_{\mathrm{mix}}(\mathcal{L}) \approx 125$ versus $t_{\mathrm{mix}}(\mathcal{L}_O) \approx 796$. No numerical accuracy floor is observed on the tested truncation and time window, and at $\eta = 0.1$ we find $\|P_S h_\eta - g_\eta\|/\|g_\eta\| \lesssim 10^{-14}$.

Algorithmic lift via generic square-root factorization. For completeness, the construction used in this example is the following three-step procedure.

Step 1 (compute slow operators). From the Galerkin matrix in (85), we compute the slow generator $\mathcal{L}_O$ and its reversible part Q_O.

Step 2 (build a generic lifted fast block). We choose $\mathcal{H}_{F,\mathrm{alg}} = \mathcal{H}_S$, $\mathcal{R}_{F,\mathrm{alg}} = -\lambda_R\,\mathrm{id}$, and $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$. This gives a 34×34 lifted generator (compared with 272×272 for the physics-adapted lift).

Step 3 (match the non-equilibrium part). We determine the remaining perturbation block by enforcing the consistency relation (76). In Langevin, $\mathcal{V}_{\mathrm{pert}} P_S = 0$ (because $F \cdot \nabla_p$ vanishes on the $n = 0$ Hermite mode), so this step reduces to a linear solve, which we compute directly.

Figure 2 compares the algorithmic and physical lifts. Unlike the boundary-driven model (Section 6.3), where both constructions coincide, here the two lifts differ: the physics-adapted lift uses the full momentum space $\mathcal{H}_F = L^2(\nu_p)$, whereas the algorithmic lift compresses it to a single copy

of the slow space. To align the protocol with the boundary-driven algorithmic-lift experiment, we benchmark the same number of *non-equilibrium* configurations: $U_0 \in \{0.01, 0.02, 0.05, 0.08\}$ and $\eta \in \{0.10, 0.15\}$ (8 combinations in total). Across this matched grid, both lifts remain faster than the overdamped generator at the level of the truncated peak gap, but the physical and algorithmic lifts display visibly different log–log trends in Panel (b). Accordingly, we use that panel only as a comparative diagnostic on this finite grid, not as a numerical identification of a universal asymptotic exponent for either lift. Denoting the NESS of the algorithmic lift by h_η^{alg}, we verify $\|P_S h_\eta^{\mathrm{alg}} - g_\eta\|/\|g_\eta\| \lesssim 10^{-15}$ at $\eta = 0.1$.

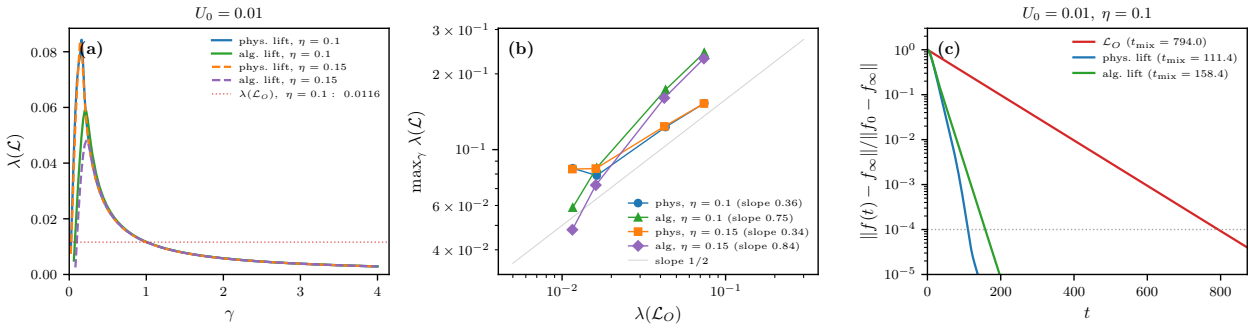

FIGURE 2. Comparison of the *physical* (600×600, full Langevin) and *algorithmic* (50×50, square-root factorization) lifted generators for Example 6.2. **(a)** Spectral gap vs. γ at $U_0 = 0.01$ for both lifts ($\eta = 0.10$ and $\eta = 0.15$); the overdamped gap $\lambda(\mathcal{L}_O)$ is shown as a horizontal reference. **(b)** Log–log plot of peak spectral gap vs. $\lambda(\mathcal{L}_O)$; fitted slopes are listed in the legend. **(c)** Mixing-time comparison ($U_0 = 0.01$, $\eta = 0.10$, $\varepsilon = 10^{-4}$): algorithmic lift, physical lift, and overdamped $\mathcal{L}_O$.

TABLE 1. Peak spectral gaps for the physical and algorithmic lifts in the Langevin model on the matched non-equilibrium grid $U_0 \in \{0.01, 0.02, 0.05, 0.08\}$, $\eta \in \{0.10, 0.15\}$. The "speedup" column is $\max_\gamma \lambda(\mathcal{L})/\lambda(\mathcal{L}_O)$.

U_0	η	$\lambda(\mathcal{L}_O)$	phys. peak	alg. peak	speedup
0.01	0.10	0.01161	0.08434	0.05884	7.27×
0.01	0.15	0.01156	0.08369	0.04805	4.16×
0.02	0.10	0.01629	0.07889	0.08467	4.84×
0.02	0.15	0.01608	0.08407	0.07229	4.50×
0.05	0.10	0.04292	0.12294	0.17295	2.86×
0.05	0.15	0.04220	0.12366	0.16042	3.80×
0.08	0.10	0.07405	0.15291	0.24270	2.06×
0.08	0.15	0.07369	0.15250	0.23015	2.07×

6.3. **Example 2: A Boundary-Driven Two-Qubit Model.** We now study a two-qubit boundary-driven model that satisfies all structural assumptions, including the kernel-compatible gap (Assumption 4.4). Its main role is to provide a finite-dimensional illustration in which the effective slow generator is non-self-adjoint and the reverse adiabatic-elimination algebra can be written out explicitly. Because $\dim \mathcal{H} = 16$, all operators are bounded and there are no domain subtleties.

Setup. Both fast and slow subsystems are qubits: $\mathfrak{h}_F \cong \mathfrak{h}_S \cong \mathbb{C}^2$. We work in $\mathcal{H} = L^2(\sigma)$ with $\sigma = \frac{1}{2}\mathbf{1}_F \otimes \frac{1}{2}\mathbf{1}_S$. The slow subspace is $\mathcal{H}_S = \mathbf{1}_F \otimes \mathcal{B}(\mathfrak{h}_S)$, with projection $P_S(X) = \mathbf{1}_F \otimes \mathrm{tr}_F(\frac{1}{2}\mathbf{1}_F \cdot X)$.

Generator decomposition. The decomposition is $\mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma\mathcal{R} + \eta\mathcal{V}_{\text{pert}}$. Here $\mathcal{R} = \mathcal{R}_F \otimes \mathrm{id}_S$ with $\mathcal{R}_F X = \Gamma \sum_{j\in\{x,y,z\}} \mathcal{D}^\dagger[\sigma_j]X$, so $\lambda_R = 4\Gamma$. The transport part uses a *two-channel* coupling,

$$\mathcal{V}_{\text{skew}} X = -i[g(\sigma_x \otimes \sigma_x + \sigma_z \otimes \sigma_z), X], \tag{86}$$

and the perturbation is $\mathcal{V}_{\text{pert}} = \sigma_z \otimes \mathcal{G}_S$, $\mathcal{G}_S(X) = \mathcal{D}^\dagger[\sigma_-]X$. Because $\mathrm{tr}(\frac{1}{2}\mathbf{1}_F \cdot \sigma_z) = 0$, both terms map $\mathcal{H}_S$ into $\mathcal{H}_F$, so centering is satisfied.

Effective generator $\mathcal{L}_O$ (explicit formula). Let $X = \mathbf{1}_F \otimes X_S \in \mathcal{H}_S$.
Step 1 (first excursion to fast space). The two-channel coupling gives

$$(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})X = \sum_{a\in\{x,z\}} \sigma_a \otimes \Big(- ig\,[\sigma_a, X_S]\Big) + \sigma_z \otimes \eta\,\mathcal{G}_S(X_S). \tag{87}$$

Step 2 (Poisson equation on the fast ancilla). Since $\mathcal{R}_F(\sigma_a) = -\kappa\,\sigma_a$ with $\kappa := 4\Gamma$ for each $a \in \{x, y, z\}$, we have $\mathcal{R}^+(\sigma_a \otimes Y) = -(1/\kappa)\sigma_a \otimes Y$. Therefore

$$\mathcal{R}^+(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})X = -\frac{1}{\kappa} \sum_{a\in\{x,z\}} \sigma_a \otimes \Big(- ig\,[\sigma_a, X_S]\Big) - \frac{1}{\kappa}\sigma_z \otimes \eta\,\mathcal{G}_S(X_S). \tag{88}$$

Step 3 (return map and projection). Using $\mathcal{V}_{\text{skew}}(\sigma_a \otimes Y) = -ig \sum_{b\in\{x,z\}} \sigma_a\sigma_b \otimes [\sigma_b, Y]$, the cross-channel products $\sigma_x\sigma_z = i\sigma_y$ and $\sigma_z\sigma_x = -i\sigma_y$ produce σ_y factors, which vanish under P_S (since $\mathrm{tr}(\sigma_y) = 0$). Only same-channel terms survive. Projecting gives

$$\begin{aligned}\mathcal{L}_O(\mathbf{1}_F \otimes X_S) = -\frac{1}{\kappa}\,\mathbf{1}_F \otimes \Big(& g^2 \sum_{a\in\{x,z\}} [\sigma_a, [\sigma_a, X_S]] \\ &+ i\eta g\,\big[\sigma_z, \mathcal{G}_S(X_S)\big] \\ &+ i\eta g\,\mathcal{G}_S([\sigma_z, X_S]) \\ &- \eta^2\,\mathcal{G}_S^2(X_S)\Big).\end{aligned} \tag{89}$$

The reference operator is the $\eta = 0$ part,

$$Q_O(\mathbf{1}_F \otimes X_S) = -\frac{g^2}{\kappa}\,\mathbf{1}_F \otimes \sum_{a\in\{x,z\}} [\sigma_a, [\sigma_a, X_S]]. \tag{90}$$

Non-self-adjointness on the slow space. Since $\pi_S \propto \mathbf{1}_S$, the $L^2(\pi_S)$ inner product is proportional to the Hilbert–Schmidt inner product, and one checks that $(\mathcal{D}^\dagger[\sigma_-])^\dagger = \mathcal{D}^\dagger[\sigma_+] \neq \mathcal{D}^\dagger[\sigma_-]$. Thus $\mathcal{G}_S$ is not self-adjoint on $L^2(\pi_S)$. Consequently, the $\mathcal{O}(\eta)$ mixed operator

$$\mathcal{B} := ig\,\big(\,\mathrm{ad}_{\sigma_z} \circ \mathcal{G}_S + \mathcal{G}_S \circ \mathrm{ad}_{\sigma_z}\,\big), \qquad \mathrm{ad}_{\sigma_z}(X) := [\sigma_z, X], \tag{91}$$

is generically not self-adjoint either (unless additional commutation/self-adjointness relations hold), so $\mathcal{L}_O$ is generically non-self-adjoint for $\eta \neq 0$.

Verification of Assumptions 4.2 and 4.4. All operators are bounded on the finite-dimensional space $\mathcal{H}$.

(1) Assumption 4.2.1 holds for $\mathcal{R}$ by construction, with $\lambda_R = 4\Gamma$.
(2) Assumption 4.2.2 holds since both $\mathcal{V}_{\text{skew}}$ and $\mathcal{V}_{\text{pert}}$ map $\mathcal{H}_S$ into $\mathcal{H}_F$.
(3) Assumption 4.2.3 holds with $M_R := \|\mathcal{R}\|$. For the flux regularity bound (20), cross-channel products ($\sigma_x\sigma_z = i\sigma_y$) produce σ_y factors, which do not vanish under P_F. Numerically, the flux regularity constant $\kappa_{\text{ell}} > 0$ is verified for all $g > 0$.
(4) For Assumption 4.2.4, decompose $\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$ by $\mathcal{K}_{\text{pert}} := \frac{1}{2}(\mathcal{V}_{\text{pert}} - \mathcal{V}_{\text{pert}}^\dagger)$ and $\mathcal{S}_{\text{pert}} := \frac{1}{2}(\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}^\dagger)$. Since $\mathcal{V}_{\text{pert}}$ is bounded, the energy bounds (23)–(24) hold with $a_1 = a_2 = 0$ and $b_1 = \|\mathcal{K}_{\text{pert}}\|$, $b_2 = \|\mathcal{S}_{\text{pert}}\|$.

The key improvement over a single-channel coupling (e.g., $H_{\text{int}} = g\sigma_z \otimes \sigma_z$): the double commutator $\text{ad}^2_{\sigma_a}$ has kernel $\text{span}\{I, \sigma_a\}$. With two channels,

$$\ker(Q_O) \cap \mathcal{H}_0 = \ker(\text{ad}^2_{\sigma_x}) \cap \ker(\text{ad}^2_{\sigma_z}) \cap \{\text{tr} = 0\} = \{0\}, \tag{92}$$

so **Assumption 4.4 holds** with $\ker(Q_O) = \mathbb{C}\mathbf{1}$ and $\lambda(Q_O) = 4g^2/\kappa = g^2/\Gamma$. This is verified numerically to machine precision (Table 2).

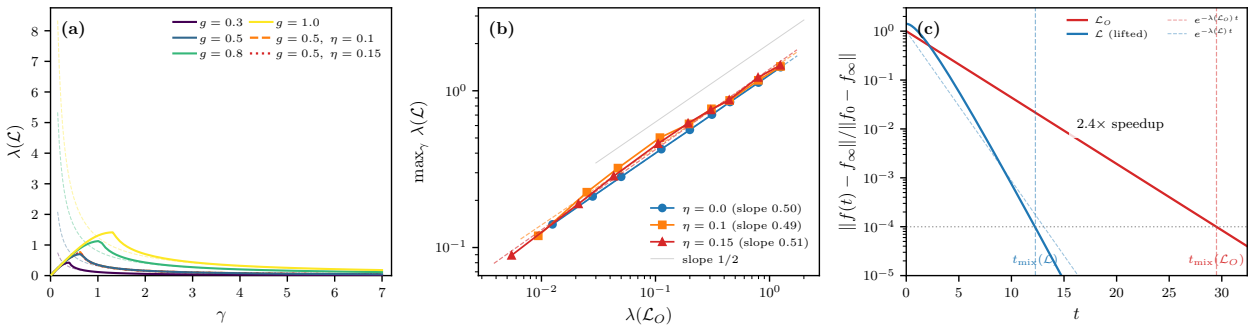

FIGURE 3. Numerical verification for Example 6.3. **(a)** Spectral gap $\lambda(\mathcal{L})$ vs. γ at $\eta = 0$ for several values of g; dashed curves show the overdamped envelope $\lambda(\mathcal{L}_O)/\gamma$. **(b)** Log–log plot of $\max_\gamma \lambda(\mathcal{L})$ versus $\lambda(\mathcal{L}_O)$. Fitted slopes are shown in the legend. **(c)** Decay of the projected slow component for the lifted dynamics at γ^* and for $\mathcal{L}_O$ ($g = 0.5$, $\eta = 0$); dashed lines show the exponential envelopes. Vertical markers indicate the ε-level crossing times ($\varepsilon = 10^{-4}$). Parameters: $\Gamma = 0.8$, $\kappa = 3.2$, $g \in \{0.1, \ldots, 1.0\}$, 53-point γ-grid over $[0.02, 7.0]$.

TABLE 2. Kernel compatibility and spectral gap for Q_O with the two-channel coupling $H_{\text{int}} = g(\sigma_x \otimes \sigma_x + \sigma_z \otimes \sigma_z)$. For all $g > 0$: $\dim\ker(Q_O) = 1$ ($= \mathbb{C}\mathbf{1}$) and $\lambda(Q_O) = g^2/\Gamma$.

g	$\dim\ker(Q_O)$	$\lambda(Q_O)$
0.3	1	0.1125
0.5	1	0.3125
1.0	1	1.2500

Numerical verification. We assemble the exact 16×16 Liouville generator and compute spectral gaps. The scan uses $g \in \{0.1, 0.15, 0.2, 0.3, 0.4, 0.5, 0.6, 0.8, 1.0\}$, $\Gamma = 0.8$ (so $\kappa = 3.2$), $\gamma \in [0.02, 7.0]$ (53 points), $\eta \in \{0, 0.10, 0.15\}$. Panel (a) shows the three-regime structure: underdamped ($\lambda \propto \gamma$), peaked, and overdamped ($\lambda \propto \lambda(\mathcal{L}_O)/\gamma$). Panel (b) shows $\max_\gamma \lambda(\mathcal{L})$ versus $\lambda(\mathcal{L}_O)$; the fitted log–log slope is close to 1/2, confirming the $\sqrt{\lambda(Q_O)}$ acceleration. Panel (c) shows a representative convergence comparison: for $g = 0.5$, $\eta = 0$, $\varepsilon = 10^{-4}$, the mixing times are approximately 12.3 for the lift and 29.5 for $\mathcal{L}_O$, yielding a 2.4× speedup. At $\eta = 0.1$, we verify for the stationary states $\|P_S h_\eta - g_\eta\|/\|g_\eta\| \sim 10^{-16}$.

Algorithmic lift via generic square-root factorization. Here we use exactly the same three-step construction as above, now in the boundary-driven toy model.

Step 1 (compute slow operators). Compute $\mathcal{L}_O$ and Q_O from the physical model.

Step 2 (build a generic lifted fast block). Set $\mathcal{H}_{F,\text{alg}} = \mathcal{H}_S$, $\mathcal{R}_{F,\text{alg}} = -\lambda_R\,\text{id}$, and $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$. This produces an 8×8 lifted generator (the physical model is 16×16).

Step 3 (match the non-equilibrium part). Solve the consistency relation (76) numerically for the perturbation block. This linear algebra problem has multiple solutions; we choose the one with smallest matrix norm. The final residual is below 10^{-13}.

Figure 4 compares algorithmic and physical lifts on the matched grid $g \in \{0.3, 0.5, 0.8, 1.0\}$ and $\eta \in \{0.10, 0.15\}$. Table 3 shows the peak spectral gaps; the algorithmic lift achieves gaps that match or exceed those of the physical lift, consistent with the fact that the algorithmic construction optimally uses the kernel-compatible Q_O. Panel (b) shows that both constructions follow the same log–log trend, and Panel (c) shows nearly identical convergence curves. For the steady state, at $g = 0.5$, $\eta = 0.1$ we get $\|P_S h_\eta^{\mathrm{alg}} - g_\eta\|/\|g_\eta\| \lesssim 10^{-16}$.

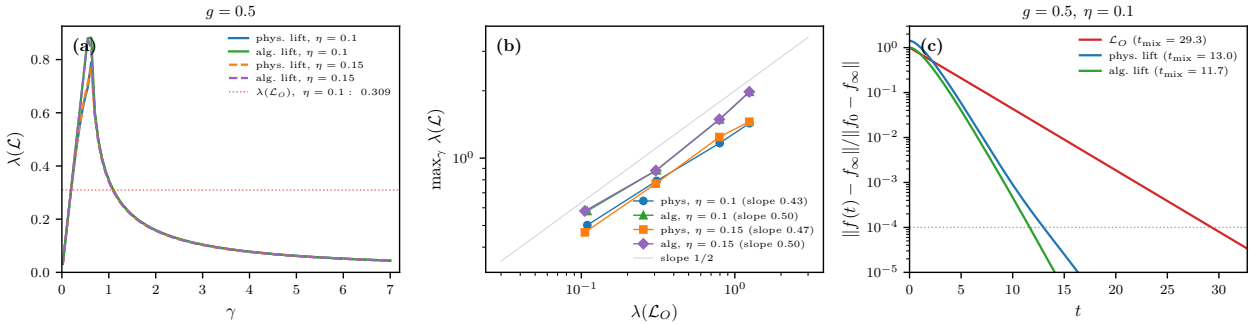

FIGURE 4. Comparison of the *physical* (16×16) and *algorithmic* (8×8, square-root factorization) lifted generators for Example 6.3. **(a)** Spectral gap $\lambda(\mathcal{L})$ vs. γ at $g = 0.5$ for both lifts ($\eta = 0.10$ and $\eta = 0.15$); the overdamped reference is $\lambda(\mathcal{L}_O)$. **(b)** Log–log plot of peak spectral gap vs. $\lambda(\mathcal{L}_O)$ for both lifts. **(c)** Convergence comparison ($g = 0.5$, $\eta = 0.10$, $\varepsilon = 10^{-4}$): algorithmic lift, physical lift, and overdamped $\mathcal{L}_O$.

TABLE 3. Peak spectral gaps $\max_\gamma \lambda(\mathcal{L})$ for the physical and algorithmic lifts (53-point γ-grid). The "speedup" column is $\max_\gamma \lambda(\mathcal{L}_{\mathrm{phys}})/\lambda(\mathcal{L}_O)$.

g	η	$\lambda(\mathcal{L}_O)$	phys. peak	alg. peak	speedup
0.3	0.10	0.1094	0.5020	0.5806	4.59×
0.3	0.15	0.1055	0.4661	0.5806	4.42×
0.5	0.10	0.3094	0.7907	0.8831	2.56×
0.5	0.15	0.3055	0.7688	0.8800	2.52×
0.8	0.10	0.7969	1.1672	1.4906	1.46×
0.8	0.15	0.7930	1.2427	1.4906	1.57×
1.0	0.10	1.2469	1.4321	1.9789	1.15×
1.0	0.15	1.2430	1.4539	1.9789	1.17×

Remark 6.6 (Bias behaviour in the two examples). Both examples show extremely small projection bias. In the boundary-driven model, we verify numerically that $\|P_S h_\eta - g_\eta\|/\|g_\eta\| \lesssim 10^{-16}$ at $\eta = 0.1$: the slow projection of the lifted NESS coincides with the overdamped NESS to machine precision. In the Langevin model, we obtain identical behaviour: $\|P_S h_\eta - g_\eta\|/\|g_\eta\| \lesssim 10^{-14}$ at $\eta = 0.1$. Hence no accuracy floor is observed in either experiment at the level of projected stationary states. For models with $P_F \mathcal{V} P_F \neq 0$, use the two-phase strategy of Remark 6.3.

7. Conclusion and open problems

The framework developed above gives explicit, gap-dependent bounds for the interplay between friction, reversible backbone gap, and non-equilibrium forcing in lifted generators. We conclude with several directions for future investigation.

(1) Sharp stability thresholds. The threshold η^* involves structural constants that may not be sharp; numerically, acceleration persists well beyond η^*. For models where forcing enhances the effective gap (e.g. optimally rotated diffusions [21]), one expects $\lambda(\mathcal{L}_O) > \lambda(Q_O)$, and a refined analysis could yield

$$\max_{\gamma>0} \lambda(\mathcal{L}_\gamma) \asymp \sqrt{\lambda(\mathcal{L}_O)} \qquad \text{(conjectured, when } \lambda(\mathcal{L}_O) > \lambda(Q_O)\text{)}. \tag{93}$$

(2) Optimal factorization. The reverse adiabatic elimination recipe lifts any target Q_O, but the structural constants depend on the auxiliary space $(\mathcal{H}_F, \mathcal{R}_F, T)$. Characterizing the factorization that maximises η^* in infinite-dimensional settings, and extending the framework to multilevel decompositions $\mathcal{H}_{S_1} \subset \mathcal{H}_{S_2} \subset \cdots$, remain open.

(3) Beyond the perturbative regime. Extending the analysis to large η may require space-time divergence methods [16, 17] or probabilistic contractivity arguments [33]. A related problem is to formulate the overdamped ceiling directly in a stationary-state-adapted norm, making the contraction hypothesis automatic.

(4) Quantum many-body NESS. The two-qubit toy model of Section 6.3 demonstrates that the abstract framework applies to boundary-driven Lindbladian generators [19, 31] when Q_O satisfies a Poincaré inequality on $L^2(\sigma)$ [29]. The challenge for many-body systems (e.g., boundary-driven XXZ chains) is that the slow Hilbert space $\mathcal{H}_S$ is exponentially large; scaling the structural constants with system size and establishing physically meaningful $\lambda(Q_O)$ lower bounds remain open.

References

1. Franz Achleitner, Anton Arnold, and Eric A. Carlen, *On multi-dimensional hypocoercive BGK models*, Kinet. Relat. Models **11** (2018), no. 4, 953–1009.
2. Dallas Albritton, Scott Armstrong, Jean-Christophe Mourrat, and Matthew Novack, *Variational methods for the kinetic Fokker-Planck equation*, Anal. PDE **17** (2024), no. 6, 1953–2010.
3. Simon Apers, Francesco Ticozzi, and Alain Sarlette, *Lifting Markov chains to mix faster: Limits and opportunities*, arXiv preprint arXiv:1705.08253, 2017.
4. Anton Arnold, Irene M. Gamba, Maria Pia Gualdani, Stephane Mischler, Clement Mouhot, and Christof Sparber, *The wigner-fokker-planck equation: Stationary states and large-time behavior*, 2012, arXiv:1010.2791.
5. Rajendra Bhatia, *Matrix analysis*, Graduate Texts in Mathematics, vol. 169, Springer, 1997.
6. Giovanni Brigati, Francis Lörler, and Lihan Wang, *Hypocoercivity meets lifts*, arXiv preprint arXiv:2412.10890, 2025.
7. Yu Cao, Jianfeng Lu, and Lihan Wang, *On explicit l^2-convergence rate estimate for underdamped langevin dynamics*, Archive for Rational Mechanics and Analysis **247** (2023), no. 5, 90.
8. Eric A. Carlen, David A. Huse, and Joel L. Lebowitz, *Stationary states of boundary driven quantum systems: some exact results*, arXiv preprint arXiv:2408.06887, 2024.
9. Eric A. Carlen, David A. Huse, and Joel L. Lebowitz, *Boundary-driven quantum systems near the zeno limit: steady states and long-time behavior*, arXiv preprint arXiv:2512.12825, 2025.
10. Sourav Chatterjee, *Spectral gap of nonreversible Markov chains*, arXiv preprint arXiv:2310.10876, 2025.
11. Fang Chen, László Lovász, and Igor Pak, *Lifting Markov chains to speed up mixing*, Proceedings of the Thirty-First Annual ACM Symposium on Theory of Computing (New York), STOC '99, ACM, 1999, pp. 275–281.
12. Helge Dietert, *L^2-hypocoercivity for non-equilibrium kinetic equations*, 2023.
13. Jean Dolbeault, Clément Mouhot, and Christian Schmeiser, *Hypocoercivity for linear kinetic equations conserving mass*, Transactions of the American Mathematical Society **367** (2015), no. 6, 3807–3828.
14. Jean Dolbeault, Clément Mouhot, and Christian Schmeiser, *Hypocoercivity for kinetic equations with linear relaxation terms*, 2009.
15. Andreas Eberle, Arnaud Guillin, Leo Hahn, Francis Lörler, and Manon Michel, *Convergence of non-reversible Markov processes via lifting and flow Poincaré inequality*, arXiv preprint arXiv:2503.04238, 2025.

16. Andreas Eberle and Francis Lörler, *Non-reversible lifts of reversible diffusion processes and relaxation times*, arXiv preprint arXiv:2402.05041, 2024.
17. ———, *Space-time divergence lemmas and optimal non-reversible lifts of diffusions on Riemannian manifolds with boundary*, arXiv preprint arXiv:2412.16710, 2024.
18. Di Fang, Jianfeng Lu, and Yu Tong, *Mixing time of open quantum systems via hypocoercivity*, Phys. Rev. Lett. **134** (2025), 140405.
19. Vittorio Gorini, Andrzej Kossakowski, and E. C. G. Sudarshan, *Completely positive dynamical semigroups of N-level systems*, J. Math. Phys. **17** (1976), no. 5, 821–825. MR 0413882
20. Martin Grothaus and Patrik Stilgenbauer, *Hypocoercivity for Kolmogorov backward evolution equations and applications*, J. Funct. Anal. **267** (2014), no. 10, 3515–3556.
21. Chii-Ruey Hwang, Sheng-Yao Hwang-Ma, and Shuenn-Jyi Sheu, *Accelerating diffusions*, The Annals of Applied Probability **15** (2005), no. 2, 1433–1444.
22. Alessandra Iacobucci, Stefano Olla, and Gabriel Stoltz, *Convergence rates for nonequilibrium Langevin dynamics*, arXiv preprint arXiv:1702.03685, 2017.
23. Marcelo Janovitch, Matteo Brunelli, and Patrick P. Potts, *Active quantum reservoir engineering: Using a qubit to manipulate its environment*, arXiv preprint arXiv:2505.16898, 2025.
24. Tosio Kato, *Perturbation theory for linear operators*, reprint of the 1980 edition ed., Classics in Mathematics, Springer-Verlag, Berlin, 1995.
25. E. M. Kessler, *Generalized Schrieffer-Wolff formalism for dissipative systems*, Physical Review A **86** (2012), no. 1, 012126.
26. Sergeı M Kozlov, *Effective diffusion in the fokker-planck equation*, Mathematical notes of the Academy of Sciences of the USSR **45** (1989), no. 5, 360–368.
27. Gabriel T. Landi, Dario Poletti, and Gernot Schaller, *Nonequilibrium boundary-driven quantum systems: Models, methods, and properties*, Rev. Mod. Phys. **94** (2022), 045006.
28. Tony Lelièvre, Francis Nier, and Grigorios A. Pavliotis, *Optimal non-reversible linear drift for the convergence to equilibrium of a diffusion*, Journal of Statistical Physics **152** (2013), no. 2, 237–274.
29. Bowen Li and Jianfeng Lu, *Quantum space-time Poincaré inequality for Lindblad dynamics*, arXiv preprint arXiv:2406.09115, 2024.
30. ———, *Speeding up quantum Markov processes through lifting*, arXiv preprint arXiv:2505.12187, 2025.
31. G. Lindblad, *On the generators of quantum dynamical semigroups*, Comm. Math. Phys. **48** (1976), no. 2, 119–130. MR 0413878
32. Pierre Monmarché and Mouad Ramil, *Overdamped limit at stationarity for non-equilibrium Langevin diffusions*, arXiv preprint arXiv:2110.01238, 2022.
33. Pierre Monmarché, *L2 geometric ergodicity for the kinetic langevin process with non-equilibrium steady states*, 2025.
34. Daniel Paulin, *Concentration inequalities for markov chains by marton couplings and spectral methods*, arXiv preprint arXiv:1212.2015, 2018, Introduces the pseudo spectral gap and relates it to multiplicative reversiblization for non-reversible Markov chains.
35. Vladislav Popkov, Carlo Presilla, and Mario Salerno, *Manifolds of exceptional points and effective Zeno limit of an open two-qubit system*, arXiv preprint arXiv:2412.13674, 2025.
36. Julien Roussel and Gabriel Stoltz, *Spectral methods for Langevin dynamics and associated error estimates*, ESAIM: Math. Model. Numer. Anal. **52** (2018), no. 3, 1051–1083, arXiv:1702.04718.
37. Mario Szegedy, *Quantum speed-up of Markov chain based algorithms*, 45th Annual IEEE Symposium on Foundations of Computer Science, IEEE, 2004, pp. 32–41.
38. Masaaki Tokieda and Angela Riva, *Time-convolutionless master equation applied to adiabatic elimination*, Physical Review A **111** (2025), no. 5, 052206.
39. Cédric Villani, *Hypocoercivity*, Mem. Amer. Math. Soc., vol. 202, American Mathematical Society, Providence, RI, 2009. MR 2562709

Appendix A. Proofs for Preliminaries

This appendix contains the proofs of the spectral bounds and hypocoercivity characterizations stated in Section 3. The arguments are carried out at Hilbert-space level with $\mathcal{H} = \mathbb{C}\mathbf{1} \oplus \mathcal{H}_0$. In the noncommutative realization, this corresponds to $\mathcal{H} = L^2(\sigma)$ and $\mathcal{H}_0 = \{X : \mathrm{tr}(\sigma X) = 0\}$.

A.1. **Proof of Lemma 3.2.** We first clarify the geometric relationship between the fixed point space $\mathcal{F}(\mathcal{L}) = \mathbb{C}\mathbf{1}$ and the kernel of the symmetric part $\mathcal{S}$.

Proposition A.1 (Kernel Inclusion)**.** *For any dissipative generator with symmetric part* $\mathcal{S} := \frac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger) \leq 0$*, one has*

$$\ker(\mathcal{L}) \subseteq \ker(\mathcal{S}). \tag{94}$$

Proof. Let $X \in \ker(\mathcal{L})$. Then $\mathcal{L}(X) = 0$. Consequently,

$$\langle X, \mathcal{S}(X)\rangle = \Re\langle X, \mathcal{L}(X)\rangle = 0. \tag{95}$$

Since $\mathcal{S}$ is a dissipative operator (negative semi-definite, $\mathcal{S} \leq 0$), the equality $\langle X, \mathcal{S}X\rangle = 0$ implies $\mathcal{S}X = 0$ (essentially due to the Cauchy–Schwarz inequality for the semi-inner product defined by $-\mathcal{S}$). Thus, $X \in \ker(\mathcal{S})$. □

Proof of Lemma 3.2. We prove the coercivity characterization and show that, if $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$, then no exponential decay estimate can hold with prefactor $C = 1$.

(⇐): Assume $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$ and $-\mathcal{S}|_{\mathcal{H}_0}$ has a positive spectral gap $\lambda_S > 0$.

$$\langle X, \mathcal{S}X\rangle \leq -\lambda_S \|X\|^2, \qquad \forall X \in \mathcal{H}_0. \tag{96}$$

For any $X_t = \mathcal{P}_t X_0 \in \mathcal{H}_0$, the evolution of the norm satisfies:

$$\frac{1}{2}\frac{d}{dt}\|X_t\|^2 = \Re\langle X_t, \mathcal{L}X_t\rangle = \langle X_t, \mathcal{S}X_t\rangle \leq -\lambda_S \|X_t\|^2. \tag{97}$$

Integrating this differential inequality yields $\|X_t\| \leq e^{-\lambda_S t}\|X_0\|$. This is an exponential decay with prefactor $C = 1$ and rate λ_S. Thus, the system is coercive, not strictly hypocoercive.

(⇒): Assume the system is coercive ($C = 1$). By definition, there exists $\nu > 0$ such that $\|\mathcal{P}_t X\| \leq e^{-\nu t}\|X\|$ for all $X \in \mathcal{H}_0$. The Hille–Yosida theorem (or simple differentiation at $t = 0$) implies that the generator satisfies the dissipativity condition:

$$\Re\langle X, \mathcal{L}X\rangle \leq -\nu\|X\|^2, \quad \forall X \in \mathcal{H}_0. \tag{98}$$

Substituting $\Re\langle X, \mathcal{L}X\rangle = \langle X, \mathcal{S}X\rangle$, we obtain:

$$\langle X, \mathcal{S}X\rangle \leq -\nu\|X\|^2. \tag{99}$$

Now, suppose there exists a non-trivial vector $Y \in \ker(\mathcal{S}) \cap \mathcal{H}_0$. Then $\langle Y, \mathcal{S}Y\rangle = 0$. However, the inequality above requires $\langle Y, \mathcal{S}Y\rangle \leq -\nu\|Y\|^2 < 0$, a contradiction. Therefore, $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$, which implies $\ker(\mathcal{S}) = \mathbb{C}\mathbf{1} = \ker(\mathcal{L})$.

Finally, if $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$ then the above argument shows that coercivity ($C = 1$ with some positive rate) is impossible. In that case, any exponential decay estimate (when it holds) must necessarily come with some prefactor $C > 1$. □

Appendix B. Proof of Lemma 4.11

This appendix establishes the structural bounds of Lemma 4.11.

Proof. Recall the definition of the reference effective dynamics $Q_O = -P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S$. By Assumption 4.2.1, the pseudoinverse is bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \leq \lambda_R^{-1}$, hence

$$\langle g, -\mathcal{R}^+ g\rangle \leq \frac{1}{\lambda_R}\|g\|^2, \quad \forall g \in \mathcal{H}_F. \tag{100}$$

The complementary (lower) control needed below is supplied by the flux conditioning assumption (18), which applies to fluxes of the form $g = g_\psi = P_F\mathcal{V}_{\text{skew}}P_S\psi$.

B.1. **Proof of Properties 1 and 2 (Boundedness and Regularity). Solver boundedness and forward regularity.** Let $f \in \mathcal{H}$ and define $u := \mathcal{A}_\varpi f \in \mathcal{H}_S$. The defining equation is $(\varpi - Q_O)u = P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F f$. Taking the inner product with u and using the skew-adjointness $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$:

$$\varpi\|u\|^2 + \langle u, -Q_O u\rangle = \langle u, P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F f\rangle = -\langle \mathcal{V}_{\text{skew}} u, (-\mathcal{R}^+)P_F f\rangle. \tag{101}$$

Let $v := \mathcal{V}_{\text{skew}} u$. Since $P_S\mathcal{V}_{\text{skew}}P_S = 0$ (Centered Coupling), we strictly have $v \in \mathcal{H}_F$. Using the definition of Q_O and the pseudoinverse bound (100):

$$\langle u, -Q_O u\rangle = \langle v, -\mathcal{R}^+ v\rangle = \|(-\mathcal{R}^+)^{1/2} v\|^2. \tag{102}$$

Substituting this into the energy identity and applying Cauchy–Schwarz yields:

$$\varpi\|u\|^2 + \|(-\mathcal{R}^+)^{1/2} v\|^2 \le \|(-\mathcal{R}^+)^{1/2} v\|\, \|(-\mathcal{R}^+)^{1/2} P_F f\|. \tag{103}$$

Writing $a := \|(-\mathcal{R}^+)^{1/2} v\|$ and $b := \|(-\mathcal{R}^+)^{1/2} P_F f\|$, we have $\varpi\|u\|^2 + a^2 \le ab$. Completing the square,

$$a^2 - ab = \left(a - \tfrac{1}{2}b\right)^2 - \tfrac{1}{4}b^2 \ge -\tfrac{1}{4}b^2, \tag{104}$$

so $\varpi\|u\|^2 \le \frac{1}{4}b^2$, hence

$$\|u\| \le \frac{1}{2}\,\varpi^{-1/2}\,\|(-\mathcal{R}^+)^{1/2} P_F f\| \le \frac{1}{2\sqrt{\lambda_R}}\,\varpi^{-1/2}\,\|f\|. \tag{105}$$

Moreover, $a \le b$. In the bounded-operator setting one has $\operatorname{dom}(Q_O) = \mathcal{H}_S$, so $u \in \operatorname{dom}(Q_O)$. Then the vector $v = \mathcal{V}_{\text{skew}} u$ is a slow-induced flux of the form g_ψ with $\psi \in \operatorname{dom}(Q_O)$; we may therefore apply the flux conditioning bound (18) to v to obtain

$$\|v\|^2 \le M_R\,\langle v, -\mathcal{R}^+ v\rangle = M_R a^2 \le M_R b^2 \le \frac{M_R}{\lambda_R}\,\|f\|^2, \tag{106}$$

so $\|v\| \le \sqrt{M_R/\lambda_R}\,\|f\|$. This proves $\|\mathcal{A}_\varpi\| \le k_{\mathcal{A}}\varpi^{-1/2}$ and $\|\mathcal{V}_{\text{skew}}\mathcal{A}_\varpi\| \le k_{\text{reg}}$, with

$$k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}, \qquad k_{\text{reg}} := \sqrt{M_R/\lambda_R}. \tag{107}$$

Dual regularity. We consider the operator $\mathcal{T} := \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} P_F$. Its norm equals that of its adjoint $\mathcal{T}^\dagger$. We compute the adjoint of the solver:

$$\mathcal{A}_\varpi^\dagger = \left((\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F\right)^\dagger = -P_F(-\mathcal{R}^+)\mathcal{V}_{\text{skew}} P_S(\varpi - Q_O)^{-1}. \tag{108}$$

(Using $Q_O^\dagger = Q_O$ and $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$). Therefore:

$$\mathcal{T}^\dagger = P_F \mathcal{V}_{\text{skew}}^\dagger \mathcal{A}_\varpi^\dagger = P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+)\mathcal{V}_{\text{skew}} P_S(\varpi - Q_O)^{-1}. \tag{109}$$

Let $g \in \mathcal{H}_S$ and $\psi := (\varpi - Q_O)^{-1} g$. Define the macroscopic flux

$$g_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi \in \mathcal{H}_F. \tag{110}$$

Then $\mathcal{T}^\dagger g = P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) g_\psi$. Invoking the **Flux Regularity** condition from Assumption 4.2.3:

$$\|P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) g_\psi\| \le \kappa_{\text{ell}}\|(-Q_O)\psi\|. \tag{111}$$

Substituting ψ back:

$$\|\mathcal{T}^\dagger g\| \le \kappa_{\text{ell}}\|(-Q_O)(\varpi - Q_O)^{-1} g\|. \tag{112}$$

Since Q_O is self-adjoint and non-positive, the operator $(-Q_O)(\varpi - Q_O)^{-1}$ is a spectral contraction ($\|\cdot\| \le 1$). Thus:

$$\|\mathcal{T}^\dagger g\| \le \kappa_{\text{ell}}\|g\|. \tag{113}$$

This proves $\|\mathcal{A}_\varpi \mathcal{V}_{\text{skew}} P_F\| \le \kappa_{\text{ell}}$.

B.2. **Proof of Property 3 (Perturbation Control).** We use the structural decomposition $\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$ from Assumption 4.2.4. We also recall the centered mapping property (40), which will be used later to sharpen the perturbation bookkeeping in the dissipation matrix. Recall the bound for the skew part (Assumption 4.2.4(a)):

$$\|\mathcal{K}_{\text{pert}} f\| \le a_1 \|(-\mathcal{R})^{1/2} f\| + b_1 \|f\|, \qquad f \in \operatorname{dom}((-\mathcal{R})^{1/2}). \tag{114}$$

For the symmetric part, we use the energy bound (Assumption 4.2.4(b)):

$$\|\mathcal{S}_{\text{pert}} f\| \le a_2 \|(-\mathcal{R})^{1/2} f\| + b_2 \|f\|, \qquad f \in \operatorname{dom}((-\mathcal{R})^{1/2}). \tag{115}$$

Reverse estimate ($\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\varpi$). Let $f \in \mathcal{H}$ and $u := \mathcal{A}_\varpi f$. The adjoint is $\mathcal{V}_{\text{pert}}^\dagger = -\mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$. Applying the triangle inequality and the bounds above:

$$\|\mathcal{V}_{\text{pert}}^\dagger u\| \le \|\mathcal{K}_{\text{pert}} u\| + \|\mathcal{S}_{\text{pert}} u\| \tag{116}$$

$$\le b_1 \|u\| + b_2 \|u\| + (a_1 + a_2) \|(-\mathcal{R})^{1/2} u\|. \tag{117}$$

Since $u \in \mathcal{H}_S = \ker(\mathcal{R})$ (by definition of $\mathcal{A}_\varpi$), one has $(-\mathcal{R})^{1/2} u = 0$, hence

$$\|\mathcal{V}_{\text{pert}}^\dagger u\| \le (b_1 + b_2) \|u\|. \tag{118}$$

Substituting the Solver Bound ($\|u\| \le k_{\mathcal{A}} \varpi^{-1/2} \|f\|$):

$$\|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\varpi f\| \le (b_1 + b_2) k_{\mathcal{A}} \varpi^{-1/2} \|f\|. \tag{119}$$

This establishes (37a) with $k_P(\varpi) := (b_1 + b_2) k_{\mathcal{A}} \varpi^{-1/2}$.

Forward estimate ($\mathcal{A}_\varpi \mathcal{V}_{\text{pert}}$). We aim to bound the operator norm of $\mathcal{W}_{\text{fwd}} := \mathcal{A}_\varpi \mathcal{V}_{\text{pert}}$. By adjoints on Hilbert spaces, $\|\mathcal{W}_{\text{fwd}}\| = \|\mathcal{W}_{\text{fwd}}^\dagger\| = \|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\varpi^\dagger\|$. Let $g \in \mathcal{H}_S$ be arbitrary with $\|g\| = 1$ and set $w := \mathcal{A}_\varpi^\dagger g$. Using the bounds on $\mathcal{K}_{\text{pert}}$ and the energy bound on $\mathcal{S}_{\text{pert}}$,

$$\|\mathcal{V}_{\text{pert}}^\dagger w\| \le \|\mathcal{K}_{\text{pert}} w\| + \|\mathcal{S}_{\text{pert}} w\|$$

$$\le (b_1 + b_2) \|w\| + (a_1 + a_2) \|(-\mathcal{R})^{1/2} w\|. \tag{120}$$

We compute the norms on the right-hand side.

(1) **Solver term ($\|w\|$):**

$$\|w\| = \|\mathcal{A}_\varpi^\dagger g\| \le \|\mathcal{A}_\varpi\| \|g\| \le k_{\mathcal{A}} \varpi^{-1/2}. \tag{121}$$

(2) **Energy term ($\|(-\mathcal{R})^{1/2} w\|$):** Using $w = \mathcal{A}_\varpi^\dagger g$ and the functional calculus for the non-positive self-adjoint operator Q_O,

$$\|(-\mathcal{R})^{1/2} w\| = \|(-\mathcal{R})^{1/2} \mathcal{A}_\varpi^\dagger g\| = \|(-\mathcal{R})^{-1/2} P_F \mathcal{V}_{\text{skew}} P_S (\varpi - Q_O)^{-1} g\|. \tag{122}$$

Setting $\psi := (\varpi - Q_O)^{-1} g \in \mathcal{H}_S$, we note the identity (for such slow-induced fluxes)

$$g_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi, \tag{123}$$

$$\|(-\mathcal{R})^{-1/2} g_\psi\|^2 = \langle g_\psi, -\mathcal{R}^+ g_\psi \rangle = -\langle \psi, Q_O \psi \rangle = \|(-Q_O)^{1/2} \psi\|^2, \tag{124}$$

hence

$$\|(-\mathcal{R})^{1/2} w\| = \|(-Q_O)^{1/2} (\varpi - Q_O)^{-1} g\| \le \frac{1}{2} \varpi^{-1/2} \|g\|. \tag{125}$$

(The last step follows from the scalar bound $\sup_{x \ge 0} \frac{\sqrt{x}}{\varpi + x} = \frac{1}{2\sqrt{\varpi}}$.)

Substituting these estimates into (120) yields

$$\|\mathcal{V}_{\text{pert}}^{\dagger}\mathcal{A}_{\varpi}^{\dagger}g\| \le \left((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\right)\varpi^{-1/2}\,\|g\|. \tag{126}$$

Since g is arbitrary, this establishes (37b) with $\tilde{k}_P(\varpi) := \left((b_1+b_2)k_{\mathcal{A}} + \frac{a_1+a_2}{2}\right)\varpi^{-1/2}$.

□

Appendix C. Proof of Theorem 5.1

This appendix derives the dissipation matrix governing the hypocoercive convergence. The computation is based on the modified Lyapunov functional with resolvent shift $\varpi > 0$ and mixing weight $\alpha > 0$:

$$\mathcal{E}_{\alpha,\varpi}(f) := \frac{1}{2}\|f\|^2 + \alpha\Re\langle\mathcal{A}_{\varpi}f, f\rangle, \tag{127}$$

where $\mathcal{A}_{\varpi} = (\varpi - Q_O)^{-1}P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F$ is the DMS corrector associated with the reference effective dynamics Q_O.

C.1. **Preliminaries.** This appendix uses the structural estimates of Lemma 4.11 (solver bound, regularity, and perturbation control) and the notation of equations (44)–(48). We set $\varpi := \varpi_* = \lambda(Q_O)$ throughout and use the effective perturbation coefficient

$$b_{2,\text{eff}} := b_2 + \frac{a_2}{2}(1+\lambda_R), \tag{128}$$

as defined in (44).

C.2. **Derivation of Differential Estimates. Standing regularity for the Lyapunov computation.** In the bounded-operator setting, for any $f_0 \in \mathcal{H}$ the solution $f_t = e^{t\mathcal{L}}f_0$ is continuously differentiable and satisfies $\partial_t f_t = \mathcal{L}f_t$. In unbounded settings, the computations below are performed on a common invariant core $\mathcal{C} \subset \text{dom}(\mathcal{L}) \cap \text{dom}(\mathcal{A}_{\varpi})$ and extended to the energy domain by density.

Lemma C.1 (Energy identity for the fast dissipator)**.** *Assume $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$. Then for all $f \in \text{dom}(\mathcal{R})$,*

$$\Re\langle\mathcal{R}f, f\rangle = \langle\mathcal{R}f, f\rangle = -\|(-\mathcal{R})^{1/2}f\|^2. \tag{129}$$

In particular, if $f = f_S + f_F$ with $f_S \in \ker(\mathcal{R})$ and $f_F \in \mathcal{H}_F$, then

$$\Re\langle\mathcal{R}f, f\rangle = \Re\langle\mathcal{R}f_F, f_F\rangle. \tag{130}$$

Proof. Since $\mathcal{R}$ is self-adjoint and non-positive, $-\mathcal{R} \ge 0$ admits a functional calculus square root $(-\mathcal{R})^{1/2}$ and $\langle\mathcal{R}f, f\rangle = -\langle(-\mathcal{R})f, f\rangle = -\|(-\mathcal{R})^{1/2}f\|^2$. Moreover $\mathcal{R}f_S = 0$ and $\mathcal{R}f = \mathcal{R}f_F$. □

Let f_t satisfy $\partial_t f_t = \mathcal{L}f_t$. By differentiating (42) along the flow and using that $\mathcal{A}_{\varpi}$ is time-independent, we obtain

$$\frac{d}{dt}\mathcal{E}_{\alpha,\varpi}(f_t) = \Re\langle\partial_t f_t, f_t\rangle + \alpha\,\Re\langle\mathcal{A}_{\varpi}\partial_t f_t, f_t\rangle + \alpha\,\Re\langle\mathcal{A}_{\varpi}f_t, \partial_t f_t\rangle \tag{131}$$

$$= \Re\langle\mathcal{L}f_t, f_t\rangle + \alpha\,\Re\langle\mathcal{A}_{\varpi}\mathcal{L}f_t, f_t\rangle + \alpha\,\Re\langle\mathcal{A}_{\varpi}f_t, \mathcal{L}f_t\rangle. \tag{132}$$

We now split the right-hand side into the "plain" dissipation coming from the generator itself and the hypocoercive correction generated by the cross term:

$$\frac{d}{dt}\mathcal{E}_{\alpha,\varpi}(f_t) = \mathcal{D}_I(f_t) + \mathcal{D}_{\text{mod}}(f_t), \tag{133}$$

where we define

$$\mathcal{D}_I(f) := \Re\langle \mathcal{L}f, f\rangle, \tag{134}$$

$$\mathcal{D}_{\mathrm{mod}}(f) := \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{L}f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{L}f\rangle. \tag{135}$$

Expanding $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ in $\mathcal{D}_I$ then yields the decomposition displayed below.

Natural dissipation ($\mathcal{D}_I$).

$$\mathcal{D}_I = \gamma\Re\langle \mathcal{R}f, f\rangle + \Re\langle \mathcal{V}_{\mathrm{skew}} f, f\rangle + \eta\Re\langle \mathcal{V}_{\mathrm{pert}} f, f\rangle. \tag{136}$$

Using Assumption 4.2.1 ($\mathcal{R}$ coercivity) and Assumption 4.2.4 (Perturbation decomposition $\mathcal{V}_{\mathrm{pert}} = \mathcal{K}_{\mathrm{pert}} + \mathcal{S}_{\mathrm{pert}}$): the fast dissipator contributes $\gamma\Re\langle \mathcal{R}f, f\rangle \le -\gamma\lambda_R\|f_F\|^2$, while $\Re\langle \mathcal{V}_{\mathrm{skew}} f, f\rangle = 0$ and $\Re\langle \mathcal{K}_{\mathrm{pert}} f, f\rangle = 0$ by skew-adjointness. For the symmetric part, using (24) and Cauchy–Schwarz,

$$\eta\Re\langle \mathcal{S}_{\mathrm{pert}} f, f\rangle \le \eta\|\mathcal{S}_{\mathrm{pert}} f\|\,\|f\| \le \eta\Big(b_2\|f\|^2 + a_2\,\|(-\mathcal{R})^{1/2} f\|\,\|f\|\Big) \tag{137}$$

$$\le \eta\Big(\big(b_2 + \tfrac{a_2}{2}\big)\|f\|^2 + \tfrac{a_2}{2}\,\|(-\mathcal{R})^{1/2} f\|^2\Big). \tag{138}$$

Combining with the dissipator contribution, note that by Lemma C.1, $\Re\langle \mathcal{R}f, f\rangle = \Re\langle \mathcal{R}f_F, f_F\rangle = -\|(-\mathcal{R})^{1/2} f\|^2$. Hence

$$\gamma\Re\langle \mathcal{R}f, f\rangle + \eta\Re\langle \mathcal{S}_{\mathrm{pert}} f, f\rangle \le -\big(\gamma - \tfrac{\eta a_2}{2}\big)\,\|(-\mathcal{R})^{1/2} f\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f\|^2 \tag{139}$$

$$\le -\big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big)\,\|f_F\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f_S\|^2, \tag{140}$$

where we used $\|(-\mathcal{R})^{1/2} f\|^2 \ge \lambda_R\|f_F\|^2$, the decomposition $\|f\|^2 = \|f_S\|^2 + \|f_F\|^2$, and the fact that $\frac{a_2\lambda_R}{2} + (b_2 + \frac{a_2}{2}) = b_{2,\mathrm{eff}}$ by definition. The fast mode carries the full effective coefficient $b_{2,\mathrm{eff}}$, while the slow mode retains only the smaller $b_2 + \frac{a_2}{2}$ (since $f_S \in \ker(\mathcal{R})$ does not benefit from the λ_R-amplification).

Thus:

$$\mathcal{D}_I \le -\big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big)\,\|f_F\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f_S\|^2. \tag{141}$$

Hypocoercive correction ($\mathcal{D}_{\mathrm{mod}}$). Recall

$$\mathcal{D}_{\mathrm{mod}}(f) = \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{L}f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{L}f\rangle. \tag{142}$$

Using $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ and linearity, we regroup $\mathcal{D}_{\mathrm{mod}}$ according to which part of the generator appears:

$$\mathcal{D}_{\mathrm{mod}}(f) = T_{\mathcal{R}}(f) + T_{\mathcal{V}}(f) + T_{\mathcal{P}}(f), \tag{143}$$

where

$$T_{\mathcal{R}}(f) := \alpha\gamma\,\Re\langle \mathcal{A}_\varpi \mathcal{R}f, f\rangle + \alpha\gamma\,\Re\langle \mathcal{A}_\varpi f, \mathcal{R}f\rangle, \tag{144}$$

$$T_{\mathcal{V}}(f) := \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\mathrm{skew}} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\mathrm{skew}} f\rangle, \tag{145}$$

$$T_{\mathcal{P}}(f) := \alpha\eta\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\mathrm{pert}} f, f\rangle + \alpha\eta\,\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\mathrm{pert}} f\rangle. \tag{146}$$

Interaction with the fast dissipator. We start from the regrouped definition

$$T_{\mathcal{R}}(f) = \alpha\gamma\,\Re\langle \mathcal{A}_\varpi \mathcal{R}f, f\rangle + \alpha\gamma\,\Re\langle \mathcal{A}_\varpi f, \mathcal{R}f\rangle. \tag{147}$$

Since $\mathcal{R}f = \mathcal{R}f_F$ and $\mathcal{A}_\varpi f \in \mathcal{H}_S = \ker(\mathcal{R})$, the second term vanishes by self-adjointness of $\mathcal{R}$:

$$\Re\langle \mathcal{A}_\varpi f, \mathcal{R}f\rangle = \Re\langle \mathcal{R}\mathcal{A}_\varpi f, f\rangle = 0. \tag{148}$$

Moreover $\mathcal{A}_\varpi \mathcal{R}f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$, hence

$$T_{\mathcal{R}}(f) = \alpha\gamma\,\Re\langle \mathcal{A}_\varpi \mathcal{R}f_F, f_S\rangle. \tag{149}$$

Using $\mathcal{A}_\varpi = (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\rm skew}(-\mathcal{R}^+) P_F$ and the pseudoinverse identity $\mathcal{R}^+\mathcal{R} = P_F$ on $\mathrm{dom}(\mathcal{R})$, we obtain

$$\mathcal{A}_\varpi \mathcal{R} f_F = (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\rm skew}(-\mathcal{R}^+)\mathcal{R} f_F = -(\varpi - Q_O)^{-1} P_S \mathcal{V}_{\rm skew} f_F. \tag{150}$$

Therefore, by skew-adjointness of $\mathcal{V}_{\rm skew}$,

$$|T_{\mathcal{R}}| = \alpha\gamma \left|\Re\langle (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\rm skew} f_F, f_S\rangle\right| = \alpha\gamma \left|\Re\langle f_F, \mathcal{V}_{\rm skew}(\varpi - Q_O)^{-1} f_S\rangle\right| \tag{151}$$

$$\le \alpha\gamma \, \|f_F\| \, \|\mathcal{V}_{\rm skew}(\varpi - Q_O)^{-1} f_S\|. \tag{152}$$

To bound $\|\mathcal{V}_{\rm skew}(\varpi - Q_O)^{-1} f_S\|$, set $\psi := (\varpi - Q_O)^{-1} f_S \in \mathrm{dom}(Q_O) \subset \mathcal{H}_S$ and denote its induced microscopic flux

$$g_\psi := P_F \mathcal{V}_{\rm skew} P_S \psi \in \mathcal{H}_F. \tag{153}$$

Since $\psi \in \mathcal{H}_S$, the centered-coupling assumption gives $P_S \mathcal{V}_{\rm skew}\psi = 0$, hence $g_\psi = \mathcal{V}_{\rm skew}\psi = \mathcal{V}_{\rm skew}(\varpi - Q_O)^{-1} f_S$. Applying the flux conditioning bound (18) yields

$$\|\mathcal{V}_{\rm skew}(\varpi - Q_O)^{-1} f_S\|^2 = \|g_\psi\|^2 \le M_R \, \langle g_\psi, -\mathcal{R}^+ g_\psi\rangle. \tag{154}$$

The "extra Q_O" comes from rewriting the right-hand side using the definition $Q_O := -P_S \mathcal{V}_{\rm skew}\mathcal{R}^+\mathcal{V}_{\rm skew} P_S$: indeed, for $\psi \in \mathcal{H}_S$,

$$\langle g_\psi, -\mathcal{R}^+ g_\psi\rangle = \langle \mathcal{V}_{\rm skew}\psi, (-\mathcal{R}^+)\mathcal{V}_{\rm skew}\psi\rangle = \langle \psi, (-Q_O)\psi\rangle = \|(-Q_O)^{1/2}\psi\|^2. \tag{155}$$

Therefore

$$\|\mathcal{V}_{\rm skew}(\varpi - Q_O)^{-1} f_S\| \le \sqrt{M_R}\, \|(-Q_O)^{1/2}(\varpi - Q_O)^{-1} f_S\|. \tag{156}$$

Finally, by functional calculus for the self-adjoint operator $-Q_O \ge 0$,

$$\|(-Q_O)^{1/2}(\varpi - Q_O)^{-1}\| = \sup_{x\ge 0} \frac{\sqrt{x}}{\varpi + x} = \frac{1}{2\sqrt{\varpi}}, \tag{157}$$

which implies

$$\|\mathcal{V}_{\rm skew}(\varpi - Q_O)^{-1} f_S\| \le \frac{\sqrt{M_R}}{2}\, \varpi^{-1/2}\, \|f_S\|. \tag{158}$$

Hence

$$|T_{\mathcal{R}}| \le \alpha\gamma \, \frac{\sqrt{M_R}}{2}\, \varpi^{-1/2}\, \|f_F\|\, \|f_S\|. \tag{159}$$

Interaction with transport. Starting from the regrouped definition

$$T_{\mathcal{V}}(f) = \alpha\, \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f, f\rangle + \alpha\, \Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\rm skew} f\rangle, \tag{160}$$

we decompose $f = f_S + f_F$ and use that $\mathcal{A}_\varpi$ maps into the slow space and annihilates the slow component: $\mathcal{A}_\varpi = \mathcal{A}_\varpi P_F$ (since P_F is on the right in (32)). Hence $\mathcal{A}_\varpi f = \mathcal{A}_\varpi f_F$. Expanding the *first* bracket gives

$$\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f, f\rangle = \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle \tag{161}$$

$$+ \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_F\rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_F\rangle. \tag{162}$$

The last term vanishes because $\mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$. Moreover, since $\mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S \in \mathcal{H}_S$, we have $\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_F\rangle = 0$. Therefore

$$\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f, f\rangle = \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle. \tag{163}$$

In the *second* bracket we use $\mathcal{A}_\varpi f = \mathcal{A}_\varpi f_F$ and expand $\mathcal{V}_{\rm skew} f$:

$$\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\rm skew} f\rangle = \Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_S\rangle + \Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_F\rangle. \tag{164}$$

Putting the pieces together, we obtain the explicit decomposition

$$T_{\mathcal{V}} = \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle \tag{165}$$

$$+\,\alpha\,\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_S\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_F\rangle. \tag{166}$$

Main Coercivity. Since $f_S \in \mathcal{H}_S$ and $P_S \mathcal{V}_{\rm skew} P_S = 0$, we have $P_F \mathcal{V}_{\rm skew} f_S = \mathcal{V}_{\rm skew} f_S$. By the definition of $\mathcal{A}_\varpi$ and Q_O this yields

$$\alpha\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle = \alpha\langle (\varpi - Q_O)^{-1} Q_O f_S, f_S\rangle = -\alpha\,\langle (\varpi - Q_O)^{-1}(-Q_O) f_S, f_S\rangle \tag{167}$$

$$\leq -\alpha\,\frac{\lambda(Q_O)}{\varpi + \lambda(Q_O)}\,\|f_S\|^2. \tag{168}$$

Here we used that $f \in \mathcal{H}_0$ implies $f_S = P_S f \in \mathcal{H}_S \cap \mathcal{H}_0 = \mathcal{H}_{S,0}$, so Assumption 4.4 applies to f_S.
Cross Terms. Since $\mathcal{A}_\varpi f_F \in \mathcal{H}_S$ (by the range of the corrector) and $\mathcal{V}_{\rm skew} f_S \in \mathcal{H}_F$ (by centering, $P_S \mathcal{V}_{\rm skew} P_S = 0$), the third inner product vanishes: $\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_S\rangle = 0$. The only surviving cross term is bounded by the dual regularity estimate:

$$\alpha\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle \leq \alpha\|\mathcal{A}_\varpi \mathcal{V}_{\rm skew} P_F\|\|f_F\|\|f_S\| \leq \alpha\kappa_{\rm ell}\|f_F\|\|f_S\|. \tag{169}$$

Fast–fast remainder. Using $\mathcal{V}_{\rm skew}^\dagger = -\mathcal{V}_{\rm skew}$ and Cauchy–Schwarz,

$$\big|\alpha\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_F\rangle\big| = \alpha\big|\Re\langle -\mathcal{V}_{\rm skew}\mathcal{A}_\varpi f_F, f_F\rangle\big| \leq \alpha\,\|\mathcal{V}_{\rm skew}\mathcal{A}_\varpi\|\,\|f_F\|^2 \leq \alpha\,k_{\rm reg}\,\|f_F\|^2. \tag{170}$$

Interaction with perturbation.

$$T_{\mathcal{P}} = \alpha\eta\Re\langle \mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f, f\rangle + \alpha\eta\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm pert} f, f\rangle. \tag{171}$$

We estimate each term using Lemma 4.11 and the centered mapping property (40).
Reverse term. Since $\mathcal{A}_\varpi f \in \mathcal{H}_S$ and $\mathcal{V}_{\rm pert}^\dagger \mathcal{H}_S \subset \mathcal{H}_F$, we have $\langle \mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f, f\rangle = \langle \mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f, f_F\rangle$. Therefore,

$$\big|\alpha\eta\Re\langle \mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f, f\rangle\big| \leq \alpha\eta\,\|\mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f\|\,\|f_F\| \leq \alpha\eta\,k_P(\varpi)\,\|f\|\,\|f_F\| \tag{172}$$

$$\leq \alpha\eta\,k_P(\varpi)\,\big(\|f_F\|^2 + \|f_F\|\,\|f_S\|\big). \tag{173}$$

Forward term. Since $\mathcal{A}_\varpi \mathcal{H} \subset \mathcal{H}_S$, we have $\langle \mathcal{A}_\varpi \mathcal{V}_{\rm pert} f, f\rangle = \langle \mathcal{A}_\varpi \mathcal{V}_{\rm pert} f, f_S\rangle$. Therefore,

$$\big|\alpha\eta\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm pert} f, f\rangle\big| \leq \alpha\eta\,\|\mathcal{A}_\varpi \mathcal{V}_{\rm pert} f\|\,\|f_S\| \leq \alpha\eta\,\tilde{k}_P(\varpi)\,\|f\|\,\|f_S\| \tag{174}$$

$$\leq \alpha\eta\,\tilde{k}_P(\varpi)\,\big(\|f_S\|^2 + \|f_F\|\,\|f_S\|\big). \tag{175}$$

C.3. **Dissipation Matrix Construction.** Summing all estimates and using Cauchy–Schwarz bounds of the form $|\langle X f_F, f_S\rangle| \leq \|X\|\,\|f_F\|\,\|f_S\|$, we obtain a closed differential inequality of the form

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\varpi}(f_t) \leq -\mathbf{v}(t)^T \mathbf{M}\,\mathbf{v}(t), \qquad \mathbf{v}(t) := \begin{pmatrix} \|P_F f_t\| \\ \|P_S f_t\| \end{pmatrix}. \tag{176}$$

Here $\mathbf{M}$ is a real symmetric 2×2 matrix whose entries depend only on the structural constants and the chosen parameters (α, ϖ). The entries of $\mathbf{M}$ are explicitly decomposed by powers of ϖ below.

Fast-Fast Block (M_{11}): From Natural Dissipation, the fast–fast transport remainder, and Reverse Perturbation.

$$M_{11} = \big(\gamma\lambda_R - \eta b_{2,\rm eff}\big) - \alpha k_{\rm reg} - \alpha\eta k_P(\varpi)$$

$$= \gamma\lambda_R - \eta b_{2,\rm eff} - \alpha k_{\rm reg} - \alpha\eta(b_1 + b_2)k_{\mathcal{A}}\,\varpi^{-1/2}. \tag{177}$$

Slow-Slow Block (M_{22}): From Main Coercivity, Natural Dissipation (gain), and both Perturbations.

$$\begin{aligned} M_{22} &= \alpha \frac{\lambda(Q_O)}{\varpi + \lambda(Q_O)} - \eta\big(b_2 + \tfrac{a_2}{2}\big) - \alpha\eta\, \tilde{k}_P(\varpi) \\ &= \alpha \frac{\lambda(Q_O)}{\varpi + \lambda(Q_O)} - \eta\big(b_2 + \tfrac{a_2}{2}\big) - \alpha\eta \left[\Big((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\varpi^{-1/2}\right]. \end{aligned} \tag{178}$$

Cross Block (M_{12}): From Mixing, Transport Cross terms, and Forward Perturbation.

$$\begin{aligned} M_{12} &= -\frac{1}{2}\alpha \left[\gamma \frac{\sqrt{M_R}}{2}\, \varpi^{-1/2} + \kappa_{\text{ell}} + \eta\big(k_P(\varpi) + \tilde{k}_P(\varpi)\big)\right] \\ &= -\frac{1}{2}\alpha \left[\kappa_{\text{ell}} + \left(\gamma \frac{\sqrt{M_R}}{2} + \eta\Big(2(b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\right)\varpi^{-1/2}\right]. \end{aligned} \tag{179}$$

This explicit form allows for direct asymptotic analysis by choosing $\varpi \sim \lambda(Q_O)$, confirming the stability for small η.

C.4. **Optimization of the Decay Rate.** We work in the unperturbed regime $\eta = 0$ to identify the dominant hypocoercive rate. Let $\mathbf{M}_0(\alpha)$ denote the dissipation matrix at $\eta = 0$. Since $\mathbf{M}_0$ is 2×2 symmetric, its smallest eigenvalue satisfies the determinant–trace bound

$$\lambda_{\min}(\mathbf{M}_0) \geq \frac{\det(\mathbf{M}_0)}{\operatorname{Tr}(\mathbf{M}_0)}. \tag{180}$$

Choice of resolvent shift. The slow-coercivity contribution in M_{22}^0 is $\alpha\lambda(Q_O)/(\varpi+\lambda(Q_O))$, while the cross term $|M_{12}^0|$ contains $\gamma\Theta\varpi^{-1/2}$. A leading-order balance (maximising $\varpi(\varpi + \lambda(Q_O))^{-2}$ in the large-γ rate) selects

$$\varpi_* := \lambda(Q_O). \tag{181}$$

At this value, $\lambda(Q_O)/(\varpi_* + \lambda(Q_O)) = \frac{1}{2}$. Using the constants $\Theta, K_{\text{geom}}, Z(\gamma)$ from (44)–(45), the unperturbed matrix entries at $\varpi = \varpi_*$ are:

$$M_{11}^0 = \gamma\lambda_R - \alpha k_{\text{reg}}, \tag{182}$$

$$M_{22}^0 = \tfrac{1}{2}\,\alpha, \tag{183}$$

$$M_{12}^0 = -\tfrac{1}{2}\,\alpha\, Z(\gamma). \tag{184}$$

Optimal mixing weight. The determinant $D(\alpha) := M_{11}^0 M_{22}^0 - (M_{12}^0)^2$ is a concave quadratic in α:

$$D(\alpha) = \tfrac{1}{2}\gamma\lambda_R\,\alpha - \big(\tfrac{1}{2}k_{\text{reg}} + \tfrac{1}{4}Z(\gamma)^2\big)\alpha^2. \tag{185}$$

It is maximised at

$$\alpha_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}, \tag{186}$$

in agreement with (46), giving

$$D(\alpha_*) = \frac{\frac{1}{4}(\gamma\lambda_R)^2}{Z(\gamma)^2 + 2k_{\text{reg}}}. \tag{187}$$

Trace upper bound and rate estimate. The exact trace is $\operatorname{Tr}(\mathbf{M}_0)|_{\alpha_*} = (\gamma\lambda_R - \alpha_* k_{\text{reg}}) + \frac{1}{2}\alpha_* = \gamma\lambda_R - \alpha_*(k_{\text{reg}} - \frac{1}{2})$. Since $k_{\text{reg}} = \sqrt{M_R/\lambda_R} \geq 1$ (the flux conditioning bound gives $M_R \geq \lambda_R$), we have $k_{\text{reg}} - \frac{1}{2} \geq \frac{1}{2} > 0$, hence

$$\operatorname{Tr}(\mathbf{M}_0)|_{\alpha_*} \leq \gamma\lambda_R. \tag{188}$$

Applying (180) yields

$$\lambda_{\min}(\mathbf{M}_0) \geq \frac{D(\alpha_*)}{\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*}} \geq \frac{\frac{1}{4}\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\mathrm{reg}}} =: \Lambda_{\mathrm{hypo}}(\gamma), \tag{189}$$

consistent with definition (47). This provides a rigorous lower bound on the unperturbed decay rate for all $\gamma > 0$.

Asymptotic scaling analysis. Equation (189) reveals the precise scaling behavior across regimes without relying on ad-hoc approximations.
Overdamped Limit ($\gamma \to \infty$). The denominator is dominated by the $\mathcal{O}(\gamma^2)$ term, giving

$$\lim_{\gamma\to\infty} \gamma\Lambda_{\mathrm{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R\,\lambda(Q_O)}{\Theta^2} = \lambda(Q_O)\,\frac{\lambda_R}{M_R}. \tag{190}$$

Thus, $\Lambda_{\mathrm{hypo}}(\gamma) = \lambda(Q_O)\,\frac{\lambda_R}{M_R}\,\gamma^{-1} + \mathcal{O}(\gamma^{-2})$. In particular, $\Lambda_{\mathrm{hypo}}(\gamma) \asymp \lambda(Q_O)\,\gamma^{-1}$.
Underdamped/Intermediate Regime. For small or order-one γ, the denominator is controlled by the γ-independent structural constants K_{geom} and $2k_{\mathrm{reg}}$. These arise from the slow–fast coupling through $\mathcal{V}_{\mathrm{skew}}$ and prevent the estimate from degenerating even though $\mathcal{R}$ does not dissipate on $\mathcal{H}_S$ directly. The transport mixes slow directions into fast dissipative directions, yielding $\Lambda_{\mathrm{hypo}}(\gamma) \asymp \gamma$ for small γ.

We now establish the stability condition by computing the operator norm of the perturbation matrix $\mathbf{E}(\gamma)$ as a function of γ, tracking the dependence on the macroscopic spectral gap $\varpi_* = \lambda(Q_O)$.

Explicit norm of the perturbation matrix. Recalling that $\mathbf{M} = \mathbf{M}_0 - \eta\mathbf{E}$, the perturbation matrix $\mathbf{E}$ has entries read off from the full $\mathbf{M}$ computation:

$$E_{11} = b_{2,\mathrm{eff}} + \alpha_*\,k_P(\varpi_*), \tag{191}$$

$$E_{22} = \big(b_2 + \tfrac{a_2}{2}\big) + \alpha_*\,\tilde{k}_P(\varpi_*), \tag{192}$$

$$E_{12} = \frac{1}{2}\,\alpha_*\,\big(k_P(\varpi_*) + \tilde{k}_P(\varpi_*)\big). \tag{193}$$

Here E_{11} contains the full effective coefficient $b_{2,\mathrm{eff}} = b_2 + \frac{a_2}{2}(1+\lambda_R)$, while E_{22} contains only the slow-mode coefficient $b_2 + \frac{a_2}{2}$; the extra $\frac{a_2\lambda_R}{2}$ absorbed into $b_{2,\mathrm{eff}}$ arises in the fast block from the bound $\|(-\mathcal{R})^{1/2}f\|^2 \geq \lambda_R\|f_F\|^2$ applied to the symmetric perturbation in $\mathcal{D}_I$ (see (141)). Since $\mathbf{E}$ is symmetric, we bound its spectral norm by the maximal absolute row sum (Gershgorin):

$$\|\mathbf{E}(\gamma)\| \leq \max\{E_{11} + |E_{12}|,\, E_{22} + |E_{12}|\}. \tag{194}$$

Using $b_{2,\mathrm{eff}} \geq b_2 + \frac{a_2}{2}$ and $k_P(\varpi_*) \leq \tilde{k}_P(\varpi_*)$, both row sums admit the same uniform upper bound:

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\mathrm{eff}} + \alpha_*(\gamma)\left[\tfrac{3}{2}\,\tilde{k}_P(\varpi_*) + \tfrac{1}{2}\,\tilde{k}_P(\varpi_*)\right] = b_{2,\mathrm{eff}} + 2\alpha_*(\gamma)\,\tilde{k}_P(\varpi_*). \tag{195}$$

Recall from Lemma 4.11 that the bounds k_P and $\tilde{k}_P$ depend on $\varpi_*^{-1/2}$. We define the *strictly constant* structural coefficients (independent of ϖ and γ):

$$C_{\mathrm{sing}} := 2(b_1 + b_2)k_{\mathcal{A}} + (a_1 + a_2). \tag{196}$$

Indeed,

$$2\tilde{k}_P(\varpi_*) = 2\Big((b_1 + b_2)k_{\mathcal{A}} + \frac{a_1 + a_2}{2}\Big)\varpi_*^{-1/2} = C_{\mathrm{sing}}\,\varpi_*^{-1/2},$$

so the bound (195) becomes

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\mathrm{eff}} + \alpha_*(\gamma)\,C_{\mathrm{sing}}\,\varpi_*^{-1/2}. \tag{197}$$

Substituting $\alpha_*(\gamma)$ from (46), the bound becomes

$$\|\mathbf{E}(\gamma)\| \le b_{2,\mathrm{eff}} + \alpha_*(\gamma)\, C_{\mathrm{sing}}\, \lambda(Q_O)^{-1/2} = B_{\mathrm{pert}}(\gamma), \tag{198}$$

consistent with definition (48).

Friction-dependent threshold. The stability condition $\lambda_{\min}(\mathbf{M}) > 0$ is ensured by $\eta\|\mathbf{E}(\gamma)\| \le \frac{1}{2}\Lambda_{\mathrm{hypo}}(\gamma)$, which yields the exact matrix threshold

$$\eta^\sharp(\gamma) := \frac{\Lambda_{\mathrm{hypo}}(\gamma)}{2\|\mathbf{E}(\gamma)\|}. \tag{199}$$

Since $\|\mathbf{E}(\gamma)\| \le B_{\mathrm{pert}}(\gamma)$, the explicit certified threshold used in Theorem 5.1 is

$$\eta^*(\gamma) := \frac{\Lambda_{\mathrm{hypo}}(\gamma)}{2B_{\mathrm{pert}}(\gamma)} \le \eta^\sharp(\gamma). \tag{200}$$

Substituting the explicit forms of Λ_{hypo} (47) and B_{pert} (48) yields

$$\eta^*(\gamma) = \frac{\frac{1}{4}\,\gamma\lambda_R}{\big(Z(\gamma)^2 + 2k_{\mathrm{reg}}\big)\big(2b_{2,\mathrm{eff}} + 2\alpha_*(\gamma)\, C_{\mathrm{sing}}\, \lambda(Q_O)^{-1/2}\big)}. \tag{201}$$

Asymptotic analysis of the certified threshold. We analyze the scaling of $\eta^*(\gamma)$ in the limiting regimes. Note that $Z(\gamma) \asymp \gamma$ as $\gamma \to \infty$.
Overdamped Limit ($\gamma \to \infty$). The numerator scales as $\mathcal{O}(\gamma)$. In the denominator, the term $2b_{2,\mathrm{eff}} Z(\gamma)^2$ scales as $\mathcal{O}(\gamma^2)$, while the skew-term scales as $\mathcal{O}(\gamma)$. If $b_{2,\mathrm{eff}} > 0$ (symmetric perturbation present), the $\mathcal{O}(\gamma^2)$ term dominates and $\eta^*(\gamma) \asymp \gamma^{-1}$. If $b_{2,\mathrm{eff}} = 0$ (purely skew perturbation, i.e. $b_2 = a_2 = 0$), the denominator is dominated by the skew term $\mathcal{O}(\gamma)$ and $\eta^*(\gamma) \asymp \mathcal{O}(1)$.
Optimal Lift ($\gamma \sim \sqrt{\lambda(Q_O)}$). At the optimal friction γ_{opt}, determined in the optimization step below, we have $Z(\gamma_{\mathrm{opt}}) = \mathcal{O}(1)$. Consider the singular term in the denominator:

$$\gamma_{\mathrm{opt}} \cdot C_{\mathrm{sing}} \lambda(Q_O)^{-1/2} \asymp \sqrt{\lambda(Q_O)} \cdot \lambda(Q_O)^{-1/2} \asymp \mathcal{O}(1). \tag{202}$$

The singularity thus cancels, and at the optimal friction the stability threshold is well-behaved:

$$\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}. \tag{203}$$

From coercivity to global decay. Having established the positivity of the dissipation matrix, we convert the algebraic estimate into a global semigroup decay bound. This step involves explicit algebraic substitution to derive the final decay rate, ensuring that the singular scaling of the perturbation is correctly propagated.

Uniform norm equivalence. Recall the distortion parameter

$$\delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|. \tag{204}$$

Using $\|\mathcal{A}_{\varpi_*}\| \le k_{\mathcal{A}}\, \varpi_*^{-1/2}$ with $k_{\mathcal{A}} = (2\sqrt{\lambda_R})^{-1}$ and the explicit weight $\alpha_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\mathrm{reg}}}$, we obtain

$$\delta_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\mathrm{reg}}}\, \frac{1}{\sqrt{\lambda_R\, \varpi_*}} = \frac{\gamma\sqrt{\lambda_R}\, \varpi_*^{-1/2}}{\big(\gamma\Theta\varpi_*^{-1/2} + K_{\mathrm{geom}}\big)^2 + 2k_{\mathrm{reg}}}. \tag{205}$$

Setting $x := \gamma\Theta\varpi_*^{-1/2}$ gives

$$\delta_*(\gamma) = \frac{\sqrt{\lambda_R}}{\Theta} \cdot \frac{x}{(x + K_{\mathrm{geom}})^2 + 2k_{\mathrm{reg}}}. \tag{206}$$

The elementary optimisation

$$\sup_{x>0} \frac{x}{(x+K_{\mathrm{geom}})^2 + 2k_{\mathrm{reg}}} = \frac{1}{2\big(K_{\mathrm{geom}} + \sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}}\big)} \tag{207}$$

therefore yields

$$\delta_*(\gamma) \le \frac{\sqrt{\lambda_R}}{\sqrt{M_R}\big(K_{\mathrm{geom}} + \sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}}\big)} \le 2^{-1/2}, \tag{208}$$

because $M_R \ge \lambda_R$ and $\sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}} \ge \sqrt{2k_{\mathrm{reg}}} \ge \sqrt{2}$ (the latter since $k_{\mathrm{reg}} = \sqrt{M_R/\lambda_R} \ge 1$). Hence the norm-equivalence prefactor satisfies

$$C_*(\gamma) := \sqrt{\frac{1+\delta_*(\gamma)}{1-\delta_*(\gamma)}} \le 1 + \sqrt{2}. \tag{209}$$

Thus, for all f,

$$\frac{1}{2}(1-\delta_*(\gamma))\|f\|^2 \le \mathcal{E}_{\alpha_*,\varpi_*}(f) \le \frac{1}{2}(1+\delta_*(\gamma))\|f\|^2. \tag{210}$$

Explicit calculation of the decay rate. Set

$$c(\gamma,\eta) := \Lambda_{\mathrm{hypo}}(\gamma) - \eta\,\|\mathbf{E}(\gamma)\|. \tag{211}$$

The algebraic dissipation estimate reads

$$\frac{d}{dt}\,\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \le -c(\gamma,\eta)\,\|f_t\|^2. \tag{212}$$

Next we use the norm equivalence from Proposition 4.13 in the convenient form

$$\frac{1}{2}(1-\delta_*(\gamma))\,\|f\|^2 \le \mathcal{E}_{\alpha_*,\varpi_*}(f) \le \frac{1}{2}(1+\delta_*(\gamma))\,\|f\|^2, \tag{213}$$

where $\delta_*(\gamma) := 2\alpha_*(\gamma)\,\|\mathcal{A}_{\varpi_*}\| \in (0,1)$. The upper bound implies

$$\|f\|^2 \ge \frac{2}{1+\delta_*(\gamma)}\,\mathcal{E}_{\alpha_*,\varpi_*}(f). \tag{214}$$

Substituting this into the dissipation inequality gives a closed differential inequality for the Lyapunov functional:

$$\frac{d}{dt}\,\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \le -\frac{2c(\gamma,\eta)}{1+\delta_*(\gamma)}\,\mathcal{E}_{\alpha_*,\varpi_*}(f_t). \tag{215}$$

By Grönwall's inequality,

$$\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \le \exp\Big(-\frac{2c(\gamma,\eta)}{1+\delta_*(\gamma)}\,t\Big)\,\mathcal{E}_{\alpha_*,\varpi_*}(f_0). \tag{216}$$

Finally, converting back to the L^2 norm using the two-sided equivalence yields

$$\|f_t\|^2 \le \frac{2}{1-\delta_*(\gamma)}\,\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \le \frac{1+\delta_*(\gamma)}{1-\delta_*(\gamma)}\,e^{-\frac{2c(\gamma,\eta)}{1+\delta_*(\gamma)}t}\,\|f_0\|^2. \tag{217}$$

Therefore

$$\|f_t\| \le \sqrt{\frac{1+\delta_*(\gamma)}{1-\delta_*(\gamma)}}\,e^{-\lambda_\eta t}\,\|f_0\|, \qquad \lambda_\eta := \frac{c(\gamma,\eta)}{1+\delta_*(\gamma)}. \tag{218}$$

We now substitute the explicit expressions derived in the previous steps. For the unperturbed rate:

$$\Lambda_{\text{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R\,\gamma}{Z(\gamma)^2 + 2k_{\text{reg}}}, \quad \text{with } Z(\gamma) = \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}. \tag{219}$$

For the perturbation norm, we use the sharp bound (197) at $\varpi_* = \lambda(Q_O)$ to handle the macroscopic singularity:

$$\|\mathbf{E}(\gamma)\| \le b_{2,\text{eff}} + \underbrace{\frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}}_{\alpha_*(\gamma)} C_{\text{sing}}\,\lambda(Q_O)^{-1/2}. \tag{220}$$

Substituting these into the expression for λ_η and using the explicit distortion bound on $\delta_*(\gamma)$:

$$\lambda_\eta \ge \frac{\frac{1}{4}\lambda_R\,\gamma}{(1+\delta_*(\gamma))(Z(\gamma)^2 + 2k_{\text{reg}})} - \frac{\eta}{1+\delta_*(\gamma)}\left(b_{2,\text{eff}} + \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}} C_{\text{sing}}\,\lambda(Q_O)^{-1/2}\right). \tag{221}$$

Equivalently,

$$\lambda_\eta \ge \frac{\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)}{1+\delta_*(\gamma)} \ge (2-\sqrt{2})\big(\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)\big), \tag{222}$$

using $1+\delta_*(\gamma) \le 1 + 2^{-1/2}$, i.e. $(1+\delta_*(\gamma))^{-1} \ge (1+2^{-1/2})^{-1} = 2-\sqrt{2}$. This is the rigorous lower bound. The potentially singular factor $C_{\text{sing}}\lambda(Q_O)^{-1/2}$ is multiplied by the mixing weight $\alpha_*(\gamma) = \gamma\lambda_R/(Z(\gamma)^2 + 2k_{\text{reg}})$, which scales as $\sqrt{\lambda(Q_O)}$ at the optimal lift, ensuring the perturbation contribution remains bounded.

Finally, converting back to the L^2 norm via Grönwall's lemma yields the global estimate:

$$\|e^{t\mathcal{L}} f\|_{\mathcal{H}} \le C_*(\gamma)\, e^{-\lambda_\eta t}\|f\|_{\mathcal{H}} \le (1+\sqrt{2})\, e^{-\lambda_\eta t}\|f\|_{\mathcal{H}}. \tag{223}$$

This completes the proof.

C.5. **Optimization over Friction γ: Kinetic Acceleration.** We determine γ_{opt} maximising $\Lambda_{\text{hypo}}(\gamma)$. From (189), setting $x := \Theta\lambda(Q_O)^{-1/2}\gamma$, the rate function reads

$$\Lambda_{\text{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R}{\Theta}\sqrt{\lambda(Q_O)} \cdot \frac{x}{(x + K_{\text{geom}})^2 + 2k_{\text{reg}}}. \tag{224}$$

Optimal friction. The function $x \mapsto x/[(x+K_{\text{geom}})^2 + C_0]$ with $C_0 := 2k_{\text{reg}}$ is maximised when $(x+K_{\text{geom}})^2 + C_0 = 2x(x+K_{\text{geom}})$, equivalently $x^2 = K_{\text{geom}}^2 + C_0$. Translating back:

$$\gamma_{\text{opt}} = \frac{\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}}}}{\Theta}\sqrt{\lambda(Q_O)} =: C_{\text{lift}}\sqrt{\lambda(Q_O)}, \tag{225}$$

where $C_{\text{lift}} := \Theta^{-1}\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}}}$.

Maximal rate. At $\gamma = \gamma_{\text{opt}}$ one has $x_* := \Theta\lambda(Q_O)^{-1/2}\gamma_{\text{opt}} = \sqrt{K_{\text{geom}}^2 + C_0}$. Substituting into the denominator: $(x_* + K_{\text{geom}})^2 + C_0 = 2(K_{\text{geom}}^2 + C_0 + K_{\text{geom}}x_*)$. Hence

$$\Lambda_{\max} := \Lambda_{\text{hypo}}(\gamma_{\text{opt}}) = \frac{\frac{1}{4}\lambda_R}{2\Theta(x_* + K_{\text{geom}})}\sqrt{\lambda(Q_O)} \asymp \sqrt{\lambda(Q_O)}. \tag{226}$$

This confirms the kinetic acceleration: $\Lambda_{\max} \asymp \sqrt{\lambda(Q_O)}$, compared to the overdamped ceiling $\Lambda_{\text{hypo}}(\gamma) \asymp \lambda(Q_O)/\gamma$ as $\gamma \to \infty$.

Certified threshold at optimal lift. The admissible perturbation range certified by Theorem 5.1 is determined as follows. From (199), the exact matrix threshold depends on $\|\mathbf{E}(\gamma)\|$, while the

theorem uses the explicit upper bound $B_{\mathrm{pert}}(\gamma)$. The dominant term involves the mixing weight $\alpha_*(\gamma)$. Evaluating at γ_{opt}:

$$\alpha_*(\gamma_{\mathrm{opt}}) = \frac{\gamma_{\mathrm{opt}}\lambda_R}{Z(\gamma_{\mathrm{opt}})^2 + 2k_{\mathrm{reg}}}. \tag{227}$$

Using $\gamma_{\mathrm{opt}} = C_{\mathrm{lift}}\sqrt{\lambda(Q_O)}$ and $Z(\gamma_{\mathrm{opt}}) = \sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}} + K_{\mathrm{geom}} = \mathcal{O}(1)$, the numerator of $\alpha_*(\gamma_{\mathrm{opt}})$ is $\gamma_{\mathrm{opt}}\lambda_R \asymp \sqrt{\lambda(Q_O)}$ and the denominator $Z(\gamma_{\mathrm{opt}})^2 + 2k_{\mathrm{reg}} = \mathcal{O}(1)$ (independent of $\lambda(Q_O)$). Hence $\alpha_*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}$, so $\alpha_*(\gamma_{\mathrm{opt}}) \cdot \lambda(Q_O)^{-1/2} = \mathcal{O}(1)$. Applying (197):

$$\|\mathbf{E}(\gamma_{\mathrm{opt}})\| \le b_{2,\mathrm{eff}} + \alpha_*(\gamma_{\mathrm{opt}})\, C_{\mathrm{sing}}\, \lambda(Q_O)^{-1/2} = \mathcal{O}(1). \tag{228}$$

Since $\Lambda_{\max} \asymp \sqrt{\lambda(Q_O)}$ and $B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) = \mathcal{O}(1)$, the certified threshold scales as

$$\eta^*(\gamma_{\mathrm{opt}}) = \frac{\Lambda_{\max}}{2B_{\mathrm{pert}}(\gamma_{\mathrm{opt}})} \asymp \sqrt{\lambda(Q_O)}. \tag{229}$$

This confirms that the accelerated regime retains an $\mathcal{O}(\sqrt{\lambda(Q_O)})$ admissible perturbation range.

Appendix D. Upper bound via singular value gap

This appendix proves Theorem 5.6.

D.1. **Exponential decay implies a singular value-gap bound.** For a (possibly unbounded) closed operator T on $\mathcal{H}_0$, define its injectivity modulus

$$s(T) := \inf_{f \in \mathrm{dom}(T)\cap\mathcal{H}_0\setminus\{0\}} \frac{\|Tf\|}{\|f\|}. \tag{230}$$

Lemma D.1 (From exponential decay to a singular value-gap bound)**.** *Assume* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$. *Assume moreover that* $e^{t\mathcal{L}}$ *is a contraction semigroup on* $\mathcal{H}_0$*, i.e.* $\|e^{t\mathcal{L}}f\| \le \|f\|$ *for all* $t \ge 0$ *and* $f \in \mathcal{H}_0$ *(equivalently,* $\mathcal{L}$ *is dissipative on* $\mathcal{H}_0$*:* $\mathrm{Re}\langle \mathcal{L}f, f\rangle \le 0$*). If there exist constants* $C \ge 1$ *and* $\nu > 0$ *such that*

$$\|e^{t\mathcal{L}}f\| \le C\, e^{-\nu t}\, \|f\|, \qquad \forall t \ge 0,\ \forall f \in \mathcal{H}_0, \tag{231}$$

then 0 *belongs to the resolvent set of* $\mathcal{L}|_{\mathcal{H}_0}$ *and*

$$\nu \le (1 + \log C)\, s(\mathcal{L}). \tag{232}$$

Proof. For $f \in \mathcal{H}_0$, define

$$u := -\int_0^\infty e^{t\mathcal{L}} f\, dt. \tag{233}$$

Set $t_* := (\log C)/\nu \ge 0$ (the time at which $Ce^{-\nu t}$ crosses 1). On $[0, t_*]$ use the contraction bound $\|e^{t\mathcal{L}}f\| \le \|f\|$; on $[t_*, \infty)$ use the exponential decay bound $\|e^{t\mathcal{L}}f\| \le Ce^{-\nu t}\|f\|$. Combining:

$$\|u\| \le \int_0^{t_*} \|f\|\, dt + \int_{t_*}^\infty Ce^{-\nu t}\|f\|\, dt = t_*\, \|f\| + \frac{C}{\nu} e^{-\nu t_*}\, \|f\| = \frac{\log C}{\nu}\, \|f\| + \frac{1}{\nu}\, \|f\| = \frac{1 + \log C}{\nu}\, \|f\|, \tag{234}$$

where we used $Ce^{-\nu t_*} = Ce^{-\log C} = 1$. Moreover, $u \in \mathrm{dom}(\mathcal{L})$ and differentiating under the integral gives

$$\mathcal{L}u = -\int_0^\infty \mathcal{L}e^{t\mathcal{L}} f\, dt = -\int_0^\infty \partial_t\big(e^{t\mathcal{L}} f\big)\, dt = f. \tag{235}$$

Thus $(\mathcal{L}|_{\mathcal{H}_0})^{-1}$ exists and satisfies $\|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\| \le (1 + \log C)/\nu$, hence $s(\mathcal{L}) = \|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\|^{-1} \ge \nu/(1 + \log C)$, i.e. $\nu \le (1 + \log C)\, s(\mathcal{L})$. □

Lemma D.2 (Relaxation time is bounded below by the singular value gap)**.** *Assume* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$*. Assume moreover that* $e^{t\mathcal{L}}$ *is a contraction semigroup on* $\mathcal{H}_0$*, i.e.* $\|e^{t\mathcal{L}} f\| \le \|f\|$ *for all* $t \ge 0$ *and* $f \in \mathcal{H}_0$*. Assume there exist constants* $C \ge 1$ *and* $\nu > 0$ *such that*

$$\|e^{t\mathcal{L}} f\| \le C\, e^{-\nu t}\, \|f\|, \qquad \forall t \ge 0,\ \forall f \in \mathcal{H}_0. \tag{236}$$

Define the L^2 *relaxation time*

$$t_{\mathrm{rel}}(\mathcal{L}) := \inf\Big\{ t \ge 0\,;\ \|e^{t\mathcal{L}} f\| \le e^{-1}\, \|f\| \text{ for all } f \in \mathcal{H}_0 \Big\}. \tag{237}$$

Then

$$t_{\mathrm{rel}}(\mathcal{L}) \ge \frac{1}{2\, s(\mathcal{L})}. \tag{238}$$

This L^2 *bound is a continuous-time analogue of discrete-time relaxation estimates based on singular values for non-reversible Markov chains; compare* [10].

Proof. By the semigroup property and the definition of $t_{\mathrm{rel}}(\mathcal{L})$, for any $t \ge 0$ write $t = n\, t_{\mathrm{rel}}(\mathcal{L}) + r$ with $n \in \mathbb{N}_0$ and $r \in [0, t_{\mathrm{rel}}(\mathcal{L}))$. Since $e^{t\mathcal{L}}$ is a contraction on $\mathcal{H}_0$, $\|e^{r\mathcal{L}}\| \le 1$. Combined with $\|e^{t_{\mathrm{rel}}(\mathcal{L})\,\mathcal{L}}\| \le e^{-1}$, the semigroup property gives

$$\|e^{t\mathcal{L}}\| \le \|e^{r\mathcal{L}}\|\, \|e^{t_{\mathrm{rel}}(\mathcal{L})\,\mathcal{L}}\|^n \le 1 \cdot e^{-n} = e^{-n}.$$

As in the proof of Lemma D.1, the inverse on $\mathcal{H}_0$ is given by $(\mathcal{L}|_{\mathcal{H}_0})^{-1} = -\int_0^\infty e^{t\mathcal{L}}\, dt$, hence

$$\|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\| \le \int_0^\infty \|e^{t\mathcal{L}}\|\, dt \le \sum_{n=0}^{\infty} \int_0^{t_{\mathrm{rel}}(\mathcal{L})} e^{-n}\, dr = \frac{t_{\mathrm{rel}}(\mathcal{L})}{1 - e^{-1}} \le 2\, t_{\mathrm{rel}}(\mathcal{L}).$$

Therefore $s(\mathcal{L}) = \|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\|^{-1} \ge (2\, t_{\mathrm{rel}}(\mathcal{L}))^{-1}$. □

D.2. **Test function and completion of the proof.**

Proof of Theorem 5.6. Let $\phi \in \mathrm{dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0$ be a normalized eigenfunction of Q_O with

$$Q_O \phi = -\lambda(Q_O)\, \phi, \qquad \|\phi\| = 1. \tag{239}$$

Define the first-order corrector

$$\psi := \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} \phi \in \mathcal{H}_F, \tag{240}$$

and the test function $f_\gamma := \phi - \gamma^{-1}\psi$. Since $\phi \in \mathcal{H}_S$ and $\psi \in \mathcal{H}_F$ we have $\langle \phi, \psi \rangle = 0$ and therefore

$$\|f_\gamma\|^2 = \|\phi\|^2 + \gamma^{-2}\|\psi\|^2 \ge 1. \tag{241}$$

Using $\mathcal{R}\phi = 0$ and $\mathcal{R}\mathcal{R}^+ = P_F$ on $\mathcal{H}$, together with centered coupling $\mathcal{V}_{\mathrm{skew}}\phi \in \mathcal{H}_F$, we obtain

$$\mathcal{R}\psi = \mathcal{R}\mathcal{R}^+ \mathcal{V}_{\mathrm{skew}}\phi = P_F \mathcal{V}_{\mathrm{skew}}\phi = \mathcal{V}_{\mathrm{skew}}\phi. \tag{242}$$

Consequently,

$$\mathcal{L} f_\gamma = \big(\gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}\big)\big(\phi - \gamma^{-1}\psi\big) \tag{243}$$

$$= \underbrace{\mathcal{V}_{\mathrm{skew}}\phi - \mathcal{R}\psi}_{=0} - \gamma^{-1}\mathcal{V}_{\mathrm{skew}}\psi \ + \ \eta\mathcal{V}_{\mathrm{pert}} f_\gamma. \tag{244}$$

We decompose the remaining transport term into slow/fast components:

$$\mathcal{V}_{\mathrm{skew}}\psi = P_S \mathcal{V}_{\mathrm{skew}}\psi + P_F \mathcal{V}_{\mathrm{skew}}\psi. \tag{245}$$

For the slow part, using $Q_O := -P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} P_S$ and $P_S\phi = \phi$, we have

$$P_S \mathcal{V}_{\mathrm{skew}}\psi = P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}}\phi = -Q_O\phi = \lambda(Q_O)\, \phi. \tag{246}$$

For the fast part, apply the flux regularity estimate (20) with the slow input ϕ. Writing $g_\phi := P_F\mathcal{V}_{\rm skew}P_S\phi = \mathcal{V}_{\rm skew}\phi$ and noting that $(-\mathcal{R}^+)g_\phi = -\psi$ on $\mathcal{H}_F$, we obtain

$$\|P_F\mathcal{V}_{\rm skew}\psi\| = \|P_F\mathcal{V}_{\rm skew}P_F\psi\| = \|P_F\mathcal{V}_{\rm skew}P_F(-\mathcal{R}^+)g_\phi\| \le \kappa_{\rm ell}\,\|(-Q_O)\phi\| = \kappa_{\rm ell}\,\lambda(Q_O). \tag{247}$$

Combining the slow and fast contributions yields

$$\|\mathcal{V}_{\rm skew}\psi\| \le (1+\kappa_{\rm ell})\,\lambda(Q_O). \tag{248}$$

Hence

$$\|\mathcal{L}f_\gamma\| \le \frac{(1+\kappa_{\rm ell})\,\lambda(Q_O)}{\gamma} + \eta\,\|\mathcal{V}_{\rm pert}f_\gamma\|. \tag{249}$$

To bound the perturbation, use Assumption 4.2.4 to estimate

$$\|\mathcal{V}_{\rm pert}f\| \le \|\mathcal{K}_{\rm pert}f\| + \|\mathcal{S}_{\rm pert}f\| \tag{250}$$

$$\le (a_1+a_2)\,\|(-\mathcal{R})^{1/2}f\| + (b_1+b_2)\,\|f\|, \qquad f\in{\rm dom}((-\mathcal{R})^{1/2}). \tag{251}$$

Since $\phi\in\ker(\mathcal{R})$ and $f_\gamma = \phi - \gamma^{-1}\psi$, we have

$$\|(-\mathcal{R})^{1/2}f_\gamma\| = \gamma^{-1}\,\|(-\mathcal{R})^{1/2}\psi\|. \tag{252}$$

Finally, using $\psi = \mathcal{R}^+\mathcal{V}_{\rm skew}\phi$ and the definition of Q_O,

$$\|(-\mathcal{R})^{1/2}\psi\|^2 = \langle(-\mathcal{R})\psi,\psi\rangle = \langle\mathcal{V}_{\rm skew}\phi,(-\mathcal{R}^+)\mathcal{V}_{\rm skew}\phi\rangle = -\langle Q_O\phi,\phi\rangle = \lambda(Q_O). \tag{253}$$

Therefore $\|(-\mathcal{R})^{1/2}f_\gamma\| = \sqrt{\lambda(Q_O)}/\gamma$, and

$$\|\mathcal{V}_{\rm pert}f_\gamma\| \le (b_1+b_2)\,\|f_\gamma\| + (a_1+a_2)\,\frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{254}$$

Dividing by $\|f_\gamma\|\ge 1$ gives

$$\frac{\|\mathcal{L}f_\gamma\|}{\|f_\gamma\|} \le \frac{(1+\kappa_{\rm ell})\,\lambda(Q_O)}{\gamma} + \eta\,(b_1+b_2) + \eta\,(a_1+a_2)\,\frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{255}$$

By definition of $s(\mathcal{L})$, this implies $s(\mathcal{L}) \le \|\mathcal{L}f_\gamma\|/\|f_\gamma\|$. Combining with Lemma D.1 yields the claimed bound on ν. □

Appendix E. Proof of Proposition 6.5

We verify Assumption 4.2.1–4 and Assumption 4.5 for the forced Langevin decomposition on the core $\mathcal{C}_{\rm Lan} := C^\infty(\mathbb{T}^d)\otimes\mathcal{S}(\mathbb{R}^d)\subset L^2(\mu_0)$. The space $\mathcal{C}_{\rm Lan}$ is invariant under P_S, P_F, $\mathcal{R}^+$, $\mathcal{V}_{\rm skew}$, and $\mathcal{V}_{\rm pert}$; moreover, for $\psi=\psi(q)$ one has

$$\mathcal{R}^+\mathcal{V}_{\rm skew}\psi = -p\cdot\nabla_q\psi \in \mathcal{C}_{\rm Lan},$$

so the compositions defining Q_O and the first-order corrector in Theorem 5.6 stay inside the core. Since $Q_O = \beta^{-1}\Delta_q - \nabla U\cdot\nabla_q$ is elliptic on the compact torus with smooth coefficients, elliptic regularity ensures that the resolvent $(\varpi - Q_O)^{-1}$ maps $C^\infty(\mathbb{T}^d)$ to $C^\infty(\mathbb{T}^d)$; hence $\mathcal{C}_{\rm Lan}\cap\mathcal{H}_S$ is invariant under the resolvent, and $C^\infty(\mathbb{T}^d)$ is a core for Q_O. The bilinear forms used in the appendices extend by density to the stated energy spaces.

E.1. **Microscopic spectral gap and centering.** The Ornstein–Uhlenbeck operator $\mathcal{R} = -p\cdot\nabla_p + \beta^{-1}\Delta_p$ acts only on p-variables and has spectral gap $\lambda_R = 1$ on $\mathcal{H}_F = \ker(\Pi_p)$ (classical; see e.g. [39]). For Assumption 4.2.2: $\mathcal{V}_{\rm skew}^\dagger = -\mathcal{V}_{\rm skew}$ (Hamiltonian flow preserves μ_0), $P_S\mathcal{V}_{\rm skew}P_S = 0$ (Gaussian centering $\int p\,d\nu_p = 0$), and $P_S\mathcal{V}_{\rm pert}P_S = 0$ (since $\nabla_p\phi(q) = 0$).

E.2. **Flux regularity (Assumption 4.2.3).** For $\psi = \psi(q) \in \mathcal{H}_S$, the flux is $g_\psi = p \cdot \nabla_q \psi \in \mathcal{H}_F$. Since $\mathcal{R}(p_i) = -p_i$, we have $(-\mathcal{R}^+)g_\psi = g_\psi$, giving $M_R = 1$.

For the elliptic regularity bound, define $Z_{ij}(p) := p_i p_j - \beta^{-1}\delta_{ij}$. A direct computation yields

$$P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+)g_\psi = \sum_{i,j} Z_{ij}(p)\, \partial_{q_i q_j}\psi. \tag{256}$$

By Cauchy–Schwarz and Gaussian moment calculations ($\mathbb{E}[p_i^4] = 3\beta^{-2}$),

$$\|P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+)g_\psi\|_{L^2(\mu_0)} \le c_{d,\beta}\, \|\nabla_q^2 \psi\|_{L^2(\nu)}, \qquad c_{d,\beta} = \frac{\sqrt{d(d+1)}}{\beta}. \tag{257}$$

It remains to control $\|\nabla_q^2\psi\|_{L^2(\nu)}$ by $\|(-Q_O)\psi\|_{L^2(\nu)}$.

Step 1 (Hessian vs. Laplacian). By Parseval's identity on $\mathbb{T}^d$, $\|\nabla_q^2\psi\|_{L^2(dq)} = \|\Delta_q\psi\|_{L^2(dq)}$ for periodic ψ. Switching to $L^2(\nu)$ with $\rho_{\min} \le \rho \le \rho_{\max}$ ($\rho \propto e^{-\beta U}$) gives

$$\|\nabla_q^2\psi\|_{L^2(\nu)} \le \sqrt{\rho_{\max}/\rho_{\min}}\, \|\Delta_q\psi\|_{L^2(\nu)}. \tag{258}$$

Step 2 (Laplacian vs. Q_O). From $\Delta_q\psi = \beta(Q_O\psi + \nabla U \cdot \nabla_q\psi)$ and the Dirichlet-form identity $\langle(-Q_O)\psi, \psi\rangle_{L^2(\nu)} = \beta^{-1}\|\nabla_q\psi\|^2_{L^2(\nu)}$, one derives via Cauchy–Schwarz and the Poincaré inequality for ν:

$$\|\nabla_q\psi\|_{L^2(\nu)} \le \beta\sqrt{C_{\mathrm{P}}(\nu)}\, \|(-Q_O)\psi\|_{L^2(\nu)}, \tag{259}$$

and hence $\|\Delta_q\psi\|_{L^2(\nu)} \le \beta(1 + \beta M_U \sqrt{C_{\mathrm{P}}(\nu)})\, \|(-Q_O)\psi\|_{L^2(\nu)}$, where $M_U := \|\nabla U\|_{L^\infty}$.

Combining Steps 1–2 with (257) yields the flux regularity bound $\kappa_{\text{ell}} = c_{d,\beta} \cdot \sqrt{\rho_{\max}/\rho_{\min}} \cdot \beta(1 + \beta M_U \sqrt{C_{\mathrm{P}}(\nu)})$.

E.3. **Perturbation energy bounds (Assumption 4.2.4).** By integration by parts in p, the $L^2(\mu_0)$-adjoint is $\mathcal{V}^\dagger_{\text{pert}} = -F \cdot \nabla_p + \beta(F \cdot p)$, giving the skew/symmetric decomposition:

$$\mathcal{K}_{\text{pert}} = F \cdot \nabla_p - \tfrac{\beta}{2}(F \cdot p), \qquad \mathcal{S}_{\text{pert}} = \tfrac{\beta}{2}(F \cdot p). \tag{260}$$

Since $\|\nabla_p f\| \le \sqrt{\beta}\, \|(-\mathcal{R})^{1/2} f\|$, the bounds reduce to controlling the multiplier $\|pf\|$ by $\|f\|$ and $\|\nabla_p f\|$. A Gaussian integration-by-parts argument with Young's inequality gives the explicit p-multiplier bound:

$$\||p|f\|_{L^2(\mu_0)} \le \sqrt{2d/\beta}\, \|f\|_{L^2(\mu_0)} + (2/\beta)\, \|\nabla_p f\|_{L^2(\mu_0)}. \tag{261}$$

Combining with $\|F\|_{L^\infty} < \infty$ yields Assumption 4.2.4(a)–(b) with constants (a_1, b_1, a_2, b_2) depending only on $(\beta, d, \|F\|_{L^\infty})$. □

Appendix F. Dual regularity of the modified DMS corrector (Lemma 2.2(iii))

Proof of Lemma 2.2(iii). We prove that $\|\mathcal{A}_m \mathcal{L}_{\text{ham}}(1 - \Pi_v)\phi\| \le \sqrt{2}\, \|\phi\|$ for all $\phi \in L^2(\rho_\infty)$, where U is convex. Set $\psi := (1 - \Pi_v)\phi \in \mathcal{H}_F$ and define $T : \mathcal{H}_F \to \mathcal{H}_S$ by $T\psi = \mathcal{A}_m \mathcal{L}_{\text{ham}}\psi$; it suffices to show $\|T\| \le \sqrt{2}$, which we do by bounding $\|T^*\|$.

Step 1: Adjoint computation. For $g \in \mathcal{H}_S$ (mean-zero), set $\tilde{h} := (m - Q)^{-1} g \in \mathcal{H}_S$. Self-adjointness of $(m - Q)^{-1}$ and skew-adjointness of $\mathcal{L}_{\text{ham}}$ give $\langle T\psi, g\rangle = \langle \mathcal{L}_{\text{ham}}\psi, \mathcal{L}_{\text{ham}}\Pi_v\tilde{h}\rangle = -\langle \psi, \mathcal{L}^2_{\text{ham}}\tilde{h}\rangle$. Since $\psi \in \mathcal{H}_F$, only the $\mathcal{H}_F$-component contributes: $T^* g = (1 - \Pi_v)(-\mathcal{L}^2_{\text{ham}}\tilde{h})$.

Step 2: Expanding $\mathcal{L}^2_{\text{ham}}\tilde{h}$. Since $\tilde{h} = \tilde{h}(x)$, $\mathcal{L}_{\text{ham}}\tilde{h} = v \cdot \nabla_x \tilde{h}$, and applying $\mathcal{L}_{\text{ham}}$ again: $\mathcal{L}^2_{\text{ham}}\tilde{h} = \sum_{i,j} v_i v_j\, \partial^2_{x_i x_j}\tilde{h} - \nabla_x U \cdot \nabla_x \tilde{h}$. The Π_v-component is $\Delta_x \tilde{h} - \nabla_x U \cdot \nabla_x \tilde{h} = Q\tilde{h}$ (using $\mathbb{E}_\kappa[v_i v_j] = \delta_{ij}$), so

$$(1 - \Pi_v)\mathcal{L}^2_{\text{ham}}\tilde{h} = \sum_{i,j} (v_i v_j - \delta_{ij})\, \partial^2_{x_i x_j}\tilde{h}. \tag{262}$$

Step 3: Computing $\|T^*g\|^2$ via Isserlis' lemma. From (262), $T^*g = -\sum_{i,j}(v_iv_j - \delta_{ij})\,\partial^2_{x_ix_j}\tilde{h}$. Expanding the square and integrating over v using the Gaussian fourth-moment formula (Isserlis' lemma), $\mathbb{E}_\kappa[(v_iv_j - \delta_{ij})(v_kv_l - \delta_{kl})] = \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}$, gives

$$\|T^*g\|^2 = 2\sum_{i,j}\int(\partial^2_{x_ix_j}\tilde{h})^2\,d\mu = 2\|\nabla^2_x\tilde{h}\|^2_{L^2(\mu)}. \tag{263}$$

Step 4: Bochner inequality on $L^2(\mu)$. For $\tilde{h} \in H^2(\mu)$ one has the Bochner identity

$$\|\nabla^2_x\tilde{h}\|^2_{L^2(\mu)} + \langle\nabla^2_xU\,\nabla_x\tilde{h}, \nabla_x\tilde{h}\rangle_{L^2(\mu)} = \|Q\tilde{h}\|^2_{L^2(\mu)}, \tag{264}$$

which is verified by writing $L = -Q = -\Delta_x + \nabla_xU\cdot\nabla_x$ on $L^2(\mu)$ and using the commutator $[\partial_{x_i}, L] = (\partial^2_{x_ix_k}U)\partial_{x_k}$ to expand $\|L\tilde{h}\|^2 = \sum_i\langle\partial_{x_i}\tilde{h}, \partial_{x_i}(L\tilde{h})\rangle_{L^2(\mu)}$. Since U is convex ($\nabla^2_xU \geq 0$), the Hessian term is non-negative, giving

$$\|\nabla^2_x\tilde{h}\|^2_{L^2(\mu)} \leq \|Q\tilde{h}\|^2_{L^2(\mu)}. \tag{265}$$

Step 5: Conclusion. Inserting (265) into (263): $\|T^*g\|^2 \leq 2\|Q\tilde{h}\|^2 = 2\|Q(m-Q)^{-1}g\|^2$. By spectral calculus, $\|Q(m-Q)^{-1}\| \leq 1$ (since $\lambda/(\varpi+\lambda) \leq 1$ for all $\lambda \geq 0$). Hence $\|T^*g\|^2 \leq 2\|g\|^2$, and since $\|T\| = \|T^*\|$, the claimed bound follows. The same constant $\sqrt{2}$ holds for every resolvent shift $\varpi > 0$: the Bochner–Weitzenböck bound is ϖ-independent, and ϖ enters the corrector only through the spectral contraction factor $\lambda/(\varpi+\lambda) \leq 1$. □

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